TFPT — The Standard Model from the Compiler

The φ_0-ladder, the flavor block from parabolic transport, and the worked closures (θ_{12}, quark c, the gap, Starobinsky M)

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What this document is about

The Standard Model as *one* φ_0-ladder mass formula (all masses, CKM, PMNS, neutrinos), the flavor matrix from parabolic transport on $\mathbb{P}^1 \setminus \mu_4$, the five worked closures (θ_{12}, quark c, the explicit mass gap, Starobinsky M, the H2 splitting), and **gravity/quantum gravity as the seam response** (the Seam Response Grammar $\xi u = c_3$, the *parameter-free* full covariant Einstein equation from entanglement equilibrium ($G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$, both coefficients fixed), the heat-kernel grounding of $R + R^2$, and the metric-coupled gap bound that *gap-decouples* the physical QG sector, leaving only the ambient projective *measure* as a mathematical-completeness item).

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1{+}i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading document 2 (Standard Model).

TFPT status at a glance — read this card the same way in every document

Two structural interfaces, one compiler. P1 is a primitive reflection-positive seam kernel with unit winding; its normalization is $c_3 = \frac{1}{8\pi}$. From this datum and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2), the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The numerical readouts c_3 and g_{car} are elementary symmetric

readouts of the anchor $a = (1, 1, 2)$. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below. Existing reductions share the four-mark geometry and anchor $a = (1, 1, 2)$; both W-bridge premises are derived at finite level (`v1014/v1015`), modulo the external MMST identification. The additional P1 determinant-response identification is separate. The Architecture chapter "Shared origin" consolidates these scopes (`REC20261105C3`).

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

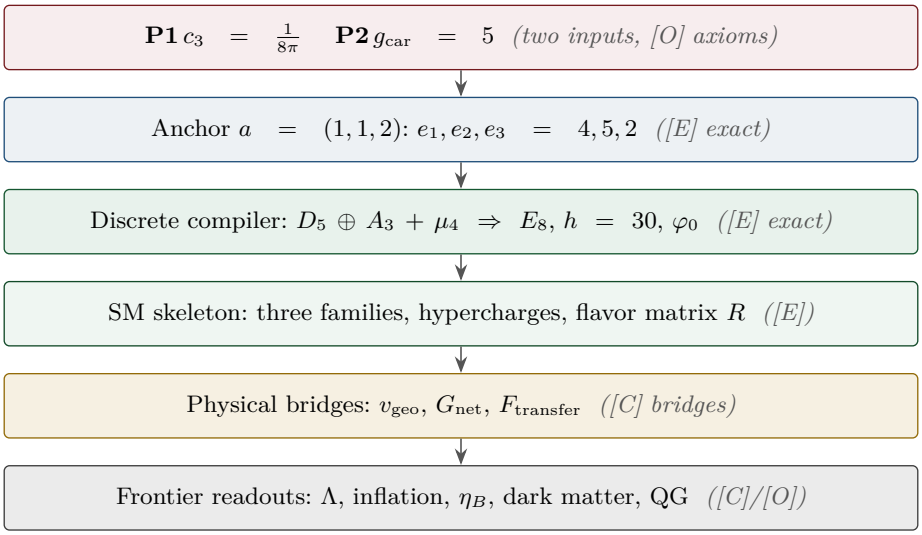
Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. The lattice and index witnesses and the cited scaling and OS theorems establish the stated conditional route; they do not leave every old prerequisite untouched. The later charged MMST manuscript proves its uniform Green, edge graph, crossed-product and relaxed TEL-B estimates and retains the named microscopic local-algebra exhaustion condition for the full source. Its P1 corollary separately requires identification of the determinant-line response with the additive transitive four-channel response. The declared operational seed already supplies state and time data. The physical question concerns applicability of these theorems to the same source. The unconditional seam parent and T1–T8 keep their ledger statuses. Under the conformal-net hypotheses, rotation invariance makes the geometric deck a consequence (`v335`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76/v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



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Part I
The Standard Model in one master formula

In one sentence. The fermion spectrum — masses, Yukawa structure, CKM, the PMNS skeleton and the neutrinos — follows from *one* master formula with *three* ingredients: the seed $\varphi_0^{\rm ret}$, the carrier base $\lambda_Y = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}$ and the *residue matrix of the compiler*. No free Yukawa *exponents* remain; the charged-lepton amplitudes are closed, the quark mass *ratios* are closed too (integer Plücker readouts, v49/v71), and only the *absolute* quark amplitude scale is a finite ($U_{\rm wall}$) anchor. **Markers:** [E] exact / machine-proven · [C] conditional (named hypotheses) · [O] open / axiom · [X] kill test — the fine per-claim type is in `verification/status_ledger.csv`.

1 The principle: one master formula instead of many Yukawa numbers

In the source-layer construction each fermion mass arises through a Yukawa coupling $y_{f,j}$ built as a transport word-length $L_{f,j}$ on a hexagon (C_6) times a holonomy amplitude. In the compiler picture this is *one* line:

Mass master formula (closed branch)

$$\hat{m}_{f,j} = \frac{v_{\rm geo}}{\sqrt{2}}\, y_{f,j}, \qquad y_{f,j} = \lambda_Y^{\,L_{f,j}}\, \Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\rm ret}\,(1-\varphi_0^{\rm ret})}$$

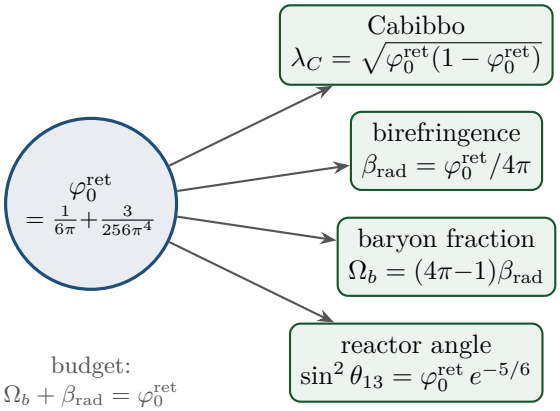
with *one* seed $\varphi_0^{\rm ret} = \dfrac{1}{6\pi} + \dfrac{3}{256\pi^4}$, the integer word-lengths $L_{f,j} = r_{f,j} + 6\,n_{f,j}$ (residue + winding) and $O(1)$ holonomy amplitudes $\Lambda_{f,j}$.

The point of the simplification: $L_{f,j}$ **is not a fit but the output of the compiler.** The residues $r_{f,j}$ form the matrix whose invariants are pure compiler numbers (Section 3); the windings $n_{f,j}$ are the anchors $\{1,1,2\}$. What used to be a separate "flavor transport" theorem is now the readout of a fixed matrix.

The seed base is the halved $(E_8)_1$ Weyl anomaly. The seed base $\frac{1}{6\pi} = \frac{4}{3}\,c_3$ carries the prefactor $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{\rm SM} = 16/12$. That same $\frac{4}{3}$ is the conformal trace anomaly of the seam: over the seam sphere S^2 ($\chi = 2$, $\int R\sqrt{g} = 8\pi$) the $(E_8)_1$ ($c = 8$) Weyl anomaly is $\frac{c}{24\pi}\int R\sqrt{g} = \frac{c\chi}{6} = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ Möbius identification to $\frac{4}{3}$. So the Cabibbo-seed prefactor is the halved $(E_8)_1$ conformal anomaly, not a free coefficient ([`verification/v172_trace_anomaly_seed.py`]).

2 The seed $\varphi_0^{\rm ret}$ as a decoder

A single seed drives four independent low-energy observables. This is the strongest compression of the flavor input:



Numerically: $\varphi_0^{\rm ret} = 0.0531720$, $\lambda_C = 0.224376$ (PDG 0.2245), $\sin^2\theta_{13} = 0.02311$ (PDG ≈ 0.0222). The exponent $\gamma = 5/6$ is the carrier trace $\mathrm{tr}_E\, Y^2$; its complement $1-\gamma = 1/6$ reappears in the neutrino ratio.

3 The flavor residue matrix is the compiler signature

The core of the simplification: the word-length matrix $L = R + 6W$ splits into the residue matrix R (the hexagon packets of the three sectors) and the winding matrix W (one +6 per first generation = anchor). R carries all the discrete flavor information — and its invariants are *exclusively* compiler numbers:

$$R = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}$$

rows = sectors u, d, e

$\mathrm{tr}\, R = 9 = N_{\rm fam}^2$

minors $2, 3, 5 \Rightarrow 2{\cdot}3{\cdot}5 = 30 = h(E_8)$

$\det R = 8 = h(D_5)$

Spectral signature of the flavor matrix (verified)

The characteristic polynomial of the residue matrix is built entirely from compiler constants:

$$\chi_R(t) = t^3 - \underbrace{9}_{N_{\rm fam}^2}\, t^2 + \underbrace{10}_{\mathcal{A}_L = \binom{5}{2}}\, t - \underbrace{8}_{h(D_5)},$$

where the linear coefficient $10 = 2+3+5$ is the sum of the three principal $2{\times}2$ minors and their *product* $2\cdot 3\cdot 5 = 30$ is the Coxeter number $h(E_8)$. The flavor matrix is therefore not an independent object — it is a *reading of the same compiler* ($N_{\rm fam}$, $\mathcal{A}_L$, $h(D_5)$, $h(E_8)$). **[E]** [`verification/v4_flavor_matrix.py`]

Concretely:

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}}_{R\ (\det 8,\ \text{minors } 2,3,5)} + 6\ \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{W\ (\text{anchor } \{1,1,2\})}, \qquad \sum L = 40 = \Omega_{\rm adm} \cdot \tfrac{5}{6}.$$

The sum $\sum L = 40$ is the transport budget number; with the Higgs contribution $N_\Phi = 1$ it gives $40 + 1 = 41 = 10\, b_1$ — exactly the coefficient the EM fixed point $\alpha_\star^{-1}$ reads.

The winding-deformation theorem: $R \to L$ in one parameter [E]

Treat the winding as a one-parameter rank-one deformation $R_s = R + s\,\mathbf{1}e_1^\top$ ($R_0 = R$, $R_6 = L$). Then

the *entire* flavor linear algebra is one line,

$$\boxed{\chi_{R_s}(t) = t^3 - (9+s)\,t^2 + (10+5s)\,t - (8+2s)},$$

with $\det R_s = 8+2s$, $\operatorname{tr} R_s = 9+s$, $\mathrm{PrinMin}_2(R_s) = \big(5,\ 3(s{+}1),\ 2(s{+}1)\big)$, and $R_s^{-1}\mathbf{1} = \frac{1}{4+s}(1,1,-1)^\top$. The physical winding value is fixed, not chosen:

$$\boxed{s = |R^+(A_3)| = 6 \text{ is } \textit{triply locked}\text{:}\quad \underbrace{6+4s=30}_{R_s^\top a \to h(E_8)},\ \underbrace{8+2s=20}_{\det \to 2\mathcal{A}_L},\ \underbrace{9+s=15}_{\operatorname{tr}\to \dim A_3}\ \Rightarrow\ s=6.}$$

At $s=6$ this gives $\chi_L = t^3 - 15t^2 + 40t - 20$ with $15 = \dim A_3$, $40 = |R(D_5)|$, $20 = |R(A_4)| = 2\mathcal{A}_L$, and $\mathrm{PrinMin}_2(L) = (5, 21, 14) = (g_{\mathrm{car}}, 3{\cdot}7, 2{\cdot}7)$ — the lift multiplies the sheet and family minors by the successor $s{+}1 = 7 = |R^+(A_3)|{+}1$ while the carrier minor 5 stays fixed. So the "+6 winding" is not "add six because hexagon": it is the unique value selected three independent ways, and the appearance of 7 in the scalaron is $s{+}1$.

Winding reality, the determinant lift, and the invariant dual anchor (machine-verified)

The rank-one winding $R \to L = R + 6\,\mathbf{1}e_1^\top$ is far from a cosmetic offset:

- **Winding makes the flavour system spectrally real.** R has a *complex* eigenpair ($\operatorname{disc}\chi_R = -2^2 \cdot 1999 < 0$, spectrum $7.857,\ 0.572 \pm 0.832i$), while L has three positive reals, with a fully compiler-factorised discriminant (two equivalent readings)

$$\operatorname{disc}\chi_L = 39200 = 2^5 \cdot g_{\mathrm{car}}^2 \cdot 7^2 = 2\,(\mathcal{A}_L \cdot \dim G_2)^2 = 2\,(10 \cdot 14)^2.$$

- **The anchor Coxeter descent.** The anchor projects onto a clean three-scale ladder,

$$R^\top a = (6, 18, 8), \qquad L^\top a = (30, 18, 8) = \big(h(E_8),\ N_{\mathrm{fam}}|R^+(A_3)|,\ h(D_5)\big),$$

  so the winding lifts $|R^+(A_3)| = 6 \to h(E_8) = 30$ (a $+24 = |W(A_3)|$ shift), with descent gaps $30 - 18 = 12 = |R(A_3)| = \dim \mathfrak{g}_{\mathrm{SM}}$ and $18 - 8 = 10 = \mathcal{A}_L$. Column-wise the lift is $\mathbf{1}^\top R = (4, 13, 5) \to \mathbf{1}^\top L = (22, 13, 5)$, adding 18 only in the first generation — which is thus literally the store of the family-hexagon winding.
- **The determinant lift is the A_3 winding over μ_4.** Since $e_1^\top R^{-1}\mathbf{1} = \frac{1}{4} = \frac{1}{|\mu_4|}$,

$$\det L = \det R\Big(1 + \tfrac{|R^+(A_3)|}{|\mu_4|}\Big) = 8\big(1 + \tfrac{6}{4}\big) = 20 = 2\mathcal{A}_L,$$

  so the D_5 selector 8 is lifted to the decuple-doublet 20 by the A_3 winding. The cokernel moves $\mathbb{Z}_8 \to \mathbb{Z}_{20}$, i.e. the winding injects the carrier factor 5 into the torsion.
- **The dual anchor is winding-invariant.** The first column is the anchor, $Re_1 = a = (1, 1, 2)$, and the dual covector is preserved by the lift,

$$a^\top R^{-1} = a^\top L^{-1} = \big(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\big), \qquad \big(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\big)\cdot\mathbf{1} = 0,$$

  so winding changes the absolute hierarchy but *not* the anchor-dual sheet balance. The uniform response keeps the sheet value fixed: $\mathbf{1}^\top \mathrm{adj}(R)\mathbf{1} = \mathbf{1}^\top \mathrm{adj}(L)\mathbf{1} = 2 = |\mathbb{Z}_2|$, while the inverse response denominator runs $|\mu_4| = 4 \to \mathcal{A}_L = 10$. **[E]**

Audit-level only: flavor linear-algebra fingerprints (not load-bearing). The following are *audit fingerprints* (exact linear algebra of R, L), kept here, not in any derivation, under the no-free-pattern rule. **[E]**

- **Cofactor seam normal.** Generations 2, 3 share columns, $c_2{=}(3,5,5)^\top$, $c_3{=}(0,2,3)^\top$; their cross product $n = c_2 \times c_3 = (5, -9, 6) = (g_{\mathrm{car}}, -N_{\mathrm{fam}}^2, |R^+(A_3)|)$ gives the determinant geometrically: $n \cdot a = 8 = \det R$, $n \cdot (a{+}6\mathbf{1}) = 20 = \det L$, $n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|$.
- **Adjugate selector (⇒ why only gen. 1 winds).** n is also the first row of both adjugates, $e_1^\top \mathrm{adj}(R) = e_1^\top \mathrm{adj}(L) = (5, -9, 6)$, so $(5, -9, 6)R = (8, 0, 0)$ and $(5, -9, 6)L = (20, 0, 0)$ — it annihilates the unchanged c_2, c_3 plane.

- **Length polynomial as an action polynomial.** $\chi_L(t) = t^3 - \dim(A_3)\,t^2 + |R(D_5)|\,t - |R(A_4)| = t^3 - 15t^2 + 40t - 20$.
- **Principal minors.** $\mathrm{PrinMin}_2(L) = (5, 21, 14) = (g_{\mathrm{car}}, 3{\cdot}7, 2{\cdot}7)$ with $\sum = 40 = |R(D_5)|$, $\prod = 1470 = 7^2 h(E_8)$.
- **Commutator is a pure winding rotation.** $[R, L]$ has rank 2, trace 0, and char. poly $\lambda(\lambda^2 + 432)$ — a single 2D rotation in the non-anchor plane with a null direction.
- **The filtration fingerprint (2026-08-06).** The row budgets of the filtration $Q \to K \to L$ are exact, *order-exact* counts of the $K_{3,2}/\mathrm{prism}/K_5/H_8$ incidence chain, with level-wise additivity mirroring $L = K + Q$; *no* entry-level reconstruction succeeds from any frozen incidence candidate (exhaustive false-positive rates 0.005–0.75% make the misses informative) — an incidence-compiler fingerprint, typed audit-only per this section's rule ([`verification/v812_flavor_graph_filtration.py`], 16/16, FILTRATION-FINGERPRINT-ONLY) **[E]**.

4 Charged leptons: completely closed in φ_0^{ret}

For the leptons the hexagonal resolvent solves the amplitudes Λ_ℓ in closed form — only pure φ_0^{ret} powers with rational carrier prefactors remain:

Absolute lepton source masses

$$(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\mathrm{geo}}\,\pi}{\sqrt{2}}\left(\tfrac{16}{7}\,(\varphi_0^{\mathrm{ret}})^5,\ \tfrac{4}{3}\,(\varphi_0^{\mathrm{ret}})^3,\ \tfrac{7}{6}\,(\varphi_0^{\mathrm{ret}})^2\right),$$

and hence the *parameter-free* ratios

$$\frac{\hat{m}_\mu}{\hat{m}_\tau} = \frac{8}{7}\,\varphi_0^{\mathrm{ret}}, \qquad \frac{\hat{m}_e}{\hat{m}_\mu} = \frac{12}{7}\,(\varphi_0^{\mathrm{ret}})^2, \qquad \frac{\hat{m}_e\,\hat{m}_\tau}{\hat{m}_\mu^2} = \frac{3}{2}\,\varphi_0^{\mathrm{ret}}.$$

Where the lepton rationals come from: $\delta = \frac{1}{2}$ resolvent + product rule [E]

The coefficients $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ are not fitted — they are *derived*, and the key is that they are *rational*, so they cannot come from the irrational pole δ_{ph}. They come from the *distinguished* transport value $\delta = \frac{1}{2}$ (rational), evaluated at the lepton leg $y = 1$ ($= |Y(e^c)|$). **The value $\delta = \frac{1}{2}$ itself is a glue-norm readout:** the two E_8-glue norms are $q(D_5) = g_{\mathrm{car}}/|\mu_4| = \frac{5}{4}$ and $q(A_3) = N_{\mathrm{fam}}/|\mu_4| = \frac{3}{4}$, so $\delta = q(D_5) - q(A_3) = (g_{\mathrm{car}} - N_{\mathrm{fam}})/|\mu_4| = |\mathbb{Z}_2|/|\mu_4| = \frac{1}{2}$ — the carrier-minus-family count over the glue index (the same pair whose *sum* is the E_8 root norm 2, architecture document), matching the harmonic-frame holonomy diagonal modulus $\frac{1}{2}$. So δ is fixed, not chosen ([`verification/v51_boundary_half_step.py`]). With that value,

$$\mathrm{amplitude}(r) = \left|\frac{1}{1 - \frac{1}{2}\zeta^r}\right|^2 = \frac{1}{\frac{5}{4} - \cos(r\pi/3)}, \qquad \zeta = e^{i\pi/3},\ r = L \bmod 6.$$

Non-anchor leptons (dressed by the winding factor $|\mu_4| = 4$, $w = \lfloor L/6 \rfloor$):

$$c = |\mu_4|^{\,w}\,\frac{1}{\frac{5}{4} - \cos(r\pi/3)}: \quad c_e = 4^1 \cdot \tfrac{4}{7} = \tfrac{16}{7}, \qquad c_\mu = 4^0 \cdot \tfrac{4}{3} = \tfrac{4}{3} \quad \text{(exact)}.$$

Anchor (heaviest lepton τ) fixed by the *sector product rule*

$$\boxed{c_e\,c_\mu\,c_\tau = \frac{2^{g_{\mathrm{car}}}}{N_{\mathrm{fam}}^2} = \frac{32}{9}} \quad \Longrightarrow \quad c_\tau = \frac{32/9}{(16/7)(4/3)} = \frac{7}{6} \quad \text{(exact)}.$$

Every ingredient is a compiler datum ($\delta{=}\frac{1}{2}$, $|\mu_4|$, hexagon angles, g_{car}, N_{fam}). The product constant reads $2^{g_{\mathrm{car}}} = 32 = \dim(S^+ \oplus S^-)$ (the full carrier Clifford space) over N_{fam}^2, and the lepton word-lengths obey $\sum_j L_j = 16 = \dim S^+$, $\sum_j (L_j - K_j) = 6 = |R^+(A_3)|$. *Edge–centre ring:* the two boundary generations close on the middle one, $c_e\,c_\tau = \frac{16}{7} \cdot \frac{7}{6} = \frac{8}{3} = |\mathbb{Z}_2|\,c_\mu$, so the lepton amplitudes form a closed algebraic ring with sheet-parity $|\mathbb{Z}_2| = 2$. [`verification/v20_lepton_c_derivation.py`]

[verification/v21_solar_product_quark.py] [verification/v37_plucker_anchor.py]

Update (the hexagon is the family monodromy, [verification/v118_hexagon_family_dictionary.py]): with the flavor-wall monodromy now exact ($\widetilde{M}_0 \in W(A_3)$, research contracts), the hexagon constants stop being separate inputs **[E]**: $\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6$ (the six sites = family spectrum + sheet twist), $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3$, and

$$c_e = \frac{|\mu_4|}{2\det(\mathbf{1} - \frac{1}{2}\widetilde{M}_0)} = \frac{16}{7}, \quad c_\mu = \frac{N_{\mathrm{fam}}}{2\det(\mathbf{1} + \frac{1}{2}\widetilde{M}_0)} = \frac{4}{3}, \quad c_\tau = \frac{|\mu_4|\det(\mathbf{1} - \frac{1}{2}\widetilde{M}_0)}{N_{\mathrm{fam}}} = \frac{7}{6},$$

with the product rule $= |\mu_4|/\det(\mathbf{1} + \frac{1}{2}\widetilde{M}_0) = \frac{32}{9}$ — the lepton 7 and 9 *are* the two family-resolvent determinants ($\times 2^3$), and $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the monodromy itself. The $e/\mu/\tau$ *leg assignment* remains the established input above; nothing is re-fitted.

*The lepton ring is an étale Frobenius algebra **[E]/[C]** ([verification/v229_lepton_frobenius_algebra.py]).* The three coefficients are the spectrum of the commutative algebra $A = \mathbb{Q}[t]/(m)$, $m = (t-c_e)(t-c_\mu)(t-c_\tau)$: A is *étale* (separable), so the trace form $(x, y) \mapsto \mathrm{Tr}(xy)$ is nondegenerate ($\det \mathrm{Gram} = \mathrm{disc}(m) \neq 0$) — the defining property of a (commutative) Frobenius algebra — and it lives over the μ_6 resolvent (C_6 has characteristic polynomial t^6-1, spectrum $\mu_6 = \mu_3$ family $\times\, \mu_2$ sheet). So the charged-lepton boundary is a genuine finite Frobenius object **[E]**; the PMNS extension (its automorphisms plus the hexagonal CM phase j=0, [verification/v220_cp_hexagonal_modulus.py]) is the natural **[C]** candidate, not a closure of the matrix.

Scope (honest). This law is lepton-specific. Applied to the down sector it fails provably: r_d=1, r_s=5 share the same amplitude, so the law forces c_d/c_s=$|\mu_4|$=4, whereas the source ratio gives 0.94. Quarks (colour, different cusp legs) are structurally different and their exact c's stay open (cf. the $U_f^\star$ reduction above).

Ratio	Formula	Prediction	PDG
$\hat{m}_\mu/\hat{m}_\tau$	$\frac{8}{7}\varphi_0^{\mathrm{ret}}$	0.06077	0.05946
$\hat{m}_e/\hat{m}_\mu$	$\frac{12}{7}(\varphi_0^{\mathrm{ret}})^2$	0.004847	0.004836
$\hat{m}_e\hat{m}_\tau/\hat{m}_\mu^2$	$\frac{3}{2}\varphi_0^{\mathrm{ret}}$	0.07976	0.08133

The factor $B = \frac{3}{2}$ in the Koide-type product is the rank ratio $\mathrm{rank}\, E_3/\mathrm{rank}\, E_2$ of the carrier — again a compiler number, not a fit. The e/μ relation hits the PDG to 0.2%.

5 Quarks: hierarchy from the same word-lengths

The quark sectors use the same master formula with $L_u = (7, 3, 0)$, $L_d = (7, 5, 2)$. Where two generations share the same word-length the mass ratio is pure transport amplitude; where they differ by 2 a factor λ_Y^{-2} appears:

Ratio	Origin	Prediction	PDG order
$\hat{m}_u/\hat{m}_d$	Λ_u/Λ_d (same L=7)	0.470	~ 0.5
$\hat{m}_c/\hat{m}_s$	$\lambda_Y^{-2}\,\Lambda_c/\Lambda_s$	13.6	~ 11–14
$\hat{m}_t/\hat{m}_b$	$\lambda_Y^{-2}\,\Lambda_t/\Lambda_b$	40.8	~ 41

The residue classes mod 6 link quarks and leptons: $L_c = L_\tau = 3$, $L_s = L_\mu = 5$ — the same hexagon position, different sectors. This is the geometric root of quark–lepton complementarity.

The φ_0^{ret}-ladder: *all* nine masses in one closed form

The quark Λ are *not* free numbers: they are evaluations of the *same* finite hexagonal resolvent D_y^{-1} (proven in closed form, [verification/v17_hexagonal_resolvent.py]) at the *same* pole δ_{ph} as the leptons.

For every fermion the *universal* closed form holds

$$\boxed{\lambda_Y^{L_{f,j}}\Lambda_{f,j} = \pi\, c_{f,j}\,(\varphi_0^{\rm ret})^{k_{f,j}} \quad\Longrightarrow\quad \hat m_{f,j} = \frac{v_{\rm geo}\,\pi}{\sqrt{2}}\, c_{f,j}\,(\varphi_0^{\rm ret})^{k_{f,j}}}$$

with carrier rationals $c_{f,j}$ and integer $\varphi_0^{\rm ret}$-order $k_{f,j}$. The leptons give $c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ *exactly*; the quark c are fixed by the same resolvent (the published Λ table is only three-digit). The $\varphi_0^{\rm ret}$-orders form the **mass ladder**

$$\underbrace{t}_{k=0} \;>\; \underbrace{b,\,c,\,\tau}_{k=2} \;>\; \underbrace{s,\,\mu}_{k=3} \;>\; \underbrace{u,\,d}_{k=4} \;>\; \underbrace{e}_{k=5},$$

each rung a factor $\varphi_0^{\rm ret} \approx 1/19$. The whole fermion hierarchy is a $\varphi_0^{\rm ret}$ power ladder with carrier-rational rungs — quarks *and* leptons in *one* family. **[E]** (leptons exact) / **[C]** (quark rationals from the resolvent)

Quark *ratio* closure: the invariant readouts of the character sector [E]

The right invariants are not the six absolute c_q (gauge-dependent through the external legs and the sector normalisation) but the three intra-sector *ratios*, which close on pure compiler building blocks. With $\Delta_Q := |R(A_3)|{+}1 = 13$, $\Omega_{\rm adm} = 48$, $N_{\rm fam} = 3$ and the anchor power sum $p_5(a) = 1^5{+}1^5{+}2^5 = 34$,

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\rm car}\cdot 11}{N_{\rm fam}^2\Delta_Q} = \frac{55}{117}, \quad \frac{c_c}{c_s} = \frac{p_5(a)}{\Omega_{\rm adm}-1} = \frac{34}{47}, \quad \frac{c_t}{c_b} = \frac{N_{\rm fam}}{2\Delta_Q} = \frac{3}{26}}$$

With the $\varphi_0^{\rm ret}$-orders $(k_u, k_d, k_c, k_s, k_t, k_b) = (4, 4, 2, 3, 0, 2)$ the mass ratios are

$$\frac{\hat m_u}{\hat m_d} = \frac{55}{117} = 0.4700855, \quad \frac{\hat m_c}{\hat m_s} = \frac{34}{47}(\varphi_0^{\rm ret})^{-1} = 13.6050, \quad \frac{\hat m_t}{\hat m_b} = \frac{3}{26}(\varphi_0^{\rm ret})^{-2} = 40.812,$$

matching the closed-branch targets (0.470085, 13.607, 40.80) to $\lesssim 0.03\%$. A rational normalisation gauge compatible with these ratios is

$$(c_u, c_d, c_s, c_c, c_b, c_t) = \Big(\tfrac{1}{2},\ \tfrac{117}{110},\ \tfrac{47}{41},\ \tfrac{34}{41},\ \tfrac{8}{3},\ \tfrac{4}{13}\Big),$$

whose $\Lambda_q = \pi c_q(\varphi_0^{\rm ret})^k/\lambda_Y^L$ all land in the $O(1)$ band ($\Lambda_{u,d,s,c,b,t} \approx 0.44, 0.93, 0.95, 0.65, 0.47, 0.97$). The denominators carry $41 = 4^2{+}5^2 = 10b_1$. The individual c_q are a gauge; the three ratios are the invariant readouts. [`verification/v24_quark_ratio_closure.py`]

Honest note on the "11". Two of the three ratios have *clean single-reading* building blocks: $c_c/c_s = p_5(a)/(\Omega_{\rm adm}{-}1)$ (the anchor power sum $p_5{=}34$) and $c_t/c_b = N_{\rm fam}/(2\Delta_Q)$. The first ratio is the odd one out: its numerator factor 11 admits *many* TFPT readings ($\dim S^+{-}g_{\rm car}$, $g_{\rm car}{+}|R^+(A_3)|$, $h(D_5){+}N_{\rm fam}$, $2g_{\rm car}{+}1$, $\mathcal{A}_\Lambda{+}1$, $\Delta_Q{-}2$, the E_8 exponent 11, …), so it is *not uniquely forced* as a loose integer. **This worry is now retired** (Plücker box below): $11 = \|\,\mathrm{Pl}(K)\|_1$ is *generated* as the L^1 norm of the anchor-plane minors of K (v42), and by Readout Rigidity (v49, with the selector stratum now *derived* — $\det R = 8$ lattice and $\mathrm{Spec}(Q_+) = \{1,2,3\}$ from D_4, v69/v71) the *ratio* $c_u/c_d = \frac{55}{117}$ is *constant* on that stratum, i.e. an integer Plücker readout that needs *no* transcendental solve. The wall-selection (U) enters only the *absolute* amplitude normalisation (an anchor), not the ratio. [`verification/v26_flavor_frontier_unification.py`] Quark masses are scheme-sensitive (esp. light quarks); we state dimensionless source-scale ratios, not pole masses (cf. FLAG 2024).

Anchor-plane Plücker localisation of the "11" (and 26) [E]

The cleanest way to retire the "many readings" worry is to stop reading loose integers and read *orientation invariants* of the one canonical plane in play — the *anchor plane* spanned by the two TFPT generators $\mathbf{1} = (1,1,1)$ and $a = (1,1,2)$. For $M \in \{R, Q, K, L\}$ form the left anchor block and its $2{\times}2$ minors (Plücker coordinates)

$$B_M = \begin{pmatrix} \mathbf{1}^\top M \\ a^\top M \end{pmatrix}, \qquad \mathrm{Pl}(M) = \big(\,[12], [13], [23]\,\big).$$

Then $\mathrm{Pl}(K) = (-N_\Phi,\ |R^+(A_3)|,\ |\mu_4|) = (-1, 6, 4)$ and

$$\boxed{\|\mathrm{Pl}(K)\|_1 = 1 + 6 + 4 = 11}$$

is the *Plücker norm of the mass-power anchor plane* — a single coordinate-free object, not a free integer. The right block $C_M = (M\mathbf{1}\ \ Ma)$ gives $\mathrm{Pl}_R(K) = (12, 12, -2)$ with $\|\mathrm{Pl}_R(K)\|_1 = 26$, and $\sum \mathrm{Pl}_R(L) = 20{+}30{+}10 = 60$ is the E_8 cascade start. **Canonical readings (** `[verification/v94_sheet_diamond.py]`**):** both quark-ratio integers live in the *same* operator K — the left block reads flavor *orientation*, the right block reads transport *propagation*:

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\mathrm{car}}\,\|\mathrm{Pl}(K)\|_1}{N_{\mathrm{fam}}^2\,\Delta_Q}, \qquad \frac{c_t}{c_b} = \frac{N_{\mathrm{fam}}}{\|\mathrm{Pl}_R(K)\|_1}}, \qquad \frac{c_c}{c_s} = \frac{\sum \mathrm{Pl}(L) - \sum \mathrm{Pl}(Q)}{1 + \sum \mathrm{Pl}(L)},$$

with the older 26-readings ($\mathrm{Pl}(L)_{13} = 26$, $2\Delta_Q = 26$) now *corollaries* of the canonical right-Plücker norm, not independent choices. **What this does and does not buy.** The Plücker values are exact identities [E], and the "11" now *lives in the mass-power matrix* K via the canonical plane rather than in a free integer pool — the best-motivated single reading. Combined with Readout Rigidity (below), this *closes the ratio*: $c_u/c_d = \frac{55}{117}$ is fixed combinatorially, no $U_f^\star$ evaluation required. What stays open is the *absolute* amplitude scale (an anchor) and a mild choice of invariant in the c_c/c_s block. `[verification/v37_plucker_anchor.py]`

Why the leg is exterior, not scalar (Exterior Leg Lemma). u and d share the hexagon-radial data $(r, w) = (1, 1)$, and the C_6 resolvent reads only (r, w) — so *any scalar* leg forces $c_u/c_d \equiv 1$. A scalar sees radius, not orientation; the u/d separation is intrinsically the oriented anchor-plane *area* $\mathrm{Pl}_{\mathbf{1},a}(K)$ (the exterior square $\Lambda^2 F$ of the $SU(3)_F$ composition), whose L^1 norm *generates* the "11". `[verification/v42_exterior_leg.py]` The norm $\|\mathrm{Pl}(K)\|_1 = 11$ also carries a topological reading: it is the dimension of the QBL-truncated ($\deg \le 2$) Fock space of the four μ_4-corner fermions, $\sum_{k\le 2}\binom{4}{k} = 1{+}4{+}6 = 11 = \dim S^+ - g_{\mathrm{car}}$ (a Witten index; the discarded high-degree complement is $\binom{4}{3}{+}\binom{4}{4} = 5 = g_{\mathrm{car}}$, the full corner Fock space $2^4 = 16 = \dim S^+$), so $c_u/c_d = 55/117$ reads the QBL-protected boundary state-space dimension (`[verification/v174_seam_fock_readings.py]`).

Readout Rigidity — the quark *ratios* are closed combinatorially [E] (residual: absolute scale [O])

The earlier "reduce c_u/c_d to the transcendental wall-selection" framing is *superseded*. The four flavor operators are integer lattice maps on $H_1(\mathbb{P}^1 \setminus \mu_4) = \mathbb{Z}^3$ with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3, 4, 8, 20)$, product $1920 = |W(D_5)|$, and the *discrete selector stratum*

$$\mathcal{S}_{8,Q} = \{\det R = 8,\ \mathrm{SNF}(R) = (1, 1, 8),\ \mathrm{Spec}(Q_+) = \{1, 2, 3\}\}$$

is now *fully derived*: $\det R = 8 = \mathrm{rank}\, E_8$ and the SNF are the lattice invariant (v70, $\det Q = 3 = N_{\mathrm{fam}}$), and $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ is the D_4-equivariant result (v69). By **Readout Rigidity** (Theorem U2, `[verification/v49_readout_rigidity.py]`) the anchor-plane readout is *constant* on $\mathcal{S}_{8,Q}$, so the quark *ratios*

$$\frac{c_u}{c_d} = \frac{g_{\mathrm{car}}\,\|\mathrm{Pl}(K)\|_1}{N_{\mathrm{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}, \qquad \frac{c_c}{c_s},\ \frac{c_t}{c_b}$$

are *pure integer Plücker readouts — independent of the continuous D_4 position, no transcendental monodromy / Hitchin solve* (`[verification/v71_simple_r_bridge.py]`). The single genuinely transcendental piece, U_{point} (full uniqueness of $\rho^\star$), enters *only* the absolute amplitude scale and is an anchor of the same nature as the one dimensionful scale — *not* a missing computation. The monodromy machinery (§ Riemann–Hilbert below) was over-engineering for the ratios; it remains the route to the *absolute* normalisation only.

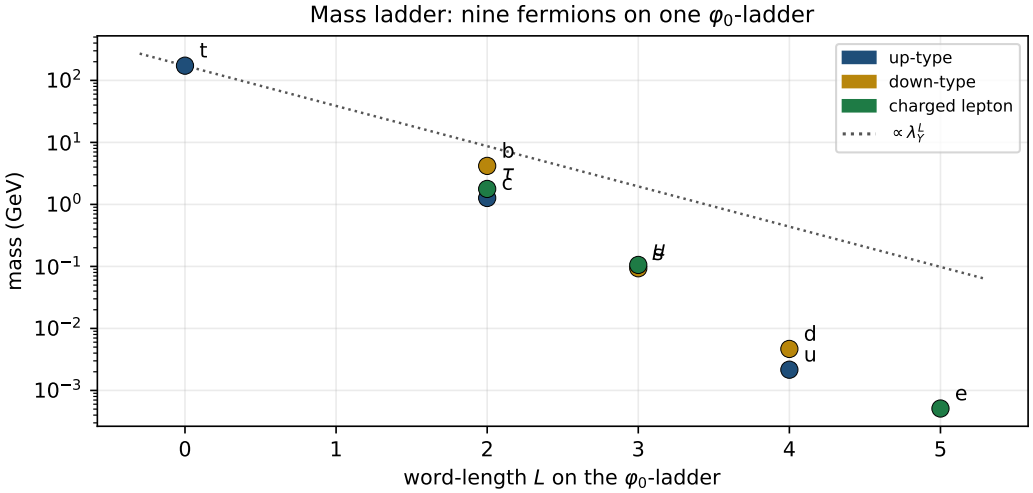


Figure 1: Mass ladder (data plot, `verification/make_figures.py`). The nine charged fermions (PDG masses) plotted against their word-length L on the φ_0^{ret}-ladder, coloured by sector; the dotted line is the geometric reference $\propto \lambda_Y^L$ with $\lambda_Y = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}$. The whole five-order hierarchy is one ladder.

6 The Sheet Diamond and the Winding Line: masses and transport from one surface

The mass-power matrix and the transport word-length matrix are *not* two independent machines: they are the two sheet-readouts of *one* parabolic projection Q acting on the residue matrix R. Collect the φ_0^{ret}-orders $k_{f,j}$ of the nine charged fermions into the **mass-power matrix**

$$K = \begin{pmatrix} 4 & 2 & 0 \\ 4 & 3 & 2 \\ 5 & 3 & 2 \end{pmatrix} \quad (\text{rows } u, d, e), \qquad Q := L - K = \begin{pmatrix} 3 & 1 & 0 \\ 3 & 2 & 0 \\ 3 & 2 & 1 \end{pmatrix}, \qquad \Sigma := \mathrm{diag}(1,-1,-1), \quad (1)$$

with Σ the *sheet* ($\mathbb{Z}_2$) involution. Then, exactly,

$$\boxed{K = R + Q\,\Sigma, \qquad L = R + Q\,(I + \Sigma)} \qquad \text{[E] [verification/v10_projection_involution.py]}, \quad (2)$$

and since $I + \Sigma = \mathrm{diag}(2,0,0)$ and Q's first column is $(3,3,3)^{\top}$, the "hand-added" first-generation winding is *forced*: $Q(I+\Sigma) = 6W = 2N_{\mathrm{fam}} \cdot \mathbf{1}e_1^{\top}$, i.e. sheet ($\mathbb{Z}_2$) × families ($\mathbb{Z}_3$) = the C_6 transport. So the φ_0^{ret}-mass powers are the Σ-odd readout, and the transport lengths are the $(I+\Sigma)$ readout, of the *same* projection Q on R.

The Sheet Diamond: all four operators are points of *one* surface [E] ([verification/v94_sheet_diamond.py])

The whole flavor algebra lives on a single two-parameter operator surface,

$$M(s,t) = R + Q\ \mathrm{diag}(s,t,t):$$
$$R = M(0,0), \qquad K = M(1,-1), \qquad L = M(2,0), \qquad F := R{+}Q = M(1,1),$$

with the two global invariants

$$\det M(s,t) = 3st^2 + 9st + 6s + t^2 + 5t + 8, \qquad \det B_{M(s,t)} = 6st + 12s + 3t^2 + 15t + 16.$$

The mass-transport pencil $K + xQ$ is the *cut* $(s,t) = (1{+}x,\ x{-}1)$, on which the anchor-block determinant factorises as $(3x{+}2)(3x{+}5)$ — the master cover of the operator-pencil sections below (unique up to Möbius reparametrisation, [verification/v85_master_cover.py]). The factorisation is **not generic**: the

surface carries its own negative controls — the diagonal cut $t{=}s$ gives $9s^2{+}27s{+}16$ with discriminant 153 (non-square; the *same* control discriminant as the R-direction pencils), and the $R{\to}L$ cut $t{=}0$ is linear (degenerate direction, no cover).

Centered normal form ([verification/v95_centered_diamond.py]). The diamond is a *centered cross*: with the center $C := M(1,0) = \frac{R+L}{2} = \frac{K+F}{2}$ and the two axes $U := Q\,\mathrm{diag}(1,0,0) = N_{\mathrm{fam}}\,\mathbf{1}e_1^\top$ (winding, rank 1, spectrum $\{3,0,0\}$) and $V := Q\,\mathrm{diag}(0,1,1)$ (sheet, rank 2),

$$\boxed{Q = U + V, \qquad R = C{-}U, \quad L = C{+}U, \quad K = C{-}V, \quad F = C{+}V} \qquad \textbf{[E]},$$

so five objects collapse to three, and the winding line below *is* the U-axis ($L = R + 2U$). The sheet axis carries exactly the cusp class: $\mathrm{Spec}(V) = \{0,1,2\} = N_{\mathrm{fam}}\{0, \frac{1}{3}, \frac{2}{3}\}$ (cf. $Q_+ = 3\,\mathrm{cusp} + 1$ with $\{1,2,3\}$, v69/v72). The center is tightly typed: $\det C = 14$, $\mathrm{SNF}(C) = (1,1,14)$, $\sum_{ij} C_{ij} = 31 = 2^{g_{\mathrm{car}}}{-}1 = 1{+}h^\vee(E_8)$ (the same 31 as the IR gap-bound numerator $\|V_{\mathrm{metric}}\| \le 31/(8\pi^2)$), $\det B_C = 28 = 2 \cdot 14$, $\mathrm{Pl}(C) = (0,14,14)$ and $\mathrm{Pl}_R(C) = 7\,(2,3,1)$ — the same ray $(2,3,1)$ as $\mathrm{Pl}_R(L) = 10\,(2,3,1)$, scalaron-7 vs decuple-10 scaling. *Typing:* the normal form and all integers are exact **[E]**; the "G_2 center" reading ($14 = \dim G_2$, also the constant term of $\chi_{[K,Q]} = t^3{+}5t{+}14$ and $\sqrt{\sum B_F}$) is an *audit* Lie-dimension match, recorded, not promoted.

Axis curvature and two ladders ([verification/v218_diamond_axis_geometry.py]). The cross has a *discrete differential geometry*: the determinant is *linear* along the winding axis and *quadratic* along the sheet axis,

$$\det(C{+}xU) = 14 + 6x, \qquad \det(C{+}yV) = 14 + 14y + 4y^2, \qquad \det B_{C+xU} = 2\det(C{+}xU),$$

so winding flows flat (A_3-driven, slope $6 = |R^+(A_3)|$) while the sheet split *curves* with second differences $8 = \mathrm{rank}\,E_8$ (determinant) and $6 = |R^+(A_3)|$ (anchor block) — the two curvatures are exactly the triply-forced 8 and the A_3 positive roots. Two exact ladders ride the same sheet axis. The **Plücker transfer ladder** $\mathrm{Pl}(K) = (-1,6,4) \to \mathrm{Pl}(C) = (0,14,14) \to \mathrm{Pl}(F) = (1,22,30)$ lifts in steps $(1,8,10) = (N_\Phi, \mathrm{rank}\,E_8, A_\Lambda)$ then $(1,8,16) = (N_\Phi, \mathrm{rank}\,E_8, \dim S^+)$ — first the decuple, then the full spinor generation, so the F_{transfer} source→pole step *is* this two-stage sheet lift **[C]**. The **spectral ramification ladder** factors the cubic discriminant of Q, K, C, F as $q(r)^2\,\mathrm{Disc}(q)$ with squares $(1,3,4,6) = (N_\Phi, N_{\mathrm{fam}}, |\mu_4|, |R^+(A_3)|)$ and kernels $(13,48,65,105) = (\Delta_Q, \Omega_{\mathrm{adm}}, g_{\mathrm{car}}\Delta_Q, N_{\mathrm{fam}} g_{\mathrm{car}} \cdot 7)$, the corner F carrying $105 = 3{\cdot}5{\cdot}7$ (family · carrier · scalaron). *Audit:* the anchor-defect spectrum $\Delta_a(Q,K,R,C,L,F) = (8,16,18,21,24,26)$ and the corner-determinant pair-sum atlas (total $64 = \dim S^+|\mu_4|$, staircase sum $81 = N_{\mathrm{fam}}^4$) are recorded fingerprints — the $2\dim G_2$ and $\dim F_4$ pair sums stay audit-only, like the G_2 center above.

The parabolic self-code and the 81 as a lattice index ([verification/v566_parabolic_anchor_selfcode.py]). The cross carries more than curvature: U and V generate the *full* 2|1 parabolic algebra $\mathcal{A}$ (dim 7, word determinant -81, multiplicatively closed) **[E]**; the anchor returns as the spectrum of $H = I + \frac{1}{2}V(V - I)$ with $\chi_H = x^3 - 4x^2 + 5x - 2$ double-coded by $(4,5,2) = (\mathrm{flag}, \mathrm{Levi}, \mathrm{radical})$, and $m = 2$ is the unique all-integer member of the parabolic family **[E]** (details and fences in Paper 1). Two findings live on *this* page: (i) the **staircase sum 81 is simultaneously a lattice index** — the saturated integral order $\mathbb{Z}\langle U, V\rangle$ has elementary divisors $(1,1,1,3,3,3,3)$ in the parabolic lattice, so $[P_{\mathbb{Z}} : \mathbb{Z}\langle U,V\rangle] = 3^4 = 81$ with cokernel $(\mathbb{Z}/3)^4$: a global determinant charge and an integral saturation index coincide **[E]** (the mark-local mechanism — one $\mathbb{Z}/3$ per μ_4 mark — stays **[O]** with its kill condition recorded); (ii) the **basic algebra of $\mathcal{A}$ is the A_2 path algebra** ($\dim \mathrm{Ext}^1 = 1$), so dimension-vector-$(1,1)$ modules fall into exactly *two* isomorphism classes — split and one nonsplit: the split/nonsplit alignment bit gains a canonical algebra *carrier* **[E]**, while its *selection* stays with geometry and reflection positivity (v528/v534) **[C]**. The anchor power compiler is the trace code $p_n(a) = 1 + \mathrm{tr}(V^n)$ (240 and $\mathrm{rank}\,E_8$ as trace readouts) — a re-encoding of the Lean AnchorLadder, typed as hardening, not as new content.

The Gaussian code bridge, and the quartic half (2026-08-03). Two exact results extend this code page to the lattice itself. *The bridge* ([verification/v689_gaussian_code_bridge.py], 26 checks, exact arithmetic): the μ_4 complex structure makes E_8 a rank-4 $\mathbb{Z}[i]$-lattice whose $(1+i)$-quotient is $\mathbb{F}_2^4$ — the information space of RM(1,3) (SNF exact, presentation-independent); the 240 roots fall 15×16 over the nontrivial classes (zero class provably empty, each class = 4 of the 60 G31 lines), the clock $\sigma = c^4$ acts as the RM(1,3) *family permutation* (3-cycle + fixed anchor, the v638 info-bit action; residual gauge of order 18), and the coordinate block is the sum-of-families label $F_1{+}F_2{+}F_3$; three must-fail controls fire **[E]** (`E8.GAUSSCODE.01`). *The quartic half* ([verification/v690_quartic_half.py],

22 checks, no floats): the degree-$\{2, 14, 18, 30\}$ vanishing under holomorphic μ_4 restriction is proven but *trivial* (μ_4-orbit factor, honest typing); the substance: $F_8|, F_{12}|, F_{20}|, F_{24}|$ are algebraically independent G31 *basic* invariants (Molien-unique degrees, product $46080 = |G31|$, Chevalley cited) — the holomorphic restriction of the $W(E_8)$ ring generates the full G31 invariant ring, and the Jacobian carries the 60 mirror lines 60/60 (`E8.QUARTICHALF.01`) [E]. No marker moves. *The orbit packet lemma (2026-08-04, [verification/v736_orbit_packet.py],* 26/26*, ORBIT-PACKET-EXACT; Lean companion OrbitPacket.lean,* 34 *theorems):* the code page's compact number family in one normal form — Burnside $7 = (15+3+3)/3$ on the 15 nonzero classes; the canonical packet split $240 = 5 \times 48$ (1 fixed block + 4 moved blocks of 12 Gaussian lines each, label-free canonical; packets $= 1 + 4 = 5 = g_{\mathrm{car}}$); the anchor $(1, 1, 2)$ from the fixed-syndrome leader weights $(0, 1, 1, 2)$ (leader pair e_6/e_7 = the v638 anchor pair); and $240 = 16 \cdot 15$, $248 = 240 + 8$. *Circularity fence:* P1/P2 come back as *exact internal reconstruction*, not derivation, and the $1 + 4$ law is σ-class-specific (free order-3 elements give Burnside 5) [E]. *The symplectic code lemma (2026-08-04, [verification/v752_projective_hamming_incidence.py],* 36/36*, PROJ-HAMMING-EXACT; Lean companion ProjectiveHamming.lean,* 32 *theorems):* the code page's quotient is a *geometry* — $h = H/4$ is $\mathbb{Z}[i]$-valued with hermitian Gram determinant 1 (the unimodular hermitian E8 lattice over $\mathbb{Z}[i]$), the reduction $\bar{h}$ makes $V = L/(1+i)L$ a symplectic $\mathbb{F}_2^4$, and the 15 nonzero classes carry the exact incidence identity $BB^{\mathsf{T}} = 4I + 3J$ (design parameters 2/3/4/7/15/16; eigenvalues $7^1(+2)^9(-2)^5$ — the multiplicities $(1, 9, 5)$ are the symplectic signature; $K^2 = (4/49)I + (45/49)\Pi_0$ for $K = B/7$). The NS/R parity of the $120 + 128$ split is the *point* $y_0 = \mathrm{FSUM}$ ($\chi_{\mathrm{NSR}} = \bar{h}(\cdot, y_0)$, σ-fixed; $112 + 128 = 240$, +8 Cartan: $248 = 120 + 128$), and the v737 seam-48 distribution is derived exactly with the 5×48 coordinatization existing via exactly one $\bar{u}$-invariant spread (kill explained, not revised) [E]. *The Arf/spinor compiler (2026-08-05, [verification/v774_arf_spinor_compiler.py],* 46/46*, ARF-SPINOR-EXACT; Lean companion ArfSpinorCompiler.lean,* 23 *theorems):* the quotient's geometry has a canonical quadratic refinement — the frozen selector picks a unique q^* among the exactly 16 refinements of $\bar{h}$, the parity lift $\iota\colon V \cong C_{\mathrm{even}}(5)$ carries $\beta = \bar{h}$ exactly, $\mathrm{Stab}(q^*) \cong S_5$ makes $16 = 1 + \bar{5} + 10$ an Arf geometry (K_6 duads; $B = I + A_{KG(6,2)}$ with exact ovoid eigenvectors), the hypercharge table $X = (-2, -2, -2, 3, 3)$ reproduces the `v310/v14` dictionary with the master-moment chain 120/40/41/240/248, the Pati–Salam split $16 = 8 + 8$ carries $B-L = 1 - 2n_c/3$ exactly, and χ_{NSR} is the anchor bit; A_3^{Fam} (family cycle on message bits) and A_3^{PS} (3+2 slot split on carrier legs) are the *same* σ in two registers — naming discipline [E]. *The fired matter fence (2026-08-05, [verification/v775_gaussian_class_d5_purity.py],* 18/18*, ROOTCLASS-MIXED):* the preregistered code→matter kill *fired* — no $D_5 \oplus A_3/SU(5)$ convention makes even one full set of 15 classes pure (≥ 7 adjoint-side classes for every embedding; 0/56 splits); the code→matter table, including the "empty mailbox = right-handed neutrino" reading, must not enter any main claim [E]. *Boundary code uniqueness (2026-08-05, [verification/v776_boundary_hamming_uniqueness.py],* 31/31*, BOUNDARY-CODE-UNIQUE):* the non-circular chain boundary → code → $E_8 \to \mathbb{F}_2^4 \to$ five slots closes with $g_{\mathrm{car}} = 5$ as *output* (census $135 \to 30 \to 14 \to 2 \to 1$; the flag datum is one deck-invariant $\mathbb{Z}_2$); honest residuals R1–R3 typed then (universe premise, semantic deck placement, anchor naming); no P2-removal claim. *Type II from the seam axioms (2026-08-06, [verification/v799_seam_code_typeii.py],* 19/19*, TYPEII-FORCED):* residual R1 is replaced by three physical axioms — exhaustively over all 308,993 subspaces of $\mathbb{F}_2^8$ (dim ≤ 4), mutual locality of the Construction-A boundary vertex words $\iff$ self-orthogonal, integer conformal spin $\iff$ doubly even, holomorphy/index one $\iff$ self-dual, with the survivor set exactly the 30 $\hat{e}_8$ copies — so the universe premise reduces to A1 (bose statistics), A2 (modular T-alignment), A3 (holomorphic index one); R2 is placed and R3 named by the metaplectic S-lift ([`verification/v798_seam_clifford_modular_s.py`]); the sharpened residual is `SEAM.CODE.TYPEII.01` [C] (the seam satisfies A1–A3); still no P2-removal claim. **The No-Carrier-Reconstruction audit, Stage A.** [`verification/v567_carrier_free_q_reconstruction.py`] The self-code's circularity question is executed as an enumeration with the carrier certificates *forbidden* (the `v11` budgets: rows $(4, 5, 6)$, columns $(9, 5, 1)$, χ_Q with its 5; monotone rows). Under the corpus' own carrier-free data — Σ, the winding normalisation $Q(I+\Sigma) = 2N_{\mathrm{fam}}W$ (`v10`), $\mathrm{Spec}\, Q_+ = \{1, 2, 3\}$, $Q_-^2 = N_{\mathrm{fam}}P$, nonneg integrality, the seam-line demand — exactly $52 \to 28$ (parabolic) reconstructions survive, and **all** 28 **code the same anchor** ($\chi_{H'} = x^3 - 4x^2 + 5x - 2$, spectrum $(1, 1, 2)$): *the anchor self-code is carrier-free* [E]. The single family invariant $81 = N_{\mathrm{fam}}^4$ (the saturation index above) pins Q to a *two-element* residual family — the compiler Q ($q_{32} = 2 = |\mathbb{Z}_2|$) and its uncoupled twin ($q_{32} = 0$), difference $2E_{32}$ — on which the anticommutator spectrum $\{6, -4, -2\}$, the commutator characteristic polynomial $t(t^2+12)$, the determinant and the saturation index all *coincide*; the one certificate that separates them is the carrier column budget $(9, 5, 1)$ — exactly the forbidden input [E]. So Q is carrier-free up to *one binary sheet-coupling choice*; the carrier-free selector of that bit is open (its resonance with the split/nonsplit

alignment bit above is a typed observation, not a claim) [O], and whether the geometric inputs themselves derive from $\mu_4/D_4/H^1$ parabolic geometry is exactly `GATE.QGEO` — the audit is *relocated*, not closed. Ablation honesty: dropping the winding input widens the bound-3 set 28 → 1376 (real selection load); the census is bound-stable ($B = 6 \to 9$).

The bit located, and its canonical witness. [`verification/v568_bit_selector_ladder.py`] The residual choice is now dissected. (i) The two 81-twins generate *different integral orders* (equal index, different lattices): the bit is integrally visible where every classical invariant coincides [E]. (ii) **The selector:** the dominant mode of the sheet direction is $v(c) = (1, 2, 2c)$ *identically* in the coupling, and demanding the *ladder self-consistency* — the dominant mode reproduces its own eigenvalue as the geometric ladder $(2^0, 2^1, 2^2) = (1, 2, 4)$ — selects $c = 2$, the compiler Q, *uniquely on all* 28 *members* of the carrier-free class, with no 81-pin and no column budget [E]; the witness is prior structure (the binary AnchorLadder steps 2^n, $2 = |\mathbb{Z}_2|$, the trace code), not a found number [C]. (iii) The *split* geometry sits at $q_{32} = 1$ — a module complement exists exactly there — and is *excluded* by the demands already imposed (its algebra collapses to dimension 5): the A_2 Ext arrow is nonzero on the whole reconstruction class, so the bit does not toggle split/nonsplit; it toggles *which* nonsplit integral order is realised [E]. (iv) Priority-2 candidates, honest both ways: the μ_4-grading fails (the order is not graded — the 81 stays a fingerprint), while left V-multiplication induces a canonical $\mathbb{F}_3$-action on the cokernel $(\mathbb{Z}/3)^4$ — the mark-local question is sharpened to a concrete module computation [O]. *Why* the dominant mode should satisfy the ladder self-consistency is the relocated open question — the same structural slot as the RP-selection of the alignment bit (v528/v534) [O].

The RP form on the twins: spectrally blind, integrally anchor-visible. [`verification/v572_rp_bit_form.py`] The kinematic half of that RP-slot question is executed. With the corpus reflection $\theta(X) = \Sigma X^{\mathsf{T}} \Sigma$ (the sheet involution) and the declared states (trace; anchor $(1, 1, 2)$; ladder $(1, 2, 4)$; unit), the RP kernel $K(x, y) = \omega(\theta(x)y)$ on each twin's integral order has *identical inertia* on both twins for every state and is never PSD — no kinematic v534-style positivity selection exists (honest negative; v534's selection was nonperturbative there too) [E]. But the *integral* Smith towers of the anchor-state Gram *separate* the twins — $(2, 2, 18, 18, 18, 18, 0)$ vs $(2, 2, 2, 18, 18, 18, 0)$: one slot, 18 vs 2, a factor $9 = N_{\text{fam}}^2$ — and the ladder state separates identically while the anonymous states (trace, unit) are blind: **the bit is visible to the RP form only through anchor-coupled states**, with the trace discriminant -81^2 on both twins (the saturation index squared) [E]. The blindness is robust under the compiler's own flow (the V-semigroup-deformed kernels keep identical inertia on both twins for every state): a dynamical selection, if any, needs genuine interaction. And the *Gibbs pencil* of the anchor Hamiltonian delivers exactly that, at the rank level: with $\omega_g \propto \mathrm{tr}(e^{-gH}\,\cdot)$, $H = I + \Pi_{\text{dom}}$ (the v566 anchor operator), the pencil determinant $\det(K_0 + sK_\Pi)$ factors $\frac{6561}{4096}(s^2{+}16s{+}16)^2(15s^2{-}16s{-}16)$ on the compiler twin against $-\frac{6561}{4096}(s^2{+}16s{+}16)^3$ on the alternative ($6561 = 81^2$, the saturation index squared) — and the compiler's extra factor has its root $s^* = (8 - 4\sqrt{19})/15$ *inside* the physical Gibbs range: the compiler twin's kernel degenerates at a finite critical temperature, the alternative twin's never does. The first one-parameter *dynamical* distinction of the bit. And the critical mode is *identified*: the null direction at s^* is rank one, supported entirely on the seam row (the radical / A_2-Ext-carrier row), fixes the seam line, is a fixed point of the sheet flow ($VX = X$) and is annihilated by the dominant projector — the Gibbs criticality lives exactly in the Ext-carrier slot, so the typed resonance with the alignment bit now has a *shared carrier*. The half-sided restrictions (Levi and radical sublattices) stay blind across the Gibbs range: the positivity selection resists every kinematic and half-sided design tried — the genuinely *interacting* construction (the v534 FK-quartet analogue) is the one remaining route. Honest typing: a rank-drop marker; the positivity selection itself stays the open RP-slot [E]/[O].

The interacting route, executed: Ext-source response separates the twins in the unit box. [`verification/v584_ext_source_response.py`] The remaining route is run. Coupling a rank-one Ext source to the Gibbs pencil, $K(s, J) = K_0 + sK_\Pi + J\,vv^{\mathsf{T}}$ with $v_i = \mathrm{tr}(\theta(x_i)N)$ and N a shared radical arrow E_{31}/E_{32}, the flip thresholds are *exact rational curves* $J_c(s) = -1/(v^{\mathsf{T}}K(s)^{-1}v)$ whose numerators are exactly the twins' pencil factors — compiler $(15s^2{-}16s{-}16)/(16(s^2{-}s{-}1))$ and $(15s^2{-}16s{-}16)/16$, alternative $(s^2{+}16s{+}16)/(16(s{+}1))$ and $-(s^2{+}16s{+}16)/16$. In the crossed phase $s \in (-1, s^*]$ the compiler twin is *healed* (inertia restored to the $(4, 0, 3)$ pattern form) by positive coupling $J_c \le 15/16 < 1$ through either arrow, while the alternative twin cannot be flipped by *any* positive coupling in the open unit box (its E_{31} threshold stays ≥ 1 on the whole physical range, saturating exactly at $s = 0$; its E_{32} threshold is negative throughout): **positive Ext coupling inside $|J| < 1$ selects exactly the compiler twin**, and negative coupling never heals — the direction matches the v534 toy's RP selection of positive seam coupling [E]. Exact anchors: $J_c(-1) = 15/16$; the mechanism is the Ext-visibility of the critical mode ($J_c(s^*) = 0$ exactly); and the compiler response pole sits on the golden polynomial

$s^2 - s - 1$ ($s = -1/\varphi = -0.618$, next to $s^* = -0.629$) — observed structure, no claim built on it. Honest typing: a *response-level* selection (inertia restoration in a declared coupling box), not an OS/RP positivity theorem — no kernel is ever PSD and the alignment-bit contract stays open [E]/[C].

The sheet involution exists unconditionally. [verification/v590_involution_existence.py] The involution-existence input of GATE.QGEO.01 is derived. The integral involutions commuting with the sheet flow V are *exactly* the spectral sign group $(\mathbb{Z}/2)^3$ — all eight sign choices on V's eigenspaces $\{0,1,2\}$ are integer matrices stabilizing the order, on both twins (a rigidity fact; generic integral operators do not have integral spectral projectors) [E]. The narrowing chain: the v574 rank theorem kills the trace $+1$ half ($6 \to 3$, all of exact Σ-signature $(1,-1,-1)$); seam reversal $Se_3 = -e_3$ narrows $3 \to 2$; fixing the zero mode ($+1$ on $\ker V$, flipping exactly the dynamical sheet pair) singles out $S_0 = [[1,-1,0],[0,-1,0],[0,0,-1]]$ uniquely — existence is *unconditional* [E]. Honest scope: the corpus RP reflection $\Sigma = \mathrm{diag}(1,-1,-1)$ (an anti-automorphism) does not commute with V — a different structure; and the alt twin admits the same classification, so the involution is not a bit selector. Deriving the two demands from the continuum geometry ($\mu_4/D_4/H^1$) is the remaining open of the gate [E]/[C]. A geometric candidate has since emerged ([verification/v597_mu4_cover_model.py]): the μ_3-cover $y^3 = x^4 - 1$ (branched at μ_4 with order-3 cusp monodromies) hits four compiler fingerprints — rank-3 Eisenstein H_1, projective D_4 via the deck, cusp-class group $(\mathbb{Z}/3)^4$ of order 81, and the unique invariant hermitian form of Lorentz signature $(1,2)$; the dictionary since stands at three exact entries ([verification/v598_cover_dictionary.py]: $U \sim G_2$, $Q_- \sim G_1 - G_3$ — the sheet coupling as the diamond-diagonal Gram difference — and the binary ladder $\sim 1 + G_1 + G_3$), with the grading half absent from 7296 canonical combinations; the real structure is the named suspect for the missing generator [E]/[C]. The suspect is now *constructed* ([verification/v599_real_structure.py]: R unique with $R^2 = +1$; the grading functor closes the gaps at char level; the invariant form has $\det J = 8$, Lorentz $(1,2)$), and the joint embedding has landed ([verification/v600_joint_embedding.py]): a Γ-even degree-2 pair realizes (U,V) exactly — $\mathbb{Z}[\omega]$-integral, conjugate to the compiler pair, integer on the saturated fixed lattice, saturation index $81 = 3^4$ with the v566 divisor pattern; canonicity stays open, the gate does not move [E]/[C]. Canonicity has since been answered ([verification/v601_equivariant_dual.py]): the J-adapted projector presentation is D_4-equivariant by construction (mark choice = winding choice), realizes the *dual* parabolic exactly, and the v590 Klein group lives integrally on the curve [E]/[C].

Stage B, first narrowing: the winding normalisation is forced. [verification/v574_stageb_winding.py] Two of the five declared geometric inputs of the Stage-A audit relocate to *consistency-forced*. The involution *type* falls to a rank theorem: any involution with a two-dimensional positive eigenspace gives a rank-≤ 1 sheet direction (char. poly. $x^2(x-t)$ identically in Q), so the binary trace code $\mathrm{Spec}\, V = \{0,1,2\}$ is unreachable — only the corpus type can carry the self-code [E]. The winding normalisation falls to a *double self-consistency*: (E) bounds $N \leq 3$ ($q_{11} = N$ must itself be an A_3 exponent); the ladder self-consistency selects exactly *one* member for every $N \in \{1,2,3\}$ ($28/16/14 \to 1/1/1$ — the mechanism is robust); and the saturation self-consistency $\mathrm{sat} = N^4$ holds *only* at $N = 3$ ($81 = 3^4$ against $32 \neq 1$ and $512 \neq 16$): the pair (ladder, saturation) pins $(N, Q) = (3, \text{compiler } Q)$ uniquely across the whole N-ablated family — equivalently, only $N = 3$ carries the binary trace code $p_n = 2 + 2^n$ [E]. The honest decomposition (the spectrum ablation, full 35-spectrum scan): freeing the exponent spectrum *itself*, the one surviving double-selfconsistent family is exactly $\{1,2,N\}$ with the compiler shape $Q(N) = [[N,1,0],[N,2,0],[N,2,1]]$ — so the self-consistency forces the *sheet* spectrum $\{1,2\}$ (the binary trace code) and the shape, while the A_3 exponent demand (E) contributes exactly *one* datum, $N = 3$: both factors are load-bearing, neither alone suffices [E]. The redundancy ablation goes further: relative to the self-code demands (parabolic algebra + ladder + saturation 81), the monodromy budget (M) and the seam-line demand (L) are *redundant* at the declared bound — dropping both widens the admissible class to 1274 members and the self-code still picks exactly the compiler Q: the declared geometric input list of the reconstruction shrinks to (S)-existence, (W)-form and (E) — and the *maximal* ablation drops (E) too: the winding column $3(1,1,1)$ plus integrality plus the self-code package alone pin the compiler uniquely (29 ladder candidates, one hit) [E]. And the winding form's content is located exactly: dropped entirely, the self-code leaves a *four*-column family $\{(0,1,1),(0,3,3),(3,3,3),(5,1,5)\}$ of which exactly one is uniform — the compiler: the winding demand *is* the uniformity choice (1 of 4) [E]. What stays declared is the involution *existence*, the column *uniformity* and $N_{\mathrm{fam}} = 3$ (independently anchored in the E_8 glue, v1): GATE.QGEO.01 is narrowed by five items, *not* closed; no $P2$ reduction, markers untouched [C].

F_{transfer} is a path, not four interfaces ([verification/v224_diamond_ftransfer_path.py]). The corners $K{=}M(1,-1)$, $C{=}M(1,0)$, $F{=}M(1,1)$ lie on the *sheet axis* $s{=}1$: the determinant there is the curved $\det M(1,t) = 4t^2{+}14t{+}14$ (values $(4,14,32)$, 2nd difference $8{=}\,\mathrm{rank}\, E_8$), while the winding axis

$\det M(s,0) = 6s{+}8$ is flat. So F_{transfer} is read as *one discrete geodesic* $K \to C \to F$ on the diamond (the Plücker steps $(1,8,10),(1,8,16)$ above), of which the Koide source→pole Möbius relaxation is the 1-D projection (v37/v183); the external QFT solver evaluates continuously *along* this known path [C]. The diagonal cut $t{=}s$ mixes the axes (negative control), so only the $s{=}1$ sheet axis is the branch-preserving transfer direction.

The center collects the three local norms ([verification/v230_center_budget_norms.py] **).** The center's row sums are $(7,11,13)$, and these are the three *local norms* of the theory: $7 = N_{\mathbb{Z}[\omega]}(3{+}2\omega)$ (the Eisenstein/hexagonal CM norm = scalaron exponent), $13 = N_{\mathbb{Z}[i]}(3{+}2i)$ (the Gaussian/square CM norm = Δ_Q), and $11 = \binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2} = 1{+}4{+}6$ (the QBL boundary Fock count on the four μ_4 corners = $\dim S^+ {-} g_{\text{car}} = 16{-}5$). So C = (hex norm, boundary Fock count, square norm): the two exceptional CM rings ([verification/v222_cm_norm_duality.py]) and the seam boundary count meet in one operator center, the ring assignment rigid [E].

The Winding Line. On $R_s = R + s\,\mathbf{1}e_1^\top$ (the $R \to L$ channel) anchor area and torsion volume run *synchronously* for all s:

$$\boxed{\det R_s = 8 + 2s, \qquad \det B_{R_s} = 16 + 4s = 2\det R_s},$$

and the physical winding $s = 6 = |R^+(A_3)|$ is fixed by *each* of three closure conditions alone: $\operatorname{tr} R_s = 15 = \dim A_3$, $\det R_s = 20 = \det L$, and the Coxeter lift $(R_s^\top a)_1 = 30 = h(E_8)$. The mechanism is the **cofactor seam normal** $n = c_2 \times c_3 = (5,-9,6)$ (cross product of the second and third *columns* of R) — the common first adjugate row of R *and* L (columns $2,3$ are winding-invariant) — with

$$n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2| \text{ (per-winding det increment)}, \quad n^\top R = (8,0,0), \quad n^\top L = (20,0,0)\,:$$

the normal annihilates the stable columns, so *generation one is the only open torsion channel* — the $+6$ winding is not sorted by hand. *Honest scope:* the winding line crosses the spectral *reality* threshold at $s^* \approx 2.83$ (not at 6), so reality is a *property* of the physical point — $\operatorname{disc}\chi_L = 39200 = 2(10\cdot 14)^2$, spectrum real positive — not its selector; and the corner $F = R{+}Q$ is an *audit* operator (the $F_4{\times}G_2$ shadow: $\det B_F = 52 = \dim F_4$, $\sum B_F = 196 = (\dim G_2)^2$), not a physical transport. *The winding quadratic (2026-08-08):* the whole line above is *one* rank-one determinant lemma ([verification/v857_simplex_fourier_winding.py], 20/20, WINDING-QUADRATIC-EXACT): $\chi_{R_s} - \chi_R = -s\,q_{\text{wind}}$ with $q_{\text{wind}} = e_1^\top \operatorname{adj}(tI - R)\,\mathbf{1} = t^2 - g_{\text{car}}\,t + |\mathbb{Z}_2|$ exact and irreducible over $\mathbb{Q}$ ($\operatorname{disc} 17$) — the keybox coefficients ($9{+}s$, $10{+}5s$, $8{+}2s$), the per-winding det increment $2 = |\mathbb{Z}_2|$ and the minor law $(5,\ 3(s{+}1),\ 2(s{+}1))$ (the carrier minor 5 s-invariant) are all readouts of this single quadratic; the triple lock re-verified (trace/det/Coxeter each solve to $s = 6$ alone, intersection $\{6\} = \{|R^+(A_3)|\}$); the locus typed exactly: $\Delta(6) = 39200 = 2\cdot 140^2 > 0$ is *not* a collision point and the sole $2{\times}$square hit in the integer window, and the reality threshold is the unique Δ-root in $(2,3)$, $s^* \approx 2.8250$ (Sturm-exact — the deployed ≈ 2.83 above); every sibling direction breaks the lock *and* changes the quadratic — $(1,-5,2)$ is direction-specific [E]. *The feedback normal form (2026-08-08, round 36):* the winding line has an exact *gain-plus-companion* normal form ([verification/v874_feedback_decoder_lean.py], 14/14, FEEDBACK-NORMALFORM-EXACT): with the integral basis $P = [\,2\cdot\mathbf{1} \mid e_3 \mid Re_3\,]$ ($\det P = -4$, SNF = $\operatorname{diag}(1,2,2)$), the *whole* line conjugates symbolically in s to $P^{-1}R_sP = \left[\begin{smallmatrix} s{+}4 & 0 & 3 \\ 0 & 0 & -2 \\ 4 & 1 & 5 \end{smallmatrix}\right]$ — a gain channel $s + |\mu_4|$, the *companion block* of q_{wind}, and the coupling pair $(3,4)$ with loop product $N_{\text{fam}} \times |\mu_4| = 12 = \dim g_{\text{SM}}$ — so $\chi_{R_s} = (t - s - |\mu_4|)\,q_{\text{wind}} - 12\,t$ identically (the remainder identity $\chi_R - (t{-}4)\,q_{\text{wind}} = -12\,t$: a pure t-tap at gain offset $|\mu_4|$), and at $s = 6$ the $(15,40,20)$ coefficients are the three gain-10 readouts $10{+}5$, $50{+}2{-}12$, 10×2. The direction $\mathbf{1}e_1^\top$ is the *only* one (against $\mathbf{1}e_2^\top$, $\mathbf{1}e_3^\top$, $e_1\mathbf{1}^\top$ — all three controls fire) whose conjugated deformation is the pure gain E_{11} and whose remainder is a pure t-tap. *The honest typing (load-bearing):* the quotient $\mathbb{Z}^3/P\mathbb{Z}^3 = \mathbb{Z}_2 \times \mathbb{Z}_2$ has *cardinality* $4 = |\mu_4|$ but exponent 2 — not the group μ_4; no identification is claimed. Kernel-checked in `TfptCarrier/FlavorFeedback.lean` (11 theorems: the action identities, the inverse-free conjugation $PF_s = R_sP$ over $\mathbb{Z}$ symbolic in s, χ_{R_s} in $\mathbb{Z}[X]$, and the SNF determinantal divisors — standard axioms, no `sorry`) [E].

The dual anchor ([verification/v134_dual_anchor.py] **).** The anchor's *inverse* response is the same on both ends of the winding line, and it is the Nariai root of the horizon paper:

$$\boxed{d := a^\top R^{-1} = a^\top L^{-1} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)}, \qquad d\cdot\mathbf{1} = 0, \quad d\cdot a = 1, \quad (1,1,-2) = -2d.$$

The invariance is structural (Sherman–Morrison): $L = R + 6\,\mathbf{1}e_1^\top$ makes a covector winding-invariant iff it kills $R^{-1}\mathbf{1} = \frac{1}{4}(1,1,-1)$; the anchor does, while $\mathbf{1}$, e_1 and n do not ($\frac{1}{4}, \frac{1}{4}, -\frac{5}{2}$ — the membership is

special). (d, n) is the dual normal pair of the flavor boundary: d the *sheet* normal (kills $\mathbf{1}$, dual to a, $\propto$ Nariai), n the *torsion* normal (first-generation channel). [E]; bridge reading [C].

Walls of the surface ([verification/v135_det_surface.py]). The surface determinant factorises,

$$\boxed{\det M(s,t) = 3s(t{+}1)(t{+}2) + t^2 + 5t + 8},$$

exposing two *iso-volume walls independent of* s: $t = -2 \Rightarrow \det = 2 = |\mathbb{Z}_2|$ and $t = -1 \Rightarrow \det = 4 = |\mu_4|$ — so $\det K = 4$ is a whole *layer* of the surface ($K = M(1,-1)$ lies on the μ_4 wall), not an isolated hit — while the winding line $t = 0$ carries $\det = 6s + 8$, i.e. $(\det R, \det C, \det L) = (8, 14, 20)$, and the fourth corner closes the pattern with $\det F = \det M(1,1) = 32 = 2^{g_{\rm car}}$. The anchor block obeys $\det B_{M(s,0)} = 2 \det M(s,0)$ *exactly* on the winding line and has its own $\mathbb{Z}_2$ wall, $\det B_{M(s,-2)} = -2$ ($t = -1$ is *not* a B-wall: $6s + 4$, recorded honestly). Row-budget axes: $\mathrm{rowsum}(C) = (7, 11, 13)$, $\mathrm{rowsum}(U) = (3, 3, 3)$ (democratic winding), $\mathrm{rowsum}(V) = (1, 2, 3) = \mathrm{Spec}(Q_+)$ verbatim; the centre budget $(7, 11, 13) = (\text{scalaron}, \|\mathrm{Pl}(K)\|_1, \Delta_Q)$ is an audit match. Two anchor micro-identities: $e_1^2 + e_2^2 = 41 = 10\, b_1$ (the anchor Pythagoras for the abelian coefficient, companion to $\Delta_Y = 25 = 9{+}16$) and $p_1^2 + p_2^2 = 52 =$ the carrier coset root count ($40{+}12$, [verification/v128_graded_hull.py]). All [E]; the "2d surface with atom walls" reading [C].

The sheet generator V is a binary internal compiler [E] ([verification/v410_sheet_generator_binary.py])

The sheet axis $V = Q\,\mathrm{diag}(0,1,1)$ is not only the rank-2 direction of the cross — its *iterates* print the binary carrier spine. The closed form (proved by the inductive step $F(n)V = F(n{+}1)$)

$$V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}, \qquad V^n\mathbf{1} = (2^{n-1},\, 2^n,\, 2^{n+1}-1),$$

gives $V^{1..4}\mathbf{1} = (1,2,3), (2,4,7), (4,8,15), (8,16,31)$ — the A_3 exponents, then $(|\mathbb{Z}_2|, |\mu_4|, 7)$, $(|\mu_4|, \mathrm{rank}\, E_8, \dim A_3)$ and $(\mathrm{rank}\, E_8, \dim S^+, 2^{g_{\rm car}}{-}1{=}248/8)$. The four bilinear families are exact in n,

$$\mathbf{1}^T V^n \mathbf{1} = 7\cdot 2^{n-1} - 1, \quad \mathbf{1}^T V^n a = 7\cdot 2^{n-1}, \quad a^T V^n a = 11\cdot 2^{n-1}, \quad a^T V^n \mathbf{1} = 11\cdot 2^{n-1} - 2,$$

so the recurring flavor integers are one operator's readouts: $(6, 7, 9, 11)$ at $n{=}1$, then $\mathbf{1}^T V^2 \mathbf{1} = 13 = \Delta_Q$, $\mathbf{1}^T V^3 \mathbf{1} = 27 = \mathbf{1}^T R a$, $\mathbf{1}^T V^4 \mathbf{1} = 55$ (the quark numerator) and $\mathbf{1}^T V^4 a = 56 = \dim \mathbf{56}_{E_7}$; the amplitude $a^T V a = 11$ sits one level below the quark Plücker norm. *Typing:* the closed form and the bilinears are exact [E]; the binary spine is *forced* by $\mathrm{Spec}(V) = \{0, 1, 2\}$ (the dominant eigenvalue 2 doubles, the eigenvalue 1 fixes the third slot), so the carrier/Lie matches are audit-typed [C], exactly like the G_2 center reading above.

The quark u/d ratio is a pure V-power readout ([verification/v411_ud_ratio_vpower.py]). The closed quark ratio re-expresses as a quotient of three bilinear readouts of the *one* generator,

$$\boxed{\frac{c_u}{c_d} = \frac{\mathbf{1}^T V^4 \mathbf{1}}{(a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})} = \frac{55}{9\cdot 13} = \frac{55}{117}},$$

identical to the established $g_{\rm car}\|\mathrm{Pl}(K)\|_1/(N_{\rm fam}^2 \Delta_Q) = 5\cdot 11/(9\cdot 13)$ — the numerator merely repackages $g_{\rm car}\cdot 11 = 55 = \mathbf{1}^T V^4 \mathbf{1}$. This is an exact *re-encoding*: the value and the u/d identification stay [C], coupled to the $(U_{\rm wall})$ readout rigidity.

The source corner J on the $\mathbb{Z}_2$ wall ([verification/v412_sheet_source_corner_J.py]). The operator on the $t{=}{-}2$ iso-volume wall (the $\mathbb{Z}_2$ layer $\det{=}2$, v135), so far unnamed, $J := M(1,-2) = C - 2V = \left(\begin{smallmatrix} 4 & 1 & 0 \\ 4 & 1 & 2 \\ 5 & 1 & 1 \end{smallmatrix}\right)$, completes the sheet axis $J, K, C, F = M(1, -2..1)$ and is tightly typed:

$$\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \;\Rightarrow\; (\mathrm{tr}, e_2, \det) = (6, 3, 2) = (|R^+(A_3)|, N_{\rm fam}, |\mathbb{Z}_2|),$$

with $a^T J a = 30 = h(E_8)$, $\det(I{+}J) = 12 = \dim \mathfrak{g}_{\rm SM} = \chi_{E_8}(i)$ and $\det(2I{+}J) = 40 = |R(D_5)|$ — the Lie readings audit-typed like the G_2/F_4 shadows.

The sheet axis is a characteristic-difference calculator

([verification/v413_sheet_characteristic_calculus.py]). On $M_t = C + tV$ the elementary-symmetric coefficients encode the atoms as *difference orders*: $e_1(t) = 3t{+}12$, $e_2(t) = 2t^2{+}15t{+}25$, $e_3(t) = \det M_t = 4t^2{+}14t{+}14$, so $\Delta e_1 = 3 = N_{\rm fam}$, $\Delta^2 e_2 = 4 = |\mu_4|$ and $\Delta^2 e_3 = 8 = \operatorname{rank} E_8$ (the e_3 curvature is the v218/v224 sheet quadratic; the full triple is new). The anchor energy is exactly linear with the amplitude $11 = a^T V a$,

$$a^T M_t a = 52 + 11t: \qquad 30 \xrightarrow{+11} 41 \xrightarrow{+11} 52 \xrightarrow{+11} 63 = (h(E_8),\ 10b_1,\ \dim F_4,\ 7N_{\rm fam}^2),$$

so 11 is the *step* of the anchor energy along the sheet axis, not only the quark Plücker norm.

The center is a resolvent portal $G_2 \to F_4 \to E_8$ ([verification/v414_center_resolvent_portal.py]). The center $C = M(1,0)$ (the $t{=}0$ value of V) has $\operatorname{tr} C = 12 = \dim \mathfrak{g}_{\rm SM} = \chi_{E_8}(i)$, $\det C = 14 = \dim G_2$ and $\sum_{ij} C_{ij} = 31 = 2^{g_{\rm car}}{-}1 = 248/8$; its shifted determinants climb the exceptional shell ladder

$$\det C = 14 = \dim G_2, \qquad \det(I{+}C) = 52 = \dim F_4, \qquad \det(2I{+}C) = 120 = |R^+(E_8)|,$$

i.e. $\det(C{+}kI)$ for $k = 0,1,2$ traces $G_2 \to F_4 \to E_8^+$ — exact [E], the portal reading [C].

The center quotient compiler: the atoms as one operator, and the CP-halved Coxeter code ([verification/v530_center_quotient_compiler.py]). The center carries a *fixed primitive line*: with $v = (-1,1,0)$ one has $Cv = v$ exactly, and in the unimodular basis $P = (v, e_1, e_3)$ the induced operator on the quotient lattice $\mathbb{Z}^3/\mathbb{Z}v$ is

$$A = \begin{pmatrix} 8 & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} \operatorname{rank} E_8 & |\mathbb{Z}_2| \\ g_{\rm car} & N_{\rm fam} \end{pmatrix}:$$

the four discrete atoms form *one integer quotient operator*, with $\operatorname{tr} A = 11$, $\det A = 14 = \det C$, $\operatorname{disc} \chi_A = 65 = 5 \cdot 13$, $\operatorname{perm} A = 34 = p_5(a)$, $\prod A_{ij} = 240 = |R(E_8)|$ and $\det(A{-}I) = 4 = |\mu_4|$. The center *self-codes*: its row sums $C\mathbf{1} = (7, 11, 13)$ equal exactly $(\det A/|\mathbb{Z}_2|,\ \operatorname{tr} A,\ \operatorname{disc} \chi_A/g_{\rm car})$, and together with the fixed eigenvalue 1 this is $(1, 7, 11, 13)$ — the positive representatives of the CP pairs $m \sim 30{-}m$ of the E_8 Coxeter exponents $(\mathbb{Z}/30)^\times$ (the CP-halved Coxeter clock, v223). The v414 resolvent ladder acquires its inner structure: $\det(C{+}kI) = (k{+}1)\det(A{+}kI)$ *identically*, with $(k{+}1) = (1,2,3,4) = (N_\Phi, |\mathbb{Z}_2|, N_{\rm fam}, |\mu_4|)$ and the quotient ladder $\det(A{+}kI) = (14, 26, 40, 56)$ (readings $\dim G_2$, $\dim V_{26}(F_4)$, $|R(D_5)|$, $\dim V_{56}(E_7)$, audit-typed [C]), first differences $(12, 14, 16) = (\dim \mathfrak{g}_{\rm SM}, \dim G_2, \dim S^+)$, and the new endpoint $\det(C{+}3I) = 224 = 248 - 24$. *Honest uniqueness:* eigenvalue 1 is *not* unique to C (K and Q share it), but the K quotient $\left(\begin{smallmatrix} 14 & 8 \\ -11 & -6 \end{smallmatrix}\right)$ and the Q quotient $\left(\begin{smallmatrix} 3 & 1 \\ 3 & 2 \end{smallmatrix}\right)$ are not the atom matrix and cannot self-code ($5 \nmid 48$; $3/2 \notin \mathbb{Z}$) — C is unique in the *conjunction* fixed line + atom quotient + CP self-code; the full selector-stratum ablation stays open [O]. Exact [E], readings [C]. *Bridge corollary [E] ([verification/v977_transfer_unistochastic_wilson.py]):* the discriminant carries a Gaussian square-sum structure the sibling quotients lack, $\operatorname{disc} \chi_A = 65 = \det(A{-}I)^2 + (\det A/2)^2 = 4^2 + 7^2$ with $-4 + 7i = (1{+}2i)(2{+}3i)$ (norms $5 \cdot 13$), and $e^{i\delta} = (-\det(A{-}I) \pm i \det A/2)/\sqrt{\operatorname{disc} \chi_A} = (-4 \pm 7i)/\sqrt{65}$ is *exactly* the phase of the unistochastic $SU(3)$ lift of the seam transport root B (Jarlskog $J = \pm 1/27$; the K/Q sibling quotients fail the square-sum identity, $48 \neq 3^2{+}2^2$ and $13 \neq 1^2{+}(3/2)^2$) — an exact transport–center bridge; the intertwiner that would force the lift *angles* from A-data stays open (research-contracts document, `TRANSFER.COHERENT.WILSON.01`).

The sheet transfer is a norm line of discriminant -7 ([verification/v533_ftransfer_disc7_norm.py]). The branch-preserving transfer path $J, K, C, F = M(1, -2..1)$ (v224/v412) carries the exact norm identity

$$D(t) = \det M(1,t) = 4t^2 + 14t + 14 = \frac{(4t+7)^2 + 7}{4} = N_{\mathbb{Q}(\sqrt{-7})/\mathbb{Q}}(\alpha_t), \qquad \alpha_t = \frac{4t + 7 + \sqrt{-7}}{2},$$

with α_t an *algebraic integer* ($-7 \equiv 1 \bmod 4$, $4t{+}7$ odd, so $\alpha_t \in \mathbb{Z}[(1{+}\sqrt{-7})/2]$) and the whole path one arithmetic *translation* $\alpha_{t+1} = \alpha_t + 2 = \alpha_t + |\mathbb{Z}_2|$ with norms $(2, 4, 14, 32) = (\det J, \det K, \det C, \det F)$. The characteristic polynomial $x^2 - (4t{+}7)x + D(t)$ has discriminant -7 for *every* t; the vertex sits at $t = -7/4$ with $D_{\min} = 7/4 > 0$ (no real zero on the branch-preserving path), and the two sheet coefficients reappear arithmetically: curvature $D'' = 8 = \operatorname{rank} E_8$, $\sqrt{-\operatorname{disc}} = \sqrt{7}$ (the scalaron as fixed quadratic discriminant, [C]). Since $-7 \equiv 1 \bmod 8$, the prime 2 *splits* in $\mathbb{Q}(\sqrt{-7})$ (class number 1): the J endpoint norm $2 = |\mathbb{Z}_2|$ is the split prime above 2 (audit). Negative controls: the winding axis $\det M(s,0) = 6s{+}8$ is linear and the anchor block $\det B(1,t) = 3t^2{+}21t{+}28$ has *positive* discriminant 105

— only the sheet determinant is a complex norm line. Whether the four external transfer solvers act on α_t by a *common* fractional-linear action is the preregistered next test (kill: four incompatible arithmetic actions); F_{transfer} itself stays an external interface [O]. Exact [E], the arithmetic frame [C]/[O].

The square CM-norm dictionary is operator-realised, and the deck is intrinsic ([verification/v415_gaussian_operator.py]). The non-commutativity of the two axes is itself a μ_4 quarter-turn: the integer commutator

$$J := \tfrac{1}{3}[U,V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}, \qquad \mathrm{Spec}(J) = \{i, -i, 0\}, \quad J^2 = -I \text{ on the rank-2 image},$$

is a complex structure born from the binary V and ternary U compilers. It lifts the Gaussian arithmetic of v222/v230 from numbers to operators: $3I{+}2J$ has spectrum $\{3{+}2i, 3{-}2i, 3\}$ and $5I{+}4J$ has $\{5{+}4i, 5{-}4i, 5\}$, so the Gaussian integers $3{+}2i$ and $5{+}4i$ are literal eigenvalues, and

$$(3I{+}2J)(3I{-}2J) = 9I - 4J^2 \sim \{13, 13, 9\}, \qquad (5I{+}4J)(5I{-}4J) = 25I - 16J^2 \sim \{41, 41, 25\},$$

i.e. $\Delta_Q = 13$ and $10b_1 = 41$ on the μ_4-plane, the real part^2 $(N_{\text{fam}}^2,\ g_{\text{car}}^2)$ on the kernel. Finally $D := -I + J - J^2$ is an *intrinsic* order-4 deck ($D^4 = I$, $\chi_D = (x{+}1)(x^2{+}1)$, $\mathrm{Spec}\{i, -1, -i\}$) whose self-conjugate χ_2-line is canonically $\ker[U,V] = (0,0,1) = a - \mathbf{1}$; this exhibits the μ_4 deck as a diamond object and sharpens the v140 $\mathbb{Z}_3$ residue to one concrete identification — the operator identities [E], the geometric-deck match [C].

The Eisenstein dual and the CP clock ([verification/v417_eisenstein_cp_operator.py]). The same lift works on the *hexagonal* side. The family rotation $P = (1\,2\,3)$ is the order-3 Eisenstein deck ($P^2 + P + I = \mathrm{ONES}$, so $\omega^2 + \omega + 1 = 0$ on the plane $\perp (1,1,1)$ — the order-3 dual of $J^2 = -I$), and it realises the hexagonal CM norm as an operator,

$$(3I + 2P)(3I + 2P^2) = 7I + 6\,\mathrm{ONES} \sim \{7, 7, 25\},$$

i.e. $7 = N_{\mathbb{Z}[\omega]}(3{+}2\omega)$ = scalaron on the ω-plane and $25 = g_{\text{car}}^2$ on $(1,1,1)$ — the exact dual of the square $\{13, 13, 9\}$. The CP phase is the order-6 clock $W = -P^2$ (the triality $\rho = \omega^2 \cdot (-1)$): $W^2 = P$, $W^3 = -I$, $\mathrm{Spec}\, W = \{-1, \zeta_6, \zeta_6^{-1}\}$, and the two measured phases are arguments of its powers, $\delta_{\text{CKM,lead}} = \arg \zeta_6 = \pi/3$ and $\delta_{\text{PMNS}} = \arg \zeta_6^4 = 4\pi/3$ (v316/v320). All four flavor-side clocks are powers of one Coxeter unit ζ_{30} ($h(E_8) = 30 = |\mathbb{Z}_2| N_{\text{fam}} g_{\text{car}}$): $\zeta_{30}^{15} = -1$, $\zeta_{30}^{10} = \omega$, $\zeta_{30}^{6} = \zeta_5$, $\zeta_{30}^{5} = \zeta_6$, while the seam $\mu_4 = i$ is *not* a power ($4 \nmid 30$) but the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30) = 8 = \mathrm{rank}\, E_8$ — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants at once. *Honest:* the carrier μ_5 has no $3{\times}3$ operator (the golden $\varphi = 2\cos\frac{\pi}{5}$ is an output-side trace, v313/v349); the operator/arithmetic identities are [E], the clock-assignment readings [C].

The Q_+ grading is cohomology: one spectrum in four readouts [E] ([verification/v137_qplus_cohomology.py])

The sheet axis is not just numerology on V: it is the cohomology of the seam curve. $b_1(\mathbb{P}\backslash\mu_4) = 3 = N_{\text{fam}}$, with holomorphic basis $\omega_k = z^{k-1}dz/(z^4 - 1)$, $k = 1, 2, 3$ (regular at infinity; the trivial character is absent), and under the μ_4 deck $z \mapsto iz$

$$\omega_k \ \longmapsto\ i^k\, \omega_k, \qquad \mathrm{res}_\zeta\, \omega_k = \tfrac{\zeta^k}{4}\ :$$

$H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ — each nontrivial μ_4 character exactly once, and the residue vector over the punctures *is* the character vector. The character degrees $(1, 2, 3)$ are exactly $\mathrm{Spec}(Q_+)$ ($Q_+ = 3\,\mathrm{cusp} + 1$, v69/v72), and the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ underneath are $\mathrm{spec}(A_0^\star)$ (v115):

$$\boxed{\text{MS weights} \ =\ \text{cusp classes} \ =\ H^1 \text{ characters} \ =\ \mathrm{Spec}(Q_+) - 1 \text{ over } N_{\text{fam}}}$$

— one spectrum in four readouts; and $(1, 2, 3)$ are the *exponents of* A_3 ($W(A_3) = S_4$, v117; $h(A_3) = 4 = |\mu_4|$): the Q_+ generation grading *is* the A_3 exponent grading, realised on the seam curve's cohomology. *Typing:* all spectrum statements [E]; what remains of `GATE.QGEO` is the *naturality* of the identification — the canonical map from the H^1 character basis to the V-axis generation basis ($\mathrm{rowsum}(V) = (1, 2, 3)$ verbatim, [verification/v135_det_surface.py]) — typed [C].

The canonical map exists and is rigid ([verification/v140_canonical_map.py]). The naturality residue collapses. Of the established μ_4 actions on generation space, the plain deck $U = \mathrm{diag}(1, i, -i)$ has characters $(0, 1, 3)$ — an invariant line, missing 2 — and *cannot* match H^1; but the **sheet-twisted** deck $-U$ (characters $(2, 3, 1)$) and the cusp exponential i^{Q_+} (characters $(1, 2, 3)$) both carry the character *set* $\{1, 2, 3\}$ — exactly the H^1 set, by the same sign twist that builds the v118 hexagon. Both sides are multiplicity-free, so by Schur the equivariant isomorphism *exists and is unique* up to one scalar per character: the canonical map is well-defined — existence and uniqueness are no longer open. Under the i^{Q_+}-map the H^1 Euler degree pulls back to Q_+ itself; under the $-U$-map to $Q_+ \circ \rho$ with ρ the (123) family rotation: the two candidates differ by *exactly one* $\mathbb{Z}_3$ rotation. *Typing:* character algebra, rigidity and grading transport [E].

The deck selection theorem: the $\mathbb{Z}_3$ choice is *derived* ([verification/v141_deck_selection.py]). The discrete choice does not survive contact with the integer model the suite already owns: the derived integer deck $G = T_A\Sigma$ ([verification/v97_sheet_conjugation_bridge.py]/[verification/v98_discriminant_dictionary.py]) has the Q_+=1 cusp line as an *exact* eigenline with eigenvalue $-1 = i^2$ — the self-conjugate character 2 — and the conjugate pair $\{1, 3\}$ on the E-plane. Of the three $\mathbb{Z}_3$ rotations only the sheet-twisted assignment $(2, 3, 1) = \mathrm{chars}(-U)$ puts character 2 on the Q_+=1 line: **the geometric boundary deck is the sheet-twisted class; the cusp exponential i^{Q_+} is excluded** (same eigenvalue multiset $\{-1, \pm i\}$, wrong grading pairing — the (-1)-eigenlines sit on Q_+ degrees 1 vs 2). An explicit Q_+=1-line-preserving conjugacy $SGS^{-1} = -U$ exists; under the relabeling $k \to -k$ only the plane orientation flips = the established sheet $\mathbb{Z}_2$ (the two Lagrangian glues, v92) — no new residual. Consequence: the H^1 Euler degree pulls back to $Q_+ \circ \rho$; cusp grading and Euler grading differ on generation space by exactly the one family rotation ρ = the coker Q triality (v72). *Typing:* selection, conjugacy and obstruction [E] (lattice-level theorem); what remains of `GATE.QGEO` is only the realisation premise the gate always carried (the integer D_4 model *is* the parabolic geometry, v69/v98) [C] — with *no* discrete freedom left.

The Möbius D_4 realisation: the premise at its floor ([verification/v146_moebius_d4.py]). The realisation premise itself now has every checkable part checked. The *full* dihedral symmetry group of the seam curve, $\langle \delta : z \mapsto iz,\ \iota : z \mapsto 1/z \rangle = D_4$, acts on the explicit cohomology basis $\omega_k = z^{k-1}dz/(z^4-1)$ exactly as the integer model: $\delta^*\omega_k = i^k\omega_k$ (the deck, v137), and

$$\boxed{\iota^*\omega_1 = \omega_3, \quad \iota^*\omega_2 = \omega_2, \quad \iota^*\omega_3 = \omega_1}$$

— the exponent duality with parity $+1$ on the self-conjugate line and $\det = -1$, *no extra phases*. In the cusp basis the anchor-forced conjugation T_A *is* this permutation (fixes the character-2 line with $+1$, $\det = -1$ = the glue-swap parity), and Σ *is* the $\delta\iota$ class (-1 on the character-2 line, $\det = +1$ = glue-fix): the v98 reflection-class split lands exactly on the geometric reflection classes. *Typing:* the dihedral match [E]; with spectra (v137), cyclic part (v141) and reflections now all matched, the premise reduces to the module identification itself — "generation space *is* H^1 of the seam curve with its parabolic weights" [C] — with no remaining checkable mismatch.

Spectra and the determinant ladder $\to |W(D_5)|$ [E]

The two operators carry the compiler atoms in their characteristic polynomials,

$$\chi_Q = (t-1)(t^2-5t+3) \ \ [\,g_{\mathrm{car}}, N_{\mathrm{fam}}\,], \qquad \chi_K = (t-1)(t^2-8t+4) \ \ [\,h(D_5), |\mu_4|\,], \tag{3}$$

with discriminants $\Delta_Q = 13 = |R(A_3)|+1$ and $\Delta_K = 48 = \Omega_{\mathrm{adm}}$. Their determinants form a torsion ladder (coker $= \mathbb{Z}_3, \mathbb{Z}_4, \mathbb{Z}_8, \mathbb{Z}_{20}$),

$$\begin{gathered}(\det Q, \det K, \det R, \det L) = (3, 4, 8, 20), \\ \boxed{\det Q \cdot \det K \cdot \det R \cdot \det L = 1920 = |W(D_5)|} \qquad [\mathbf{E}],\end{gathered} \tag{4}$$

tying flavor residues, mass powers, transport and the D_5 Weyl group (the same 1920 that appears in the horizon Hawking-power denominator). The sheet decomposition is sharp: $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ are the A_3 exponents ($\det Q_+ = 6 = |R^+(A_3)|$), $\chi_{Q_-} = t(t^2-3)$ (a square root of N_{fam}), and $\chi_{K_-} = t(t^2-8)$ (a square root of $h(D_5)$). Finally the EM budget sits in the K anchor block: $a^\top Ka = 41 = 10b_1$ with $a = (1, 1, 2)$, and $\det\begin{pmatrix} \mathbf{1}^\top K\mathbf{1} & \mathbf{1}^\top Ka \\ a^\top K\mathbf{1} & a^\top Ka \end{pmatrix} = \det\left(\begin{smallmatrix} 25 & 29 \\ 35 & 41 \end{smallmatrix}\right) = 10 = \mathcal{A}_\Lambda$.

[verification/v10_projection_involution.py], [verification/v12_mass_generation_polynomials.py]

Mass volume and the EM budget are *one* anchor operation ([verification/v74_compiler_micro_lemmas.py]). The two diagonal entries of B_K are the two physical readouts: $\mathbf{1}^\top K\mathbf{1} = 25 = g_{\rm car}^2$ is the grand *mass-volume* exponent ($\det M_{\rm SM} \sim (\varphi_0^{\rm ret})^{25}$), and $a^\top K a = 41 = 10b_1$ is the *EM budget*. Their difference is one generation,

$$\boxed{a^\top K a - \mathbf{1}^\top K\mathbf{1} = 41 - 25 = 16 = \dim S^+},$$

i.e. *EM budget = mass volume + one generation*. So the number 41 that the EM fixed point reads is not "40+1" but the anchor quadratic form of the mass-power matrix, tied to the mass volume $g_{\rm car}^2$ by a single half-spinor $\dim S^+$.

Mass and transport are one pencil: $\det(K+xQ)$ carries the scalaron sequence [E]

Because $L = K + Q$, the mass-power matrix K and the transport matrix L are the two endpoints ($x = 0, 1$) of a *single* one-parameter compiler pencil $P(x) = K + xQ$, whose determinant is

$$\boxed{\det(K+xQ) = 3x^3 + 7x^2 + 6x + 4}, \qquad (3,7,6,4) = \big(N_{\rm fam},\ \Omega_{\rm adm} - 10b_1,\ |R^+(A_3)|,\ |\mu_4|\big),$$

i.e. the coefficient sequence is $N_{\rm fam}$, the *scalaron exponent* 7, the A_3 positive roots and the glue index — one curve linking flavor, the Starobinsky scalaron, the family and the glue. The pencil is anchored at four integer nodes by compiler atoms: $P(-1) = 2 = |\mathbb{Z}_2|$ (the $K{-}Q$ sheet endpoint, with $\det(K{-}Q) = |\mathbb{Z}_2|$, $\mathrm{tr}(K{-}Q) = N_{\rm fam}$), $P(0) = 4 = \det K = |\mu_4|$, $P(1) = 20 = \det L$, and $P(2) = 68 = 2\,p_5(a)$ (twice the anchor power sum $p_5 = 34$). [verification/v52_pencil_endpoints.py]

And the consecutive *differences* are the sheet→matter chain ([verification/v74_compiler_micro_lemmas.py]). The node values $(2, 4, 20, 68)$ step by

$$\boxed{P(0){-}P(-1) = 2 = |\mathbb{Z}_2|, \quad P(1){-}P(0) = 16 = \dim S^+, \quad P(2){-}P(1) = 48 = \Omega_{\rm adm}}$$

i.e. $2 \to 16 \to 48$ — *sheet → one generation → three generations* — so the single determinant pencil already encodes the passage from the $\mathbb{Z}_2$ sheet to the one-family half-spinor to the full admissible matter packet. The (K, Q, L) operators are therefore not separate artefacts but one curve. The non-commutativity of the mass→transport step is a fixed G_2-sized obstruction: $[K, L] = [L, Q] = [K, Q]$ (forced by $L = K + Q$) with $\chi_{[K,Q]}(t) = t^3 + 5t + 14$, reading $g_{\rm car} = 5$ and $\dim G_2 = 14$ (audit-level [O]). [verification/v37_plucker_anchor.py]

Operator-pencil geometry: the anchor plane as a small area in operator space [E] ([verification/v80_operator_pencil_geometry.py])

Read the four operators through the *anchor block* $B_M = \begin{pmatrix} \mathbf{1}^\top M\mathbf{1} & \mathbf{1}^\top Ma \\ a^\top M\mathbf{1} & a^\top Ma \end{pmatrix}$. **(i) Anchor Singularity Theorem.** On the pencil $P(x) = K + xQ$,

$$\boxed{\det B_{K+xQ} = 9x^2 + 21x + 10 = (N_{\rm fam}x + |\mathbb{Z}_2|)(N_{\rm fam}x + g_{\rm car}) = (3x+2)(3x+5)} \qquad [\mathbf{E}],$$

so the oriented anchor area *degenerates* at exactly $x = -|\mathbb{Z}_2|/N_{\rm fam} = -\frac{2}{3}$ and $x = -g_{\rm car}/N_{\rm fam} = -\frac{5}{3}$. The first zero is the Koide target / transport-gap base ($\lambda_2 = (\frac{2}{3})^6$); the second is the D_5/A_3 glue asymmetry. **So $\frac{2}{3}$ is not just a recurring fraction — it is the first singular point of the mass↔transport anchor plane**, a structural *location* where an oriented area collapses. **(ii) Type checker.** The anchor-block determinants classify the operators: $\det B_Q = 9 = N_{\rm fam}^2$ (family square), $\det B_K = 10 = \mathcal{A}_\Lambda$ (pair sector), $\det B_R = 16 = \dim S^+$ (one generation), $\det B_L = 40 = |R(D_5)|$ (carrier root budget) — each a determinant of a fixed block, not a loose integer. **(iii) Buildup chain.** The pencil coefficients $(3, 7, 6, 4)$ have partial sums $(3, 10, 16, 20) = (N_{\rm fam}, \mathcal{A}_\Lambda, \dim S^+, \det L)$, so the scalaron exponent is the internal flavor transition $7 = \mathcal{A}_\Lambda - N_{\rm fam}$.

Audit layer (exact values [E], decorative reading [C]). $D(x) = \det(K{+}xQ)$ and its derivatives at $x = -1, 0, 1, 2$ give $(2, 4, 20, 68), (1, 6, 29, 70), (-4, 14, 32, 50)$ (29 the top E_8 exponent, $32 = 2^{g_{\rm car}}$, 50 the $A_4{\times}A_4$ off-block); and the missing corner $F = R{+}Q$ has $\det B_F = 52 = \dim F_4$ with $248 = 52{+}14{+}26{\cdot}7$ — the genuine $E_8 \supset F_4{\times}G_2$ branching $(52,1)\oplus(1,14)\oplus(26,7)$ as an operator *shadow*. These are exact identities; their atlas/embedding significance is kept audit-level ([C]), since the coefficients are themselves

atoms.

The Koide point is one branch of a double cover [E] ([verification/v81_singular_pencil_matrices.py])

The anchor-block determinant $\det B_{K+xQ} = (3x+2)(3x+5)$ is *quadratic*, so

$$\boxed{y^2 = \det B_{K+xQ} = (3x+2)(3x+5)}$$

is a 2:1 cover of the pencil line, *ramified exactly over its two zeros*. Hence the Koide/gap point $x = -|\mathbb{Z}_2|/N_{\rm fam} = -\frac{2}{3}$ and the carrier point $x = -g_{\rm car}/N_{\rm fam} = -\frac{5}{3}$ are **the two branch points of one double cover**: *Koide is literally "the other side" of the carrier point.* The deck involution is $y \mapsto -y$ of degree $2 = |\mathbb{Z}_2|$ — the sheet itself — which is the line-bundle reflection of the $\mathrm{Spin}(10) \to \mathrm{SO}(10)$ half-spinor cover the carrier lives on. Three rigid readings:

- **Branch locus = spine ends / family.** The ramification is at $-2/3$ and $-5/3$, i.e. $-(|\mathbb{Z}_2|, g_{\rm car})/N_{\rm fam}$ — the *two ends* of the spine $(2,3,4,5)$ over $N_{\rm fam}$.
- **Discriminant** $= N_{\rm fam}^4 = 81$ (a perfect square ⇒ rational, split branch points); $\sqrt{81} = 9 = N_{\rm fam}^2 = \det B_Q$.
- **Separation** $= 1 =$ **one transport period** (the $K \to L$ step $x : 0 \to 1$); the deck-translation $x \mapsto x+1$ carries the carrier point onto the Koide point.
- **Branch divisor = (scalaron, $\mathcal{A}_\Lambda$).** Sum of branch points $= -7/3 = -7/N_{\rm fam}$, product $= 10/9 = \mathcal{A}_\Lambda/N_{\rm fam}^2$; the integer labels $\{|\mathbb{Z}_2|, g_{\rm car}\} = \{2,5\}$ have sum $= 7$, prod $= 10 = \mathcal{A}_\Lambda$, diff $= 3 = N_{\rm fam}$. **So the scalaron is the *trace* of the branch divisor** — its several readouts ($\Omega_{\rm adm} - 10b_1$, $\mathcal{A}_\Lambda - N_{\rm fam}$, $g_{\rm car} + |\mathbb{Z}_2|$, the mixed anchor area) collapse to this one geometric origin.
- **The sheet lives on the cut.** The fibre $y^2 = \det B$ is $\mathcal{A}_\Lambda$ at K and $4\mathcal{A}_\Lambda$ at L, so transport multiplies y by $|\mathbb{Z}_2|$; the $\mathbb{Z}_2$ sheet endpoint $x = -1$ sits *between* the branch points where $y^2 = \det B_{K-Q} = -|\mathbb{Z}_2| < 0$ — on the imaginary branch cut joining Koide and the carrier.

At each branch point the pencil clears its 3-denominator into an integer compiler matrix. *Koide* $C_{2/3} = 3K - 2Q$ reproduces the carrier glue: $\mathrm{tr} = 15 = \dim A_3$, $\sum = 45 = \dim D_5$, $\det = 60 = D_{\rm start}$, and the (now forced rank-one) anchor block $B_{C_{2/3}} = \left(\begin{smallmatrix} 45 & 55 \\ 63 & 77 \end{smallmatrix}\right) = \binom{g_{\rm car}}{7}\left(\begin{smallmatrix} N_{\rm fam}^2 & 11 \end{smallmatrix}\right)$ with $\sum = 240 = |R(E_8)|$. *Carrier* $C_{5/3} = 3K - 5Q$ is the charge-*neutral* side ($\sum_{ij} = 0$) with $\chi(\lambda) = (\lambda + |\mathbb{Z}_2|)(\lambda^2 + \lambda + |R^+(A_3)|) = (\lambda+2)(\lambda^2+\lambda+6)$, splitting off the sheet eigenvalue $-|\mathbb{Z}_2|$. So the recurring $\frac{2}{3}$ is not a quotient that "appears everywhere" — it is one ramification point of the mass↔transport double cover. The sharper open question is then *why the leptonic source-to-pole transfer renormalises onto a branch point, and which sheet ($y \to \pm$) the parity selects* — a cleaner target than "why 2/3". (The D_5/A_3 and E_8 labels on the clearing matrices are exact; their per-entry atlas reading stays audit-level [C].) *Gravitational twin:* the same split $N_{\rm fam}$-linear double-cover structure reappears in classical gravity — the Schwarzschild–de Sitter mass line has disc $\propto (1-3m)(1+3m)$ (branch points $\pm 1/N_{\rm fam}$, separation $2/3$, deck = horizon swap), the Nariai roots are the traceless anchor $(1,1,-2)$, and $2/3$ is the exact Nariai/de-Sitter entropy bound (Appendix H; [verification/v101_horizon_anchor.py]).

The Koide branch is a *forced* attractor, and the clean split is non-generic [E]/[C] ([verification/v82_koide_attractor_splitting.py])

This sharpens the open question above. Work in the branch coordinate $q = -3x$ (Koide $q = 2$, carrier $q = 5$). **(A) The source-to-pole map is forced, not postulated.** Any branch-divisor-preserving Möbius map of the pencil line that fixes *both* branch points is, in the cross-ratio coordinate $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$ — hence *unique* once μ is fixed. The multiplier is not a new input: $\mu = (2/3)^6$ is the subleading eigenvalue λ_2 of the *already-established* unique gapped boundary transfer operator (spectrum $\{1, (2/3)^6, (1/3)^6\}$, gap $6\log\frac{3}{2} > 0$; [verification/v54_seam_horizon_keystones.py], [verification/v56_unique_attractor.py]), and the Koide branch $x = -2/3 = -|\mathbb{Z}_2|/N_{\rm fam}$ is precisely its cusp weight $2/3 = \{0, 1, |\mathbb{Z}_2|\}/N_{\rm fam}$ (the carrier maps to Koide under the deck period $q \mapsto q - N_{\rm fam}$, the same cusp class). With $\mu = (2/3)^6$ the unique map is the integer Möbius map

$$\boxed{F(q) = \frac{2(569q - 3325)}{665q - 3517}}, \qquad F(2) = 2, \quad F(5) = 5, \quad F'(2) = (2/3)^6 < 1, \quad F'(5) = (3/2)^6 > 1,$$

so Koide is the **attractor** and the carrier the **repeller**, and $q_n \to 2$ i.e. $x \to -2/3$ at the *same* rate that governs the SM flavor gap and horizon recovery. The three structural postulates needed for a Koide derivation thus collapse to *one* physical identification — *source-to-pole transfer = branch-preserving relaxation of the established transfer operator* — after which rate, uniqueness and direction all follow. That single identification stays [C]; the rest is [E]. **(B) The clean rational split is non-generic.** The splitting-type exponents $(2,1,1)$ (the anchor $a=(1,1,2)$) placed across the three family slots give three different covers: $(1,1,2)$ gives $\det B_{K+xQ} = (3x+2)(3x+5)$, disc $= 81 = N_{\rm fam}^4$ (*split*, rational labels $\{2,5\}$); $(2,1,1)$ gives $(x+2)(9x+11)$, disc $= 49 = 7^2$ (*split*); $(1,2,1)$ gives $5x^2+10x+3$, disc $= 40 = |R(D_5)| = \det B_L$, which is *not* a square — a **non-split** cover with Galois-conjugate branch points $-1 \pm \sqrt{10}/5$ centred exactly on the sheet $x=-1$ ($q=3$), separation $2\sqrt{\mathcal{A}_\Lambda}$. (Negative control, as for the cover itself: the R-pencil has disc $= 153$, also not a square.) So Koide and the carrier appearing as *separate rational* singularities is a special feature of the canonical anchor placement, not a generic one — a hardening of anchor-first. The physical fermion-sector identification of the three placements is not claimed and stays [C].

Branch kernels select the sectors; one deck involution [E] ([verification/v96_branch_kernel_selection.py], [verification/v97_sheet_conjugation_bridge.py])

The "which sheet does parity select" question above had a blocker: no *independently motivated* sector→sheet map. The cover supplies one itself. At each branch point the anchor block is **rank one**, with canonical *integer* kernels:

$$\text{Koide } x=-\tfrac{2}{3}: \quad w = 11\,\mathbf{1} - 9a = (2,2,-7), \quad v = 7\,\mathbf{1} - 5a = (2,2,-3);$$
$$\text{carrier } x=-\tfrac{5}{3}: \quad w = \mathbf{1}, \quad v = 8\,\mathbf{1} - 7a = (1,1,-6),$$

— at the carrier point the block annihilates the *democratic vector itself*. **Collapse lemma [E]:** rank one forces $P(x_0)\,w \perp \mathrm{span}\{\mathbf{1}, a\}$, hence proportional to the antisymmetric direction $(-1,1,0)$ of the anchor's two *equal* slots — on both sides:

$$P(-\tfrac{2}{3})\,w = \tfrac{20}{3}\,(1,-1,0), \qquad P(-\tfrac{5}{3})\,w = \tfrac{2}{3}\,(-1,1,0), \qquad v^\top P(x_0) \propto (-1,1,0) \text{ at both points.}$$

Reading the rows as the sectors (u,d,e): *up and down are the deck-odd pair* ($f_u = -f_d$; convention-free, $f_u f_d < 0$ at both points), and the *lepton pairing vanishes* — the leptons sit *on* the ramification, consistent with Koide being a charged-lepton relation. Indeed the lepton pairing forms *are* the two det-B factors: the **sector zero ladder** reads lepton $\to \{-\frac{2}{3}, -\frac{5}{3}\}$ (the branch points), up $\to -\frac{3}{2} = -N_{\rm fam}/|\mathbb{Z}_2|$ *doubly* (the whole up-row of $P(-\frac{3}{2})$ *is* the odd direction $-\frac{1}{2}(1,-1,0)$ — the previously unexplained GL(2) rung), down $\to \{0\ (=K),\ -\frac{9}{5} = -N_{\rm fam}^2/g_{\rm car}\}$. The deck translation flips the orientation $(+,-) \to (-,+)$ with magnitude ratio $10 = \mathcal{A}_\Lambda$. *Controls:* anchor $(2,1,1)$ moves the collapse to $(0,1,-1)$; $(1,2,1)$ is non-split (disc 40), no rational kernels at all.

The bridge to the glue chiralities ([verification/v97_sheet_conjugation_bridge.py]). The two E_8 glues are swapped by *either single* spinor conjugation and fixed by the simultaneous one ([verification/v92_glue_uniqueness.py]); on the family side, conjugation negates the cusp classes (0 fixed, $\frac{1}{3} \leftrightarrow \frac{2}{3}$ on the Q_+ eigenspaces $e_1=(0,0,1)$, $e_2=(0,1,2)$, $e_3=(1,0,0)$). That conjugation has a *unique anchor-compatible* integer realisation: $T\mathbf{1} = \mathbf{1}$ and $Ta = a$ *both* force the same scale, giving

$$\boxed{T_A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 2 & -2 & 1 \end{pmatrix}, \quad T_A^2 = I, \quad \det T_A = -1}, \qquad \text{and} \qquad \boxed{a = e_2 + e_3}$$

— *the anchor is the conjugation-symmetrised vector of the swapped pair* (the two alternative pairings admit *no* anchor-compatible scale). On the quotient by the anchor plane, $T_A = -1$ exactly like the sector transposition: **one deck action where the cover lives**. The parity matches the glue arithmetic ($\det = -1$ single/swapping vs $\det \Sigma = +1$ simultaneous/fixing); $\langle T_A, \Sigma\rangle$ closes the *same* $D_4 = \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ as the Q-geometry, with $\chi_{T_A\Sigma} = (t+1)(t^2+1)$ identical to the μ_4 homology matrix; and the **sheet-index lemma**: the generation-μ_4 and the homology-μ_4 are rationally but *not* $\mathrm{GL}(3,\mathbb{Z})$-conjugate — every intertwiner has *even* determinant, minimum $|\det| = 2 = |\mathbb{Z}_2|$ (one sheet doubling; independent of $\det Q = N_{\rm fam}$).

The dictionary is derived, not assumed ([verification/v98_discriminant_dictionary.py]). The generation space carries an *integer* μ_4: $G := T_A\Sigma$ acts on the cusp basis as

$$\boxed{G\,e_1 = -e_1, \qquad G\,e_3 = e_2, \qquad G\,e_2 = -e_3} \qquad (G^4 = I, \quad \chi_G = (t+1)(t^2+1)),$$

i.e. the $B_1 \oplus E$ decomposition of the D_4-equivariant Q-geometry realised *integrally*: e_1 spans the B_1 line (G-eigenvalue -1 = the self-conjugate $\mathbb{Z}_4$ class 2), and span$\{e_2, e_3\}$ is the E-plane (eigenvalues $\pm i$ = the conjugate class pair $\{1,3\}$). **So the cusp-0 line *is* the self-conjugate discriminant line, and the swapped cusp pair *is* the conjugate class pair** — the dictionary *"Q_+ grading = A_3 discriminant grading"* holds inside the model. Consistently: $T_A G T_A^{-1} = G^{-1}$ (the discriminant conjugation $k \mapsto -k$), $q_{A_3}(1) = q_{A_3}(3) = \frac{3}{8}$ (equal on the swapped pair) with $q_{A_3}(2) = \frac{1}{2}$ fixed, and the two D_4 reflection classes separate exactly the glue arithmetic: $T_A e_1 = +e_1$, det $= -1$ (glue-*swapping*) vs $\Sigma e_1 = -e_1$, det $= +1$ (glue-*fixing*), with $\Sigma = T_A G$. *Honest scope:* the sheet question therefore carries **no separate [C] residual any more** — what remains is exactly the already-open Q-geometry realisation gate (GATE.QGEO, the parabolic construction that realises the documented Q on $\mathbb{P}^1 \backslash \mu_4$); conditional on that established typing, the sector→sheet selection is *closed*: up/down are the deck-odd pair, the leptons sit on the ramification, and the deck involution is the A_3-side spinor conjugation realised as T_A.

Grand Mass Volume: the absolute mass-scaling normalisation [E]

The individual c_q are a gauge, but the *absolute* normalisation of the mass spectrum is fixed by the sector mass-matrix *determinants* — each a clean φ_0^{ret}-power whose exponent is the K-row sum:

$$\det M_{\mathrm{up}} \sim (\varphi_0^{\mathrm{ret}})^6 = (\varphi_0^{\mathrm{ret}})^{|R^+(A_3)|}, \quad \det M_{\mathrm{down}} \sim (\varphi_0^{\mathrm{ret}})^9 = (\varphi_0^{\mathrm{ret}})^{N_{\mathrm{fam}}^2}, \quad \det M_{\mathrm{lep}} \sim (\varphi_0^{\mathrm{ret}})^{10} = (\varphi_0^{\mathrm{ret}})^{\mathcal{A}_\Lambda},$$

$$\boxed{\det M_{\mathrm{SM}} \sim (\varphi_0^{\mathrm{ret}})^{6+9+10} = (\varphi_0^{\mathrm{ret}})^{25} = (\varphi_0^{\mathrm{ret}})^{g_{\mathrm{car}}^2} = (\varphi_0^{\mathrm{ret}})^{\Delta_Y}}$$

the *Grand Mass Volume*. The Q-row sums $(4,5,6) = (|\mu_4|, g_{\mathrm{car}}, |R^+(A_3)|)$ total $15 = \dim A_3$, and $25 + 15 = 40 = \sum L = |R(D_5)|$ (mass volume + shift volume = transport budget). So the absolute scaling is an [E] determinant identity (each sector = its K-row sum); the only genuine scale input is the overall v_{geo}, not a per-coefficient freedom. [verification/v46_grand_mass_volume.py]

Uniqueness (not fitted). K and Q are the *unique* nonnegative-integer 3×3 matrices with their row sums, column sums, characteristic polynomial and per-sector monotone hierarchy — verified by exhaustive enumeration [verification/v11_unique_KQ.py]. They are forced by small compiler budgets plus spectrum, not chosen.

Theorem Q — the Σ-split as a parabolic projection (sharper uniqueness) [E]/[C]

The sheet involution $\Sigma = \mathrm{diag}(1,-1,-1)$ splits Q into two parabolic pieces:

$$Q_+ = \tfrac{1}{2}(Q + \Sigma Q \Sigma) = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 2 & 1 \end{pmatrix}, \quad Q_- = \tfrac{1}{2}(Q - \Sigma Q \Sigma) = \begin{pmatrix} 0 & 1 & 0 \\ 3 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix},$$

$$\chi_{Q_+}(t) = (t-1)(t-2)(t-3) \;\Rightarrow\; \mathrm{Spec}(Q_+) = \{1,2,3\} \ (A_3 \text{ exponents}),$$

$$\chi_{Q_-}(t) = t(t^2-3) \;\Rightarrow\; Q_-^2|_{\mathrm{supp}} = 3 = N_{\mathrm{fam}}.$$

Uniqueness: Q is the *unique* nonnegative-integer matrix with rows $= (4,5,6)$, cols $= (9,5,1)$, $\chi_{Q_+} = (t-1)(t-2)(t-3)$ and $\chi_{Q_-} = t(t^2-3)$ (exhaustive enumeration, [verification/v50_q_geometry.py]). So Q_+ is the A_3 exponent grading, Q_- the μ_4 sheet coupling, and $K = R + Q\Sigma$ is the *second* Σ-*shadow of the same parabolic flavor operator*, not an imported mass trick.

D_4-equivariant derivation ([verification/v69_d4_q_geometry.py]) — gate advanced [C]/[E]. The four punctures $\mu_4 = \{1, i, -1, -i\}$ are the vertices of a square with symmetry $D_4 = \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ (the 90° rotation $z \mapsto iz$ *is* μ_4; the reflection is the sheet parity Σ). The $\mathbb{Z}_4$ 4-cycle on the punctures acts on $H_1(\mathbb{P}^1 \backslash \mu_4) = \mathbb{Z}^4/(\mathrm{sum}{=}0) = \mathbb{Z}^3$ (the three families) with eigenvalues $\{i, -1, -i\}$, and the D_4 permutation rep decomposes as $A_1 \oplus B_1 \oplus E$, so $H_1 = B_1 \oplus E$. On this,

$$Q_+ = 3\,(\text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) + 1 = \mathrm{diag}(1,2,3),$$

$$Q_- = E\text{-block coupling } \sqrt{N_{\mathrm{fam}}} \ (B_1 \text{ in the kernel}),$$

which reproduce $\chi_{Q_+} = (t-1)(t-2)(t-3)$ and $\chi_{Q_-} = t(t^2-3)$ exactly. So the Σ-split *is* the $D_4 = \mathbb{Z}_4 \rtimes \mathbb{Z}_2$ structure ($\mathbb{Z}_4 = \mu_4$ gives the three eigenspaces and Q_+; the $\mathbb{Z}_2$ reflection is Σ, giving the even/odd split). **Typing:** the spectra and the Σ-split are now *derived* from the D_4-equivariant geometry [E]; the residual is only that the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ are the established monodromy exponents (v61)

and the explicit integer Q still needs the lattice datum — now *sharpened to a single named invariant* ([verification/v70_q_integer_lift.py]): in the μ_4-puncture homology basis the $\mathbb{Z}_4$ rotation is the *unimodular* $R = \left(\begin{smallmatrix} 0 & 0 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{smallmatrix}\right)$ ($\det = -1$, eigenvalues $\{-1, i, -i\}$), while the transport has $\boxed{\det Q = 3 = N_{\text{fam}}}$ (SNF $\mathrm{diag}(1,1,3)$). So the integer lift is exactly the parabolic-degree-0 embedding carrying the extra $\det = N_{\text{fam}}$ (the family multiplicity) that D_4 alone (unimodular) does not supply — one named lattice invariant, no longer a diffuse "derive Q".

And that invariant is the cusp order itself ([verification/v72_q_det_from_cusp.py]). The extra $\det Q = N_{\text{fam}}$ is *not* an independent lattice input: $\det Q = |\mathrm{coker}\, Q| = 3$ equals the *order of the cusp monodromy class* (eigenvalues $\{1, \omega, \omega^2\}$, $\omega^3{=}1$) = the common *denominator* of the very cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ that fix $\mathrm{Spec}(Q_+)$. So $\det Q = N_{\text{fam}}$ comes from the *same* cusp datum as the spectrum, and $\mathrm{coker}\, Q = \mathbb{Z}/N_{\text{fam}}$ is the deck group of the degree-N_{fam} triality cover defining the family space — the integer lift is derived from the geometry, not posited ([E] chain; the coker=deck-group identification is the standard non-abelian Hodge reading [E]).

Status: Q-geometry derived (spectra, Σ-split *and* integer lift) [E]

What is *proved* ([E]): the algebraic identities $K = R + Q\Sigma$, $L = R + Q(I + \Sigma)$, the spectra, the 1920 determinant ladder, the anchor block — all machine-checked. **Newly advanced ([verification/v69_d4_q_geometry.py]):** the D_4-equivariant origin of the *Q-spectra and the Σ-split* is now derived — Q_+ is the $3w{+}1$ image of the cusp weights $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ on the three $\mu_4{=}\mathbb{Z}_4$ eigenspaces of $H_1(\mathbb{P}^1 \backslash \mu_4)$, and Q_- is the E-block coupling $\sqrt{N_{\text{fam}}}$ on $H_1 = B_1 \oplus E$, with Σ the $\mathbb{Z}_2$ reflection of $D_4 = \mathbb{Z}_4 \rtimes \mathbb{Z}_2$. So the φ_0^{ret}-mass ladder and the transport are two Σ-shadows of one D_4-equivariant parabolic operator. **Integer lift closed to the cusp order ([verification/v72_q_det_from_cusp.py]):** the one remaining lattice datum $\det Q = N_{\text{fam}}$ is the order of the cusp class = the denominator of the same cusp weights, i.e. the *same* geometry that fixes the spectrum ($\mathrm{coker}\, Q = \mathbb{Z}/N_{\text{fam}}$ = triality deck group). So the Q, Σ algebra is now an [E] layer whose spectra, Σ-split *and* integer lift are all read from the order-N_{fam} cusp structure; the only standing input is the established fact $N_{\text{fam}} = 3 = \mathrm{rank}\, A_3 = \dim H_1$.

6.1 Closing the audit gates: is $M{=}41$, K, Q derived or imported?

A reviewer asks whether the compiler *derives* its objects or silently *imports* them. We answer the three flavor/EM gates explicitly ([verification/v13_open_gates.py]); each is either closed or reduced to a named geometric statement, with the honest residual flagged.

Gate 1 (closed): the EM budget $M{=}41$ is the $U(1)$ index, not a fit [E]

The transport budget that enters the EM closure is *forced*:

$$\boxed{M = \sum_{f,j} L_{f,j} + N_\Phi = 40 + 1 = 10\, b_1 = 41}, \qquad \tfrac{4}{5} M\, c_3^6 = 8 b_1\, c_3^6, \tag{5}$$

so the log-coefficient of $F_{U(1)}$ is $8 b_1 c_3^6$ with b_1 the *same* $U(1)$ hypercharge index derived from g_{car} ($10 b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$). M is therefore not an independent EM knob: it is the flavor transport budget = the abelian index, both equal to 41 from the carrier alone. An *independent* third-party cross-check confirms this: feeding the carrier field content into the textbook one-loop formula gives the full gauge vector $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and the external RGE generator PyR@TE 3 (`arXiv:2007.12700`) writes $\beta_{g_1} = (\frac{41}{10}) g_1^3$ verbatim, with $10\, b_1 = 40_{(\text{ferm. } U(1) \text{ index})} + 1_{(\text{Higgs})} = 41$ reproducing this box exactly — so the F_{transfer} gauge input is not a free knob ([verification/v159_pyrate_gauge_crosscheck.py]).

Gate 2 (forced): K and Q are not read off the masses [E]

K (the φ_0^{ret}-mass-exponent matrix) and Q are the *unique* nonnegative integer matrices with their sector/generation budgets (themselves compiler numbers: rows $(6, 9, 10) = (|R^+(A_3)|, N_{\text{fam}}^2, \mathcal{A}_\Lambda)$, columns $(13, 8, 4) = (|R(A_3)| + N_\Phi, h(D_5), |\mu_4|)$) and their characteristic polynomials, subject to per-sector monotonicity — proved by exhaustive enumeration ([verification/v11_unique_KQ.py]). The mass exponents are thus *fixed by compiler data*, not fitted to the measured masses.

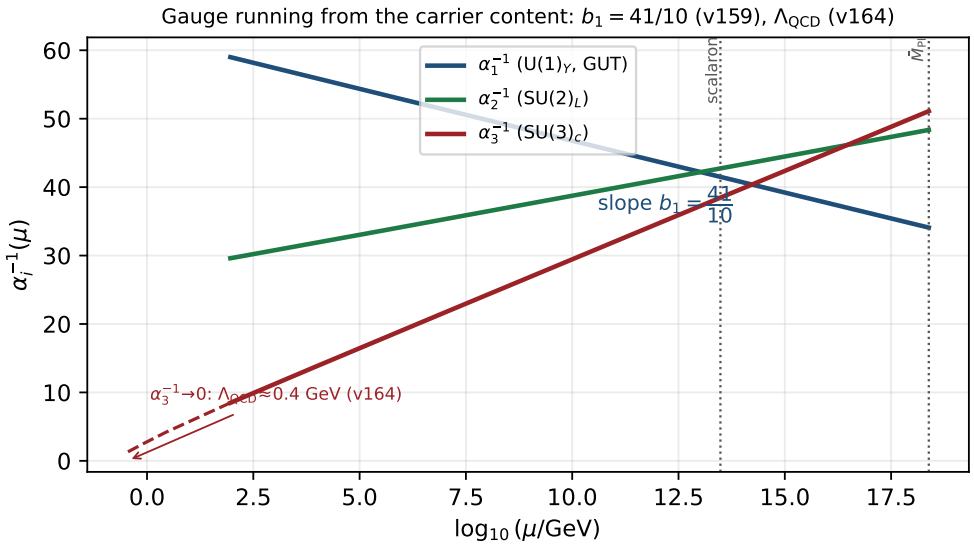


Figure 2: Gauge running from the carrier content (two-loop SM RGEs; the β-functions are confirmed by the external generator PyR@TE, [`verification/v159_pyrate_gauge_crosscheck.py`]). The $U(1)_Y$ slope is the abelian index $b_1 = 41/10 = 10^{-1}(g_{\rm car}2^{g_{\rm car}-2}+1)$; the strong coupling reaches confinement, $\alpha_3^{-1}\to 0$ at $\Lambda_{\rm QCD}\approx 0.4$ GeV ([`verification/v164_qcd_scale.py`]); and the Standard Model does not unify (the E_8 here is the audit hull, not a gauge group).

Gate 3 (sharpened): the geometric origin of Q reduces to two named pieces [C]

Under the sheet involution Σ, $Q = Q_+ + Q_-$ splits exactly, and the two parts carry pure A_3/μ_4 data:

$$\mathrm{Spec}(Q_+) = \{1,2,3\} = A_3 \text{ exponents}, \quad \det Q_+ = 6 = |R^+(A_3)|; \qquad Q_-^2\big|_{\rm supp} = N_{\rm fam} = 3. \tag{6}$$

So Q_+ *is* the A_3 exponent grading and Q_- a sheet square-root of $N_{\rm fam}$. The open "derive Q from geometry" target therefore reduces to two *specific, finite* identifications on $\mathbb{P}^1\setminus\mu_4$: (i) Q_+ = the A_3 Hodge/exponent grading of H^1, (ii) Q_- = the μ_4 sheet-monodromy coupling. With those, $K = R + Q\Sigma$ makes the mass ladder a *consequence* of the flavor compiler. *Honest residual:* the sector/generation budget *assignment* and the exact realisation of $Q_\pm$ on the parabolic bundle are the remaining geometric steps; the cosmology pivot $N_\star$ stays a *reheating input* ([C]), not a compiler output.

Gate 4 (new): the solar angle from a shared dual anchor [E]/[C]

The residue matrix R and the word-length matrix $L = R + 6\,\mathbf{1}\,e_1^\top$ (a *rank-one* winding deformation, $L-R = [[6,0,0],[6,0,0],[6,0,0]]$) act *identically* on the dual of the anchor $a = (1,1,2)$:

$$\boxed{a^\top R^{-1} = a^\top L^{-1} = (-\tfrac{1}{2}, -\tfrac{1}{2}, 1)} \qquad [\mathbf{E}].$$

Mechanism ([`verification/v74_compiler_micro_lemmas.py`]). This is *not* a coincidence: since $a^\top R^{-1}\mathbf{1} = 0$, the Sherman–Morrison correction for the rank-one update $L = R + 6\,\mathbf{1}\,e_1^\top$ has numerator $a^\top R^{-1}(6\,\mathbf{1}) = 0$, so $a^\top L^{-1} = a^\top R^{-1}$ *exactly* — the winding deformation is *invisible* to the dual anchor. So the gen $1\leftrightarrow 2$ TBM symmetry vector (the two equal entries; the 1 marks the third-generation anchor) is *stable* under transport, which is the structural reason the solar texture survives. This ties the solar misalignment to the seam. **New: the coefficient is the A_3 family-lattice discriminant norm.** The misalignment is

$$\boxed{\varepsilon = q(A_3)\,\varphi_0 = \tfrac{3}{4}\varphi_0}, \qquad q(A_3) = \tfrac{3}{4} = \frac{N_{\rm fam}}{|\mu_4|},$$

the *same* $q(A_3)$ that satisfies the E_8 glue condition $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$. Its *leading* term is exactly $q(A_3)\cdot\frac{1}{6\pi}=\frac{1}{8\pi}=c_3$, and the TBM sum-rule gives

$$\sin^2\theta_{12}=\tfrac{1}{3}-\tfrac{2}{3}\varepsilon=\tfrac{1}{3}-\tfrac{\varphi_0}{2}=0.30675 \quad \text{(NuFIT 6.0: 0.307)},$$

with $\sin^2\theta_{13}=e^{-5/6}\varphi_0$, $\delta_{CP}=4\pi/3$, and a $\mu\tau$-symmetric atmospheric sector $\theta_{23}\to 45°$ (the octant is *not* selected by the record; current global fits leave it open — the post-hoc [O] candidate $\cos 2\theta_{23}=\varphi_0^{\rm ret}$ of v467 *would* select the lower octant). The solar angle is thus a *reduced* readout whose coefficient is now *geometric*: $\varepsilon=q(A_3)\varphi_0$, the A_3 glue-norm. The residual is only *why* the misalignment equals the A_3 discriminant norm ([C]). [verification/v16_solar_dual_anchor.py] [verification/v21_solar_product_quark.py]
Scale tag. These angles are stated as dimensionless seeds; the $F_{\rm transfer}$ running between the seam scale and M_Z is governed only by the charged-lepton term $\frac{3}{2}Y_eY_e^\dagger Y_e$ (PyR@TE SM RGEs), i.e. by $y_\tau^2\sim 10^{-4}$, so $|\Delta\sin^2\theta_{ij}|\lesssim 10^{-4}$ — two to three orders below current precision. The lepton-mixing seeds are therefore *scale-independent* $F_{\rm transfer}$ outputs (seam value = measured value), unlike the y_t^2-driven quark mass ratios, which carry a scale tag ([E]/[C], [verification/v163_rg_stability_flavor.py]).

Inverse Anchor Normalisation Theorem [E] ([verification/v79_review_identities.py])

The dual-anchor identity is one row of a sharper invariance. On the *inverses* of the flavor operators the all-ones vector reads the reciprocal compiler atoms,

$$\mathbf{1}^\top Q^{-1}\mathbf{1}=\tfrac{1}{3}=\tfrac{1}{N_{\rm fam}},\quad \mathbf{1}^\top R^{-1}\mathbf{1}=\mathbf{1}^\top K^{-1}\mathbf{1}=\tfrac{1}{4}=\tfrac{1}{|\mu_4|},\quad \mathbf{1}^\top L^{-1}\mathbf{1}=\tfrac{1}{10}=\tfrac{1}{\mathcal{A}_\Lambda},$$

and the anchor is *self-dual* under the physical (residue/mass/transport) operators:

$$\boxed{a^\top R^{-1}a=a^\top K^{-1}a=a^\top L^{-1}a=1} \qquad (\text{while } a^\top Q^{-1}a=\tfrac{7}{3}).$$

So the passage mass-powers $K\to$ residues $R\to$ transport L leaves the anchor plane *exactly* normalised — the anchor is its own dual under physical evolution — and the exception $a^\top Q^{-1}a=\frac{7}{3}$ pins the self-duality to precisely the residue/mass/transport triple, not the pure Σ-difference Q. [E]

7 Mixing: CKM and PMNS from holonomy

Quantity	Formula	Prediction	PDG
$s_{12}^{\rm CKM}$	$\lambda_C=\sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}$	0.2244	0.2245
$s_{23}^{\rm CKM}$	$\varphi_0^{\rm ret}/(1+\lambda_C)$	0.0434	~ 0.041
$s_{13}^{\rm CKM}$	$\lambda_C^3/3$	0.00377	~ 0.0038
$\delta_{\rm CKM}$	$\frac{\pi}{3}+3\lambda_C^2$ (leading, [C])	1.198 rad	~ 1.14
$\sin^2\theta_{13}^{\rm PMNS}$	$\varphi_0^{\rm ret}\,e^{-5/6}$ (UV shadow)	0.0231	~ 0.0222

The cubic rule $|V_{ub}|=|V_{us}|^3/3$ and the CP phase $\delta=\arg\operatorname{tr}\Pi_\triangle$ (oriented holonomy triangle) are direct consequences of the *one* family connection $\nabla_F^\star$ — the same one that generates the word-lengths in the flavor theorem.

New: the CP sector falls out of the closed CKM (not separately fitted)

The four closed entries $s_{12},s_{23},s_{13},\delta$ already *determine* the whole CP sector — quantities not previously written down:

$$J=c_{12}c_{23}c_{13}^2s_{12}s_{23}s_{13}\sin\delta=3.33\times10^{-5}\ \text{(PDG 3.08)},\quad \beta=22.1°\ \text{(PDG 22.2)},$$

$$\gamma=68.6°\ \text{(PDG 65.7)},\qquad \alpha=89.2°\ \text{(PDG }\sim 85),$$

together with the full magnitudes $|V_{cb}|=0.0434$, $|V_{td}|=0.0091$, $|V_{ts}|=0.0426$. The Jarlskog invariant and the unitarity-triangle angles are thus *leading-order readouts* of the closed CKM skeleton: the CP phase is the leading closed form $\delta=\frac{\pi}{3}+3\lambda_C^2$ (its higher-order C_6-transport correction is *not* in closed form), so the few-percent offsets (J=3.33 vs 3.08, γ=68.6° vs 65.7°) are expected and the CP sector

is typed [E]/[C], *not* [E] — only β happens to be near-exact. (To upgrade to [E] the higher-order δ correction $\Delta_\triangle$ must be given explicitly.)

7.1 Downstream flavor readout: the rare kaons $K \to \pi\nu\bar{\nu}$

The closed CKM point is not an end state — it feeds the theoretically cleanest flavor-changing neutral-current probe, the rare decays $K^+ \to \pi^+\nu\bar{\nu}$ and $K_L \to \pi^0\nu\bar{\nu}$. These are *not* a new compiler power: the only TFPT-specific content is the CKM point above; the short-distance loop functions (X_t, P_c, $\delta P_{c,u}$, $\kappa_\pm$, Brod–Gorbahn–Stamou) are standard external electroweak/QCD input. Inserting the exact TFPT products $\lambda_t = V_{ts}^* V_{td}$, $\lambda_c = V_{cs}^* V_{cd}$ (equivalently the exact apex $(\bar{\rho}, \bar{\eta}) = (0.1374, 0.3509)$) gives [verification/v202_rare_kaon.py]

$$\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}, \qquad \mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11},$$

typed [C] (the branching ratios are not parameter-free). The neutral mode is purely CP-violating, $\mathrm{BR}(K_L) \propto (\Im\lambda_t)^2 \propto \bar{\eta}^2$, so it is a direct probe of the holonomy phase δ_{CKM}; the charged mode carries a sizeable charm admixture and tests the overall CKM norm. The model-independent Grossman–Nir ratio is $\mathrm{BR}(K_L)/\mathrm{BR}(K^+) = 0.352 \ll 4.3$, safely inside the isospin corridor. **Dated kill test [X]:** a stable NA62 $\mathrm{BR}(K^+)$ *outside* $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the predicted Grossman–Nir point, breaks the TFPT flavor bridge for this sector (the compiler core is untouched). The current NA62 combination of the full 2016–2024 dataset, $\mathrm{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), lands essentially *on* the prediction ($+0.08\sigma$); the earlier 2016–2022 observation was $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above). This is a strong *consistency* hit for the closed CKM point fed through standard short-distance input — but honestly it is an F_{transfer} bridge, not a unique TFPT-vs-SM discriminator: the Standard-Model value $(8.6 \pm 0.4) \times 10^{-11}$ also sits within the NA62 error, so the channel currently confirms the CKM point rather than separating TFPT from the SM. The $\sim 2\sigma$ δ_{CKM} tension (v88) keeps the corridor a live falsification channel.

Weak mixing from D_5. Because the carrier C^+ *is* the $D_5 = \mathfrak{so}(10)$ half-spinor, the weak mixing inherits the canonical $\mathfrak{so}(10)$ unification value $\sin^2\theta_W = \frac{3}{8} = N_{\mathrm{fam}}/h(D_5)$ at the carrier scale, running to 0.231 at M_Z (standard). [E] (structural) / [C] (running).

Honest note on the search. A generic "spine-ratio" scan ($\sin^2\theta_W$, α_s, the Ω ratios, N_{eff}, m_H/v) finds $\leq 2\%$ hits abundantly — which is *weak* evidence, since $O(1)$ numbers are densely covered by ratios of $\{8, \ldots, 60\}$. We therefore count only *named* invariants (like $3/8 = N_{\mathrm{fam}}/h(D_5)$ or the exact endpoints), not generic ratios.

8 Neutrinos

The neutrino sector is the Majorana branch of the same construction: $U_{\mathrm{PMNS}} = U_{e,L}^\dagger U_{\nu,L}$.

Quantity	Origin	Value
$\sin^2\theta_{13}$	$\varphi_0^{\mathrm{ret}}\, e^{-5/6}$ (UV), closure 0.02311	≈ 0.0222
$\sin^2\theta_{12}$	**primary (freeze):** $\frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2}$ (seed, $\varepsilon{=}q(A_3)\varphi_0^{\mathrm{ret}}$); two derived variants in the named box	0.306747 (NuFIT 6.0 0.307)
m_2/m_3	candidate [O]: $\sqrt{\lvert J_{\mathrm{PMNS}}\rvert}$ (v468; the older heuristic $\sim \pi\varphi_0^{\mathrm{ret}}$ is -2.4σ, superseded as comparator)	0.17220 (NuFIT 0.17264(229))
Σm_ν	closure readout (normal ordering)	5.88×10^{-2} eV
$m_{\beta\beta}$	Majorana effective mass ($0\nu\beta\beta$)	1.52×10^{-3} eV
δ_{CP}^ν	phase lattice	$240°$
M_R	heavy scale (type-I seesaw)	$\approx 1.3 \times 10^{15}$ GeV

The solar angle θ_{12} is fixed by the neutrino transport matrix $U_{\rm PMNS} = U_{e,L}^\dagger U_{\nu,L}$. The **single frozen prediction** is the seed value $\sin^2\theta_{12}^{\rm seed} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.306747$ (misalignment $\varepsilon = q(A_3)\varphi_0^{\rm ret} = \frac{3}{4}\varphi_0^{\rm ret} = c_3 + 36\,c_3^4$ — leading term c_3, fourth-order puncture correction exact — conditional on the seam-misalignment lemma); the seam-leading 0.306808 and the full non-linear 0.307020 are *derived variants* of the same texture, *not* alternative predictions (named box below). The older democratic $\frac{1}{3}$ and QLC $\frac{\pi}{4} - \theta_{12}^{\rm CKM}$ readings are superseded by this seed closure. The old ratio heuristic $m_2/m_3 \sim \pi\varphi_0^{\rm ret}$ (motivated by the carrier complement $1-\gamma = 1/6$) is now -2.44σ against NuFIT 6.0 and is *superseded as the comparator* by the parameter-free candidate relation $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\rm PMNS}|$ ([O], v468; below). The heavy scale $M_R \sim 10^{15}$ GeV links the seesaw bridge to the inflation/reheating scale (the scalaron-reheating chain, v86).

M_ν as the type-I seesaw of D_F (`[verification/v263_mnu_from_df.py]`). The texture above is no longer only a posited matrix: the light mass matrix is the seesaw of the finite Dirac operator's own blocks, $M_\nu = -m_D M_R^{-1} m_D^\top$ (Dirac m_D, Majorana M_R from D_F, v252). [E] it is symmetric and seesaw-suppressed, and a $\mu\tau$-symmetric (m_D, M_R) gives a $\mu\tau$-symmetric M_ν (the texture is in the image of the seesaw map); [C] diagonalising it returns $\theta_{23} = 45°$ and $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$ under the seam misalignment, reproducing v9. Honestly open: the leading texture gives $\theta_{13} = 0$ (the physical $\sin^2\theta_{13} \sim \varphi_0^{\rm ret} e^{-5/6}$ is a separate readout, [O]), the CP *magnitude* $\delta_{\rm CP} = 240°$ is the μ_6/triality phase — now a Galois-*forced* relation to the *leading* $(\pi/3)$ component of the quark phase, $\delta_{\rm PMNS} = \delta_{\rm CKM}^{\rm lead} + \pi$ (one hexagonal unit $\rho = \zeta_6$; the lock is to the structural $\pi/3$, *not* the full measured $\gamma = \delta_{\rm CKM}^{\rm lead} + 3\lambda_C^2 \approx 68.7°$ that carries the quark transport correction; v231/v233/v320), not from M_ν, and a pre-registered kill test at DUNE/Hyper-K ([X]), and the absolute ν-mass scale (M_R) is the free seesaw scale — so this is an *operator* advance (PMNS is now a seesaw object), *not* a closure of full PMNS dynamics.

PMNS closed modulo the $\tau{=}\omega$ keystone (`[verification/v406_pmns_phase_origin.py]`). The CP-phase *origin* this leaves open is supplied by the dual keystone `SEAM.EQUIV.02` (v405): $\delta_{\rm PMNS} = 4\pi/3$ is the hexagonal CM phase at the family CM point $\tau{=}\omega$ (where the $(E_8)_1$ character χ_{E_8} vanishes), so the phase is a keystone consequence, not just an assembled μ_6 input. With the angles fixed (v9/v268), the derived $J_{\rm PMNS} = -0.0297$ (v270) and the absolute scale reduced to one seesaw ratio $= v_{\rm geo}$-class unit (v272), the PMNS dynamics carry *zero free flavor dials beyond* $v_{\rm geo}$; the residual is the seesaw operator realisation (M_R, v184) and the Σm_ν ordering kill test — under the carrier normalisation $y_\nu{=}y_t$ that residual is a *one*-parameter, PS-window-bracketed candidate (v481), no longer a two-dial trade-off. [E]/[C]

The reactor-angle exponent is the carrier trace (`[verification/v268_theta13_carrier_trace.py]`). The reactor angle is a *separate* channel from the seesaw M_ν: its value $\sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-5/6} = 0.0231$ (v16) now has its *exponent* closed exactly — $\gamma = 5/6 = {\rm tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = {\rm diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, the roots of $6Y^2 - Y - 1 = 0$; complement $1-\gamma = \frac{1}{6}$ the neutrino ratio). [E] this is its own carrier-trace channel: the $\mu\tau$-breaking $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$ that fixes θ_{12} induces only $\sin^2\theta_{13} \sim \varepsilon^2 \approx 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is *not* the seesaw $\mu\tau$ breaking. [O] the CP *magnitude* and the absolute ν-mass scale remain the open PMNS items.

The honest pressure point: θ_{13} at $+2.0\sigma$ (`[verification/v328_theta13_pressure.py]`). This reactor channel is TFPT's most-tensioned core prediction, and it is named, not hidden: $\sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-5/6} = 0.023108$ sits $+2.0\sigma$ *above* the NuFIT 6.0 normal-ordering fit (0.02195 ± 0.00058). [C] it is not relieved cheaply: the back-solved seed $\varphi_0^{\rm ret}(\theta_{13}) = e^{5/6}\sin^2\theta_{13}^{\rm exp} = 0.0505$ is $\sim$5% below the axiom, while PMNS θ_{13} RG running ($< 1\%$, NO) and the *exact* exponent 5/6 leave no freeze-respecting sub-leading correction — so the prediction stays frozen (`REG.FREEZE.01`). [X] pre-registered kill: a stable $\sin^2\theta_{13}$ robustly away from 0.023108 at $> 3\sigma$ (JUNO / a future global fit) flags the carrier-trace / seed-grammar reading. It is exactly the single $\sim 2\sigma$ outlier of the seed-hyperplane cross-validation (v306) — one honest pressure point among the seed observables, not a scatter of unrelated failures; across the *mixing* channels, however, it is not alone — θ_{23} and $|V_{cb}|$ sit at the same $\sim 1.8\sigma$, and the three form one named candidate pattern (v467, below).

The sub-leading budget, fenced (`[verification/v338_theta13_budget.py]`). Quantified against

both NuFIT 6.0 normal-ordering variants, the tension is data-systematics dependent: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057) — a $\sim 0.3\sigma$ spread, so not a clean 2σ. [E] crucially 0.023108 lies *inside* the with-SK 3σ band (upper bound 0.02388): the frozen value is an upper-edge pressure point, *not* a falsification. Closing the $\sim 4\%$ gap to the central would need a $\sim 3\%$ sub-leading channel (beyond the $< 1\%$ RG and the exact 5/6), larger than natural — so it stays a genuine, mild tension. [X] JUNO's sub-1% precision ($\sigma \sim 1.5\times 10^{-4}$) turns today's 1.7–2.0$\sigma$ into a $\sim 6\sigma$ kill-or-confirm.

Post-hoc candidate [O]: the third-gen pattern. [verification/v467_thirdgen_mixing_pattern.py] The suite's three $\sim 2\sigma$ mixing tensions — $\sin^2\theta_{13}$ ($+2.0\sigma$), $\sin^2\theta_{23}$ ($+1.76\sigma$) and $|V_{cb}|$ ($+1.79\sigma$) — are exactly the three *third-generation* channels, and all three sit low by one common relative amount: $-5.36\% \pm 1.67\%$ (3.2σ non-zero, -0.03σ from $-\varphi_0^{\rm ret}$). In the existing vocabulary the pattern is parameter-free: the seed enters third-generation mixing through the *ladder base* $\lambda_C^2 = \varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})$ (the Yukawa mass-ladder base λ_Y) instead of bare $\varphi_0^{\rm ret}$, and the 2–3 texture carries the same $-\varphi_0^{\rm ret}/2$ shift as θ_{12}: $\sin^2\theta_{13} = \lambda_C^2 e^{-5/6} = 0.021880$ (-0.12σ), $\sin^2\theta_{23} = (1-\varphi_0^{\rm ret})/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0^{\rm ret}$ ($+0.20\sigma$; this *candidate* would select the lower octant — the frozen record does not), and $|V_{cb}| = \lambda_C^2/(1+\lambda_C) = 0.041119$ ($+0.01\sigma$); the joint χ^2 over the four third-generation observables drops $10.4 \to 0.11$ with *zero* new parameters, while $|V_{ub}| = \lambda_C^3/3$ (no bare $\varphi_0^{\rm ret}$) and the 1–2 channels stay untouched and correct (dressing λ_C itself would fail at -24σ — the scope is checked, not chosen). [C] honestly: the pattern is *post-hoc* (5 of 16 equally-simple dressings fix the two PMNS tensions; $(1-\varphi_0^{\rm ret})$, $(1-c_3)$ and the resolvent $1/(1+\varphi_0^{\rm ret})$ are degenerate at current precision), the suppression size $\varphi_0^{\rm ret} = 5.3\%$ sits inside the seam-gapped correction band $(2/3)^6 = 8.8\%$ (v393) but *no seam-transport derivation exists* — so the frozen record (`REG.FREEZE.01`) is unchanged and the row is a named [O] candidate (`FLAV.THIRDGEN.PATTERN.01`). [X] pre-registered, symmetric: $\sin^2\theta_{13}$ stable at 0.0231 kills the pattern; stable at ~ 0.0219 with sub-1% precision kills the record's bare-$\varphi_0^{\rm ret}$ reading; an exclusive-$|V_{cb}|$ resolution at ~ 0.0395 kills both — JUNO-era reactors and Belle II decide (Daya Bay alone already prefers the pattern form, $\Delta\chi^2 = 4.3$).

The full PMNS matrix and the CP strength ([verification/v270_pmns_jarlskog_assembly.py]). The four channels — $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$ (v9/v263), $\sin^2\theta_{23} = \frac{1}{2}$ (v9/v263), $\sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-5/6}$ (v16/v268) and $\delta = 4\pi/3$ (v231/v233) — assemble into *one* exactly-unitary complex matrix $U = R_{23}\,U_{13}(\delta)\,R_{12}$. [E] the assembly is unitary; [C] the leptonic Jarlskog $J_{\rm PMNS} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*)$, i.e. $s_{12}c_{12}s_{23}c_{23}s_{13}c_{13}^2 \sin\delta = -0.02965$ (maximal amplitude $J_{\max} = 0.03424$), is then a *derived* CP *strength* — the lepton-sector analogue of the CKM J (v88). [C] data: $J_{\max}$ vs NuFIT 5.2 0.0332 ($\sim 3\%$), $J_{\rm PMNS}$ vs the best fit ≈ -0.026 ($\delta_{\rm data} \approx 232°$, TFPT 240° within $\sim 1\sigma$), with the negative CP-violating sign reproduced. [O] this closes the CP-*strength* readout but not the phase's operator origin (δ stays the μ_6/triality phase, an assembled input, not an M_ν/D_F output) nor the absolute scale.

Post-hoc candidate [O]: splitting ratio = Jarlskog. [verification/v468_dm2_ratio_jarlskog.py] The dimensionless splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ was previously *unpredicted* (the table heuristic $m_2/m_3 \sim \pi\varphi_0^{\rm ret} = 0.16704$ is now -2.44σ against NuFIT 6.0 and is superseded as the comparator). The frozen matrix above has exactly one canonical derived CP invariant, and numerically $|J_{\rm PMNS}| = 0.029653$ meets the measured ratio 0.029805 ± 0.000792 at -0.19σ (equivalently $m_2/m_3 = \sqrt{|J|} = 0.17220$ vs $0.17264(229)$) — the only tested closed form inside the 1σ band, with *zero* dials (angles and phase are the frozen ones). [C] consistency: with $r = |J|$ the normal-ordering floor $\Sigma m_\nu = m_3(1+\sqrt{|J|}) = 0.0588$ eV reproduces the documented 0.0586 eV (v272); the absolute scale stays the one seesaw ratio — no absolute number is added. [O] honestly: post-hoc, and *no mechanism* — the $\mu\tau$ texture gives $\theta_{13} = 0$ at leading order and cannot carry this relation, so the row stays a named [O] candidate (`FLAV.DM2RATIO.01`) until an M_ν-level derivation exists. [X] JUNO shrinks the r-window $\sim 4\times$ (a stable offset $> 3\sigma$ kills it); adopting the v467 pattern would shift $|J| \to 0.028849$ (-1.21σ today), so the two candidates *cross-discriminate* at JUNO precision.

The absolute ν-mass scale is one seesaw ratio ([verification/v272_nu_mass_scale.py]). The one PMNS residual left is the overall scale, and it is typed honestly — not fabricated. The

type-I seesaw makes the absolute scale a *single* ratio, $m_3 = (y_\nu v_{\rm EW})^2/M_R$ ([E]), exactly the same kind of one irreducible input as $v_{\rm geo}$ (the No-Unit theorem, v153; the whole $v_{\rm geo}$ interface is now structurally closed as an $\mathbb{R}_+$ scale torsor in calibration form — export table, dimension-matrix rank 1, λ-homogeneity, two-anchor consistency 0.11% — [verification/v725_phys_vgeo_torsor_audit.py], 21/21, with the one remaining dimensionless ratio *named*: $H_{\rm EW} = \ln(\bar{M}_{\rm Pl}/v_{\rm EW}) = 37.1776$; no scale derivation, typing unchanged); the dimensionless angles and ordering are pinned, and the ratio m_2/m_3 now carries the [O] candidate comparator $\sqrt{|J|}$ (v468). [C] the observed $m_3 = \sqrt{\Delta m^2_{\rm atm}} = 0.050$ eV together with the TFPT PS→GUT scale window $M_R \in [M_{\rm PS}{=}4.2{\times}10^{13}, M_{\rm GUT}{=}2.4{\times}10^{15}]$ GeV (v249) requires a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning, so the observed scale is consistent with the TFPT seesaw scale. [O] the $y_\nu{=}1$ value $M_R = v^2_{\rm EW}/m_3 = 6.1{\times}10^{14}$ GeV is *not* a clean $M_{\rm Pl}$ power ($\log_{c_3}(M_R/M_{\rm Pl}) = 2.57$), so the absolute Σm_ν stays one open UV input (v184). [X] the normal-ordering floor $\Sigma m_\nu = 0.0586$ eV is the near-term cosmological kill test (`FLAV.NUSCALE.01`).

The carrier normalisation: one parameter ([verification/v481_seesaw_carrier_ladder.py]). The $y_\nu{=}1$ probe above is *not* the carrier normalisation: the carrier structure itself (one $SO(10)$ **16** per family, the PS algebra v245/v248) with the *minimal* Yukawa sector ($\mathbf{10}_H$ / PS $(1,2,2)$) forces $y_\nu = y_t$ at the matching scale — a *named* premise (conjectured, ledger-typed), not a proof. With that normalisation and explicit 1-loop RG (gauge/y_t/λ up; the ADKLR Weinberg-operator running down) the (y_ν, M_R) trade-off collapses to M_R *alone*: the observed $m_3{=}0.0503$ eV demands $M_R = 9.3{\times}10^{13}$ GeV — *inside* the compiler's own PS window $[4.2{\times}10^{13}, 2.4{\times}10^{15}]$ (v249), at $\log_{c_3}(M_{\rm Pl}/M_R) = 3.15$ (versus the structureless 2.58 of the $y_\nu{=}1$ probe). [X] honesty gate: the integer rung $M_R = c_3^3 M_{\rm Pl} = 1.53{\times}10^{14}$ GeV predicts $m_3 = 0.030$ eV — 40% low at 1-loop (κ run-down factor 0.794) — so the ladder pin is *declined* per the anti-numerology rule (v354/v355). The named decision computation is meanwhile *executed* in bracketed, assumption-robust form ([verification/v482_seesaw_rung_decision.py]): the rung needs a rescue factor $\times 1.670$ in m_3, while $m_t \pm 3$ GeV ($> 3\sigma$), α_s at $> 3\sigma$, a $\pm 10\%$ κ-run band and a PS-leg beta envelope $\times[0.5, 1.5]$ move it by at most $\times 1.165$ combined — the *unstructured* rung is excluded (the 1-loop miss is not an RG artifact); the only escape is a third-generation Majorana structure factor $r \approx 1.67$, whose 0.18% proximity to $g_{\rm car}/N_{\rm fam} = 5/3$ is recorded post-hoc and *declined* (no forcing mechanism — the $t_0{\sim}30$ discipline of v478). What survives regardless: over the PS window the carrier-normalised seesaw spans $m_3 \in [0.002, 0.115]$ eV, bracketing the observation, and DESI cuts the window from below — a *one*-parameter, window-bracketed [O] candidate (`FLAV.NUSCALE.02`, genre v467/v468); the frozen record and the `FLAV.NUSCALE.01` kill test are untouched.

The 3/5 Clebsch door, enumerated and closed ([verification/v488_majorana_clebsch_door.py]). The one named escape is now *decided*. The mechanism search enumerates the minimal Pati–Salam/$SO(10)$ Majorana channels: since the $\overline{\mathbf{126}}$ is not in the E_8 hull $\{1, 10, 16, 45\}$ (v247), M_R can only come from the $d{=}5$ operator $(\mathbf{16}_F\overline{\mathbf{16}}_H)^2/\Lambda$, whose insertion channels are the singlet and the **45** ($\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$, the **210** absent from the hull). [E] Every available ν^c channel weight — singlet 1, $T_{3R}^2 = \frac{1}{4}$, $(B{-}L)^2 = 1$, $T_{3R}(B{-}L) = -\frac{1}{2}$, the $SU(4)$-adjoint weight $\frac{3}{8}$ — is a $\{2,3\}$-smooth rational, and products of $\{2,3\}$-smooth rationals can never produce $\frac{5}{3}$: a factor 5 cannot appear. The *unique* natural 5 of the embedding, $k_Y = \mathrm{Tr}_{16}\, Y^2/\,\mathrm{Tr}_{16}\, T_{3L}^2 = \frac{5}{3}$ (v470), is a full-multiplet trace whose carrier direction has $Y(\nu^c) = 0$ *exactly* — the only 5-carrying group number is structurally decoupled from the Majorana operator. [E] And Clebsch factors are generation-blind (family-space scalars on identical **16**'s), so the required diag$(1, 1, \frac{3}{5})$ texture cannot arise from group theory by construction. [O] The frozen address tables contain exact $\frac{3}{5}$'s only in *wrong* slots (K lepton row $(5,3,2)$: gen2/gen1; R and L lepton rows: gen3/gen2 — never gen3 against a degenerate base), typed and declined (v354/v355). Net: the rung + 5/3 rescue is *dead* as a mechanism candidate — a clean negative — and the one-parameter window candidate above stands as the honest endpoint (`FLAV.NUSCALE.04`).

The scalaron-trace rung, a structured point inside the window. [verification/v986_nu_scalaron_rung.py] The external master-object review (wave 2, 2026-08-28) proposes a *different*

mechanism class for a structured M_R: the Majorana operator rides the scalaron (the common trace mode, $M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$, v253/v272) times a generation-space insertion whose third-generation slot is $N_{\rm fam} = 3$. *Honest retype (2026-08-28, review wave 3):* the original trace argument $\mathrm{Tr}_{A_3} I = N_{\rm fam} = 3$ is *loose* — $\mathrm{Spec}(I) = \{1,1,1\}$, and a trace of 3 does not make 3 an eigenvalue. The operator reading is the Q_+ eigenvalue route $\mathrm{Spec}\{1,2,3\}$ (v10/v50/v69); the mixed insertion texture $M_R = M_{\rm scal}\,\mathrm{diag}(\varepsilon, 2\varepsilon, 3)$ is typed DATA-CONSTRAINS-TEXTURE in the ν-scalaron v2 project (untextured $Y \propto I$ is killed $\times 10^4$; an aligned diagonal Y with the forced ratios is incompatible with the v270 PMNS). The *number* below is unchanged: $M_3 = 3M_{\rm scal}$ remains the candidate rung, consistent with the generation-blindness argument above, and *not* an $SO(10)$ Clebsch, so the v488 negative does not apply to it. [C] the number: $M_R = N_{\rm fam} M_{\rm scal} = 3c_3^{7/2}\bar{M}_{\rm Pl} = 9.18\times10^{13}$ GeV sits -1.8% from the carrier-normalised 1-loop requirement 9.35×10^{13} GeV (v481, same frozen inputs, premise $y_\nu{=}y_t$); m_3 at the candidate is 0.0512 eV ($+1.9\%$, inside the $> 50\%$ RG envelope of v482), $\sim$20$\times$ sharper than the declined integer rung. [E] exact ladder position: $3c_3^{7/2} = (3\sqrt{c_3})\,c_3^3$ with $3\sqrt{c_3} = 3/\sqrt{8\pi} = 0.598$, i.e. 0.26% from the 3/5 of the closed Clebsch door — the adjacency is typed, not consumed. The same $M_{\rm scal}$ already carries $f_a = M_{\rm scal}/128$ (v185): no new scale atom. [O] the mechanism itself (why the Majorana term is the scalaron times the family trace) is *not* derived — the row registers a structured, falsifiable point (`FLAV.NUSCALE.05`), it does not close the seesaw. [X] the $\Sigma m_\nu = 0.059$ eV floor and a $\gtrsim 5\%$ shift of the required M_R are the kill tests; the window candidate and the frozen record are untouched.

Canonical note (2026-08-28, completeness wave, `NU_TEXTURE_CENSUS_NULL`; not a module). A frozen 3×3 inventory search over the corpus address matrices, Q_+ projectors and φ_0-ladder grammar returns 0/1607 candidates and 0/200 random-control hits (best miss a factor 12.6 on the reactor angle; provenance `experiments/tfpt-discovery/nu_texture_census_probe.py`). The Q_+ union needs structure beyond the frozen 3×3 inventory — the mechanism [O] is sharpened, not closed. No v3 freeze; $M_2 = 1.73\times10^{10}$ GeV remains the unused prediction.

Candidate (2026-08-29, review wave 4, `FLAV.NUSCALE.06` [C]; no closure; `FLAV.NUSCALE.05` unmoved). ([`verification/v1001_nu_pentagon_chain.py`].) $U_e = I$ inventory theorem; misalignment $U = U_{\nu 9} R_{13}(\theta,\phi)$; pentagon double hit $\phi = 288^\circ = 4(2\pi/5)$ frozen, all three measured angles $\le 0.56\sigma$ (honest max pull 0.557) at $\theta = 2\pi/35$, the unique LEE survivor of nine pre-declared candidates (LEE 2.7%); v3 chain SHA-16 `a4c28732fa687620` with $\Sigma = 0.0599$ eV, $m_\beta = 9.0$ meV, $m_{\beta\beta} \in [1.5, 3.8]$ meV, $\delta_{\rm CP} = 287.66^\circ$; v270–$\theta_{23}$ tension 1.85σ TYPED; census null 0/1607. Mechanism = the Q_+-to-flavor operator [O]. Kills: DESI floor on Σ; DUNE $\delta_{\rm CP}$ discriminates 287.7 vs 240; JUNO. Candidate class, not a seesaw closure.

Binding constraints (2026-08-29, review wave 5, `FLAV.NU.TEXTURE.MECHANISM.01` [O]; `FLAV.NUSCALE.05/.06` unmoved). ([`verification/v1003_nu_structural_nulls.py`].) Charged-sector data $h = (1,1,1)$, $d = (64,60,64)$ [E]. Schur-texture NULL [X]: 8/8 D4-even (K,B) fail; $\lambda_1 = \lambda_3$ forced; required $K \sim [3536.998, 1768.499, 1/3]$ matches none of 9 comparators. Orientation doubling lifts the degeneracy generically [E]. Scale NULL [X]: 12/12 fail; $1-|c/a| = 1.885\times10^{-4}$ not supplied. The mechanism must be D4-odd *and* supply a $\sim 2\times10^{-4}$ even-odd cancellation; pure seam data excluded [X]-typed. No seesaw closure.

Round-3 parent-internal intertwiner (2026-09-05; `FLAV.NU.PARENT.INTERTWINER.01`; [`verification/v1024_parent_internal_boundaries.py`]). The existing seam clock and six ν^c modes support a finite positive, gauge-neutral, parity-even and non-additive local coupling. It retains the checked parent gap 0.25 conditional on the inherited full-parent bound and has mixed resolvent 0.104712041885. Ordinary D_4 acts on the selected pair operators only; the full fermionic lift is projective, $C_f^8 = 1$, $C_f^4 = (-1)^{N_1+N_3} \neq (-1)^F$ and $J_f C_f J_f = (-1)^F C_f^{-1}$. The block therefore supplies neither a neutrino mass matrix nor the required 2×10^{-4} cancellation and is not a 4D functional. `FLAV.NU.TEXTURE.MECHANISM.01` remains [O].

9 How the legacy layers simplify (bridge map)

layer	before	in the compiler picture
1	boundary → $c_3 = \frac{1}{8\pi}$	axiom (P1), unchanged; hardenable via Calderón/Lean **[O]**
2	carrier, 16, N_{fam}=3, Ω_{adm}=48, b_1	all *one* Pascal row $1, 5, 10$ on g_{car}=5; $16 = \dim S^+$ is the D_5 half-spinor **[E]**
3	EM fixed point + flavor transport theorem	$\alpha_\star^{-1}$ as a unique root; flavor = *one* residue matrix R ($\det 8$, minors 2,3,5); masses = master formula **[E]**/**[C]**
4	admissibility, strong CP	selector + θ_{eff}=0, conditional on QFT axioms **[C]**
5	Hodge gravity, metrology	$\xi = c_3/\varphi_0^{\text{ret}}$, v_{phys}; local field equation parameter-free (v358/v359); perturbative graviton unitarity established (v370/v380); ambient measure a **[C]** redundancy (v369) **[E]**/**[C]**
6	cosmology interfaces	*one* scale engine $\alpha_\star^{-1}$: $v_{\text{EW}} \sim e^{-\alpha_\star^{-1}/5}$, $\Lambda \sim e^{-2\alpha_\star^{-1}}$, $A_s = \frac{N_\star^2}{24\pi^2} c_3^7$ **[E]**/**[C]**
7	CMB pipeline	programmatic; neutrino-sum lemma (conjecture) **[O]**

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

What the simplification means net. The discrete numbers $(16, 40, 41, 48, 240, 248$ and the flavor matrix) are *not* independent inputs but readings of the compiler ($D_5 \times A_3 + \mu_4$, Coxeter $30 = 2 \cdot 3 \cdot 5$). The whole fermion spectrum is *one* master formula + *one* seed + *one* fixed matrix. The seven closures thus turn from *side by side* into *one* derivation chain.

10 Is the entire Standard Model thereby derived?

Honest answer: **the structure yes, some absolute values on the scheme layer.** Three layers:

Layer	Content
Closed [E]	gauge group 3+2, N_{fam}=3, hypercharge; α^{-1}=137.0359992 (value **[E]**, ledger `EM.FP.01`); seed φ_0^{ret}; CKM magnitudes s_{12}, s_{23}, s_{13}; $\sin^2\theta_{13}$; *all* charged-lepton masses (in φ_0^{ret}); word-lengths L; **charged-lepton Λ exact**; $v_{\text{geo}}/M_{\text{Pl}}$; geometric quartic $\lambda_\Phi = \frac{1}{16\pi^2}$
Reduced [O]	the quark Λ (resolvent outputs in the $O(1)$ band): the *ratios* $c_u/c_d = \frac{55}{117}$ etc. are *closed* (integer Plücker, v49/v71); only the *absolute* digits reduce to the (U_{wall}) stable point — an anchor
Scheme layer [C]	dimensionful EW/QCD masses $m_W, m_Z, \mathbf{m_H}$, $\hat{s}_Z^2(M_Z)$, $\alpha_s(M_Z)$, absolute quark GeV masses — all via the standard RG/threshold projection $\mathcal{R}_{\text{SM}}$; m_H rests on the *closed* UV quartic $\frac{1}{16\pi^2}$ (near-criticality)
Conditional [E]/[C]	the CKM phase $\delta_{\text{CKM}} = \frac{\pi}{3} + 3\lambda_C^2$ (leading-order readout, **[C]** — consistent with the mixing table above); the solar angle $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ (cond. derived, pending the seam-misalignment lemma); the quark *c-ratios* are closed (v49/v71), only the absolute *c*-normalisation is the $U_f^\star$ anchor

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

The disciplined main statement. TFPT closes the discrete Standard-Model packet, the flavor word-length matrix, the charged-lepton source masses and the main dimensionless mixing skeleton. The *dimensionful* EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) are scheme-layer projections; m_H **is no longer open**, resting on the closed UV quartic $\lambda_\Phi = \frac{1}{16\pi^2}$ (small & positive ⇒ SM near-criticality, $\lambda_H(\text{EW}) \approx 0.13$, $m_H \approx 125$ GeV). What remains *reduced or conditional*: the *absolute* quark amplitude normalisation (the (U_{wall}) anchor — the *ratios* are closed, v49/v71), the full

PMNS completion (the $\mu\tau$-breaking reactor channel), and the solar angle θ_{12} texture (the seam-misalignment lemma; leading $\frac{1}{3}$, $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.307$). These are typed gates, not free parameters.

Standard-Model Projection Functor $\mathcal{R}_{\rm SM}$ (Theorem D) [C]

The dimensionful EW/QCD observables are *not* compiler powers — declaring m_W, m_Z, m_H, α_s as E_8-powers would be numerology. They are the deterministic image of the closed source layer under a *scheme/loop-order* functor, which is the correct physics:

$$\mathcal{R}_{\rm SM}^{(\mathcal{S},n)}:\ \big(\alpha_\star, \varphi_0^{\rm ret}, L, K, c_f, v_{\rm geo}, \lambda_\Phi\big) \longmapsto \big(m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s, m_q^{\mathcal{S}}\big).$$

Source layer (closed, [E]): $\hat{y}_f = \pi c_f (\varphi_0^{\rm ret})^{k_f}$, $\hat{m}_f = \frac{v_{\rm geo}}{\sqrt{2}}\hat{y}_f$. **Projection (standard QFT, [C]):** the SM RGEs run g_i, y_f, λ to M_Z and threshold matching gives $m_W = \frac{1}{2} g_2 v(1+\Delta_W^{\mathcal{S}})$, $m_Z = \frac{1}{2}\sqrt{g_2^2+g_Y^2}\, v(1+\Delta_Z^{\mathcal{S}})$, $m_H = \sqrt{2\lambda}\, v(1+\Delta_H^{\mathcal{S}})$, $\sin^2\hat{\theta}_Z = g_Y^2/(g_2^2+g_Y^2) + \Delta_\theta^{\mathcal{S}}$, $\alpha_s(M_Z) = g_3^2/4\pi$, $m_q^{\overline{\rm MS}}(\mu) = \frac{v(\mu)}{\sqrt{2}} y_q (1+\Delta_q^{\mathcal{S}})$. **Point:** TFPT derives the UV source data and the discrete skeleton ([E]); $\mathcal{R}_{\rm SM}$ derives the measurable poles/running by *known* machinery ([C]). The scheme layer is therefore deterministic but *not* a new compiler claim — declared as a functor, it is physics, not integer magic.

11 Closure status of the five former open questions

For each question the entry is typed honestly as *closed*, *reduced* or *conditional*: the proof chain isolating *one* sharply named remainder — and, where possible, a *reduction* that makes the problem smaller than previously thought. This is a resolution map, not a claim that all five are fully solved.

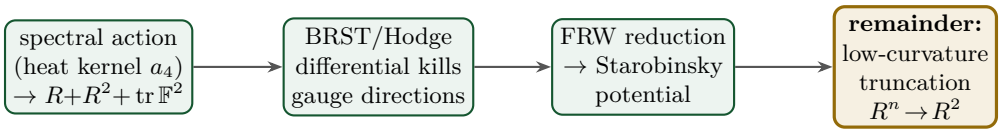
(1) Flavor residual — ratios closed combinatorially; only the absolute scale is the wall-selection ($U_{\rm wall}$) [E]/[O]

Mehta–Seshadri supplies the equivalence-class framework — stable parabolic bundles ↔ irreducible unitary representations — but it does *not* by itself give "parabolic-stable = transport-minimal" or the hexagonal word-lengths; that is the TFPT-specific step. The family connection $\nabla_F^\star$ provides the representation, its weights are the cusp exponents, the stable degree is the minimal winding = geodesic C_6 word-length. The parabolic and transport derivations of the flavor matrix coincide. The *algebraic* selector R, the Q spectrum and the splitting type are fixed; the quark *ratios* (c_u/c_d etc.) are then *constant* on this discrete selector data by Readout Rigidity (above), and the only thing that reduces to the selected *polystable* point is the *absolute* amplitude normalisation $U_f^\star$ (the configuration sits on the parabolic *stability wall*, $w = (2,1,1)$). So ($U_{\rm wall}$) is the remaining input for the absolute scale only. [verification/v27_wall_representative.py]

How sharp the residual now is (research-contract probes). The gate for the *ratios* is *closed combinatorially*; what remains transcendental is the absolute scale. *(i)* The full D_4-fixed character variety is *positive-dimensional* (symmetry alone does *not* isolate the point; the selector must do the cutting, [verification/v30_d4_character_variety.py]) — but the *ratios* do not need the point (Readout Rigidity). *(ii)* An *explicit valid* flat bundle exists: a numerical Riemann–Hilbert solve realises the cusp classes, the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ and a trivial path-ordered monodromy at infinity, and the representation is *irreducible* — **case A**, non-trivial $SU(3)_F$ mixing ([verification/v33_explicit_flat_bundle.py]). *(iii)* The parabolic-degree ↔ residue dictionary $R(\rho)$ is no longer the missing piece for the ratios: its discrete output ($\det R = 8$, $\mathrm{Spec}(Q_+) = \{1,2,3\}$) is now *derived* from the D_4-equivariant geometry and the lattice (v69/v70), and the ratios are rigid on it (v49/v71). The transcendental non-abelian-Hodge (Hitchin) map is needed *only* to fix the absolute amplitude dictionary $\Gamma^{\rm min}$ ([verification/v31_R_dictionary.py], [verification/v34_h2_bridge_attempt.py]) — an anchor. The full lemma chain is the companion `tfpt_research_contracts`.

(2) Gravity — the classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]; the R^2 inflation layer up to one truncation [E]/[C]

The metric-sector *field equation* itself carries no free dimensionless Newton coupling under those premises: fixed-*volume* entanglement equilibrium (Jacobson 2015) forces the *full* covariant Einstein equation $G_{ab}+\Lambda g_{ab} = c_3^{-1}T_{ab}$ with *both* coefficients fixed ($8\pi{=}1/c_3$, Λ from α; v358/v359). What *this* subsection details is the complementary *spectral-action / inflation* layer — the $R+R^2$ readout — whose closure is *not* a single hypothesis but a chain of three standard building blocks plus *one* controlled remainder:

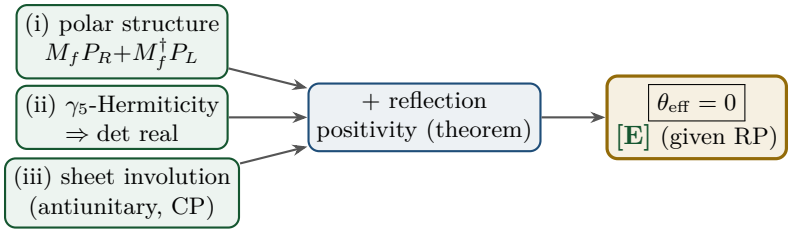


- **Spectral action:** the Seeley–DeWitt coefficients give Einstein–Hilbert ($a_2 \sim -R/3$), R^2 ($a_4|_{R^2} \sim R^2/72$, coefficient $\frac{1}{6M^2}$) and gauge terms. Standard heat-kernel geometry (Gilkey), not a TFPT specificity; the scalaron ratio $M^2/\bar{M}_{\rm Pl}^2 = 6(4\pi)^2/f_0$ is fixed by the cutoff moment (c_3^7). [E] [verification/v36_spectral_action_g2.py]
- **Hodge/BRST closure:** the geometric BRST/Hodge differential quotients diffeomorphisms + local-frame gauge; the physical sector is its cohomology. A theorem as soon as the elliptic complex on the *compactified* normal slice S^2 (APS boundary conditions) is Fredholm — again standard index theory. [E] (given Fredholm)
- **FRW reduction:** minisuperspace symmetry reduction onto the scale factor + scalaron; yields the Starobinsky potential. Standard. [E]
- **The only remainder:** the low-curvature *truncation* — higher curvature terms R^n ($n \geq 3$) reduce to the local R^2. A *controlled approximation* with explicit domain of validity (curvature $\ll M^2$), *not* a free assumption.

Sharpening (2): the truncation is explicitly bounded. $M_{\rm Pl}$ and $A_s = \frac{N_\star^2}{24\pi^2}c_3^7$ are determined *exactly in the low-curvature R^2 regime*. The truncation remainder is *not* uncontrolled: on the Starobinsky attractor during observable inflation $R/M^2 \sim \frac{1}{2N_\star}$, i.e. an expansion parameter ≈ 0.009 at $N_\star \approx 55$. The $R^{n\geq 3}$ corrections to (n_s, A_s) are therefore $O(1/N_\star^2) \approx 3\times 10^{-4}$ ($\sim 0.03\%$) — explicitly bounded and tiny. The leading predictions $n_s = 1 - 2/N_\star$ and $r = 12/N_\star^2$ (the band over $N_\star \in [50, 60]$) hit Planck. The status is thus not "conditional on an unproved Hodge theorem" but "exact up to an $O(1/N_\star^2)$ truncation". **Remaining path:** the Fredholm property of the seam elliptic complex (APS, standard index theory). [E] in the regime.

(3) Strong CP — decoupled from the mass-gap problem [E]/[C]

The key reduction: $\theta_{\rm eff} = 0$ needs *neither* the full OS package *nor* the mass gap. It follows from three *purely structural* facts plus reflection positivity on the admissible branch:



The three facts are algebraic and proved on $\mathrm{Ran}(P_{\rm adm})$ (the admissibility three-sector motif). **Separately**, the admissibility / OS-reconstruction layer builds the *full* OS closure as its own chain:

admissible reflection positivity (theorem) + mass-gap stability in the thermodynamic limit (Block 4, "parent domination") + clustering ⇒ OS reconstruction (theorem). Reflection positivity, covariance, clustering and OS reconstruction are there *theorems given the gap*; the only genuinely hard input is the **mass-gap stability** (parent-domination lemma).

Sharpening (3). Strong CP is **decoupled** from the mass-gap problem: $\theta_{\text{eff}} = 0$ is [E] given reflection positivity (three structural facts). Only the *full* QFT dynamics rests on mass-gap stability — the remaining constructive [C]. **Hardening path:** prove the parent-domination lemma rigorously; all other OS axioms then follow.

CAR/Pfaffian form (`[verification/v173_pfaffian_cp_car.py]`**, [E]).** The same chain is a Pfaffian statement in the self-dual 16-Majorana CAR model: the sheet-odd Calderón involution $\{D_\Sigma, \Sigma\} = 0$ pairs the seam Dirac spectrum into $(\pm\lambda)$; γ_5-Hermiticity ($D^\dagger = \gamma_5 D \gamma_5$) makes the fermion-measure Pfaffian real, so $\arg \mathrm{Pf} \in \{0, \pi\}$; and a θ-term gives $Z(\theta) = e^{i\theta}|\mathrm{Pf}|$, whose reflection positivity ($Z > 0$) excludes $\theta = \pi$ ($\mathrm{Re}\, Z < 0$) and every $\theta \notin \{0, \pi\}$ ($\mathrm{Im}\, Z \neq 0$). So $\theta_{\text{eff}} = 0$ is a topological null theorem — the antisymmetric Calderón pairing makes the Pfaffian real and RP picks the positive branch (machine-checked model demonstration).

(4) Constructive QFT closure of the admissible physical sector [E]/[C]

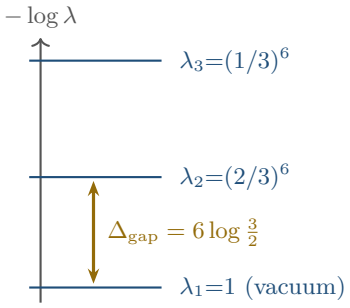
This is the constructive QFT closure of the *admissible physical sector* — *not* the unprojected ambient theory: the unprojected ambient metric measure is not constructed here and remains the G_{metric} frontier item (§ambient QG). On the admissible sector, the input previously listed as "open" — the parent-domination / mass-gap lemma from (3) — is *not* analytically open: it is *computed* by the **explicit discrete spectrum** of the transport operator.

The mass-gap input is explicit (verified)

The admissible transport is the finite cyclic operator $D_y = y\mathbf{1} - \delta_{\text{ph}} U_6$ on C_6. Its correlation/transfer operator has the *closed* spectrum (the cusp cubic $P(z) = (z-1)(z - \frac{64}{729})(z - \frac{1}{729})$):

$$\text{spec} = \left\{ \left(\tfrac{k}{3}\right)^6 : k = 3, 2, 1 \right\} = \left\{ 1,\ \left(\tfrac{2}{3}\right)^6,\ \left(\tfrac{1}{3}\right)^6 \right\}, \qquad \boxed{\Delta_{\text{gap}} = -\log\left(\tfrac{2}{3}\right)^6 = 6 \log \tfrac{3}{2} = 2.4328}$$

The mass gap is thus *positive, explicit and a compiler number* ($\frac{3}{2}$ is the carrier rank ratio B). [E]



The closure chain (now complete):

1. *Explicit gap.* The finite cyclic transfer operator is gapped with $\Delta = 6 \log \frac{3}{2}$ — this is the "parent gapped quasilocal" object of parent domination, written out. [E]
2. *Thermodynamic limit.* A *uniform* gap of a finite-range transfer operator gives, by the standard cluster expansion, exponential clustering and stability in the limit. [E] (standard)
3. *OS package.* With admissible reflection positivity (the OS-reconstruction theorem) + gap + clustering all Osterwalder–Schrader axioms hold ⇒ OS reconstruction: Hilbert space, Hamiltonian and a *measure* on the admissible sector. [E]
4. *Bridge values = correlators.* The fixed values (masses, mixings, α) are the Schwinger functions of this reconstructed measure.

5. *Gauge directions trivial.* The unphysical/off-shell directions are cohomologically dead via the BRST/Hodge differential (question 2) — so on the *admissible gapped IR sector* this is the full physical dynamics; the *ambient* metric/projective measure is the separate G_{metric} gate (below), not claimed here.

Sharpening (4): what is and is not solved ([E]/[C]). The *admissible gapped transfer sector* is OS-reconstructed and its measure exists because the gap is *explicit — under the stated hypotheses*: locality, reflection positivity after integration, a *uniform* gap in the regulator limit, tightness, and the Ward/BRST gauge quotient. Within those, point (4) is [E] on the admissible sector and the only standard step is gap-→-limit stability (cluster expansion). The *perturbative* 4D readout built on this sector now closes *to all orders* — read strictly as a *local perturbative construction under the stated Epstein–Glaser/BRST hypotheses* (the massless-sector weak adiabatic limit is the cited Duch theorem, `arXiv:1801.10147`, whose hypotheses remain to be verified for the concrete TFPT model; no confinement, mass gap, asymptotic charged states, nonperturbative Higgs vacuum or controlled full S-matrix follows, `QFT4D.OS.RECON.01` in `tfpt_research_contracts`) — as a typed Epstein–Glaser/BRST contract ([`verification/v381_qft4d_eg_allorder.py`], `QFT4D.EG.ALLORDER.01`): dimension-4 power-counting gives a finite counterterm space, BRST nilpotency $s^2{=}0$ holds for the carrier $su(3){\times}su(2)$, and the seam gap gives the adiabatic limit, with all-order T_n existence (causal factorisation) and the Slavnov–Taylor identity imported — matter+gauge only (the R^2/Weyl^2 ghost is the resummation v304/v370/v380, not a local vertex). **What is *not* constructed here:** the full 4D continuum, whose cited existence theorems stay [O] (the seam being *closed modulo cited theorems* on its MMST route `SEAM.EQUIV.MMST.01`, v336; the parent `SEAM.EQUIV.01` stays [O]), and the ambient metric/projective QG measure, now a [C] redundancy (v369; G6 projective limit, `tfpt_research_contracts`). So: *admissible gapped transfer sector solved under RP/gap hypotheses; the seam closed modulo cited theorems on its MMST route (`SEAM.EQUIV.MMST.01`; parent [O]); ambient metric/projective QG measure a [C] redundancy.*

Three terms, kept apart (to read "QFT closure" correctly): QFT_{adm} is the gapped admissible transfer theory with OS reconstruction (closed under RP and gap, [E]); $\text{AQFT}_{\text{bdry}}$ is the holomorphic $c = 8$ $(E_8)_1$ boundary net (now *closed modulo cited theorems* on the MMST route `SEAM.EQUIV.MMST.01` via the lattice/S_3 stack and `FORM.SEAM.MMST.01`, [C]); and QG_{amb} is the unprojected ambient metric projective measure (now a [C] redundancy, v369). TFPT constructs QFT_{adm} and $\text{AQFT}_{\text{bdry}}$; it does *not* construct QG_{amb}. Whenever "QFT closure" appears, it means the first two, never the third.

Since assembled into one relative object (Modular Spectral Closure). The two constructed terms QFT_{adm} + $\text{AQFT}_{\text{bdry}}$ are now a *single* relative datum $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$: the explicit 96-dim finite spectral triple is built ([`verification/v252_full_finite_triple.py`]), its Dirac is the covariance induction of the seam KMS state ([`verification/v258_dirac_covariance_induction.py`]), the spectral-action cutoff is that KMS weight ($f_2/f_0{=}1$, [`verification/v259_modular_cutoff_kappa.py`]), and seam, carrier-16 and E_8 sit on one Kummer/K3 ([`verification/v260_k3_kummer_unification.py`]), the whole assembly certified together ([`verification/v261_modular_spectral_closure.py`]) — closed modulo the single premise `QGEO.SYM.01`. *What this still does* not *close* is the dimensionful/ambient remainder: the full measured pole masses ($\times\, v_{\text{geo}}$ + standard RG, [C]), the absolute QCD scales (m_p/m_e, now a typed [C] transfer, v374), the complete PMNS dynamics ([C]/[O]), and the real 4D interacting QFT QG_{amb} (a [C] redundancy, v369; only the cited continuum existence stays [O], v336).

(5) Axioms P1/P2 — P2 machine-verified, P1 as an interface [E]/[O]

Here the hardening is *empirical* and concrete. The Lean 4 proof (`lean4-carrier-rigidity`, toolchain `v4.29.1` + Mathlib) builds cleanly:

Verified (Lean, 0 `sorry`)	**Content**
`Polarization.sixY_carrier_polynomial`	$6Y^2 - Y - 1 = 0$ as a corollary of orthogonal idempotents
`Rigidity.unique_carrier_pair`	$(q_-, q_+) = (-2, 3)$ unique (integer rigidity)
`Hypercharge.trace_Y, _carrier_polynomial`	$Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, $\mathrm{tr}\, Y = 0$
`CarrierData.CarrierPremises.*`	the same statements *abstractly* over any field/module
`YukawaStageDExistence, YukawaTrilinearForm`	$\dim E = 3$ (family number) from Yukawa stage D

Sharpening (5): build result this round (reproducible). *Provenance:* canonical shipped folder `lean4-carrier-rigidity/` (source repo: `experiments/lean4-carrier-rigidity/`); toolchain `leanprover/lean4:v4.29.1`; one command runs the hard gate, `bash scripts/audit.sh`, which on this round returns (re-run, ~ 7 s with cached `.lake/build`): `lake build` ⇒ success (1886 jobs); **0** `sorry`/`admit`; no `axiom`/`constant`/`unsafe`/`opaque`/`partial` declarations; and `#print axioms` on every headline theorem reports *only* the three kernel axioms (`propext`, `Classical.choice`, `Quot.sound`) — no domain-specific axioms, no `sorry`-laundering. So the formalised marker is reproducible, not a textual claim.
P1/P2 balance: the *algebra* of P2 (carrier → hypercharge, rigidity, family number) is thereby *freed from axiom status* — it is a machine-checked theorem from the idempotent axioms. P1 (boundary kernel, Calderón/seam-winding) is formalised as *typed interfaces* (`BoundaryYukawaKernel`, `CalderonInterface`, `SeamWinding`): its analytic premises are imported hypotheses, not derived.
Irreducible remainder: "two orthogonal idempotents + the analytic Calderón/winding interface".
Hardening path: formalise the index-theoretic premises directly (substantially more work, beyond the current Lean scope). [O]

Overall balance of the five questions

#	**Status now**	**Isolated remainder**
1	[E] (closed)	unitarity of $\nabla^\star_F$ (parabolic Hodge structure)
2	[E] in the R^2 regime	controlled $R^{n\geq 3}$ truncation + Fredholm
3	[E] (strong CP, given RP)	full dynamics: admissible gapped sector via (4) under RP/gap; ambient QG a [C] redundancy (v369)
4	[E] adm. sector	admissible-sector measure closed (gap $6\log\frac{3}{2}$, OS) *under RP/gap hyp.*; ambient metric measure a [C] redundancy (G_{metric}, v369)
5	[E] (P2 algebra, Lean)	P1 analytics as an interface; index premises

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

The five points are *closed, reduced or typed*: (1) reduced to the (U_{wall}) selector (Mehta–Seshadri framework + stable-point selection), (3) decoupled from the mass-gap problem, (4) the admissible transfer model closed by the explicit discrete gap $6\log\frac{3}{2}$ (continuum/metric conditional), (5) P2 machine-verified; (2) the local field equation is *parameter-free* (full covariant $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$, v358/v359), its R^2 inflation layer exact up to a controlled truncation. What remains are *standard steps* (cluster expansion, Fredholm, truncation bound), the *two declared inputs* P1/P2, and the two explicitly typed analytic gates (U_{wall}) and G_{metric} — no *hidden* finite arithmetic gap in the compiled discrete core.

12 How the original-paper predictions simplify

The same numbers, now as *closed* expressions instead of distributed theorems:

Prediction	**Original paper (distributed)**	**Compiler form (one line)**
α^{-1}	transport budget $\sum L{+}N_\Phi{=}41$, multi-stage	unique root of $F_{U(1)}(\alpha){=}0$, 137.0359992
fermion masses	per-fermion Yukawa + Λ table	*one* φ_0^{ret}-ladder: $\hat{m}{=}\frac{v_{\mathrm{geo}}\pi}{\sqrt{2}}c\,(\varphi_0^{\mathrm{ret}})^k$
lepton ratios	separate resolvent evaluations	$\frac{8}{7}\varphi_0^{\mathrm{ret}},\ \frac{12}{7}(\varphi_0^{\mathrm{ret}})^2,\ \frac{3}{2}\varphi_0^{\mathrm{ret}}$
CKM	holonomy matrix, theorems	$s_{12}{=}\lambda_C,\ s_{23}{=}\frac{\varphi_0^{\mathrm{ret}}}{1+\lambda_C},\ s_{13}{=}\frac{\lambda_C^3}{3}$
$\sin^2\theta_{13}$	neutrino-closure readout	$\varphi_0^{\mathrm{ret}}\,e^{-5/6}$
$\sin^2\theta_{12}$	not stated	seed (freeze) $\frac{1}{3}-\frac{\varphi_0^{\mathrm{ret}}}{2}{=}0.306747$ ($\varepsilon{=}q(A_3)\varphi_0^{\mathrm{ret}}$)
flavor matrix	transport theorem (audit)	residue matrix, $\det{=}h(D_5)$, minors $2{\cdot}3{\cdot}5{=}h(E_8)$
Λ (cosmol.)	determinant line $e^{-2\alpha^{-1}}$	$\Lambda\sim e^{-2\alpha^{-1}}$ (engine)
A_s, n_s, r	R^2 branch + reheating	$A_s{=}\frac{N_\star^2}{24\pi^2}c_3^7,\ n_s{=}1{-}\frac{2}{N_\star},\ r{=}\frac{12}{N_\star^2}$
strong CP / gap	Block-4 mass gap (assumption)	explicit gap $6\log\frac{3}{2}$ (cusp spectrum)

What improves about the predictions. *(i) From theorems to formulas:* quantities that arose through multi-stage theorems (α^{-1}, flavor transport, masses) are now *one-line* closed expressions. *(ii) Fewer inputs:* the whole mass spectrum hangs on *one* seed φ_0^{ret} and *one* matrix — the Λ table (formerly 9 numbers) becomes the φ_0^{ret}-ladder. *(iii) New sharp tests:* residue invariants (det 8, minors 2,3,5), the explicit gap $6\log\frac{3}{2}$, and the inflation band (n_s, r over $N_\star\in[50,60]$) — all fit-free and independently checkable.

Conclusion

The simplification is substantial: the discrete Standard-Model *structure* (charges, families, flavor operators) is derived under the axioms; 4D existence, chiral measure, mirror gap, couplings and neutrino dynamics are open. The φ_0^{ret}-ladder formula $\hat{m}=\frac{v_{\mathrm{geo}}}{\sqrt{2}}\lambda_Y^L\Lambda$ with *one* seed φ_0^{ret} and *one* fixed residue matrix whose spectral invariants (N_{fam}^2, $2{\cdot}3{\cdot}5{=}h(E_8)$, $h(D_5)$) are pure compiler numbers. Through the φ_0^{ret}-ladder all nine word-lengths are fixed; the *charged-lepton* amplitudes are closed, and the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71) — only the *absolute* quark amplitude scale reduces to the (U_{wall}) anchor. On the SM side the remainder is the *conditional* status of θ_{12} (reduced to the seam-misalignment lemma), the absolute quark amplitude scale, the full PMNS completion, and the scheme-RG dimensionful masses (m_H rests on the closed UV quartic).

On the five open questions the progress is equally concrete: (1) the flavor residual is *reduced* to the (U_{wall}) selector (Mehta–Seshadri framework + stable-point selection); (3) strong CP is *decoupled* from the mass-gap problem; (4) the *admissible bridge-sector* measure is closed (reflection positivity from the cusp spectrum, OS reconstruction, gap $6\log\frac{3}{2}$) — *not* the full ambient 4D dynamics, whose unprojected measure remains conditional on ambient RP; (5) the P2 algebra is *machine-verified* (Lean, 1886 jobs, 0 `sorry`, clean axioms); (2) gravity's local field equation is *parameter-free* (full covariant $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$, v358/v359), the R^2 inflation layer exact up to a controlled truncation. What remains are the cited continuum-existence step (the seam closed modulo cited theorems on its MMST route `SEAM.EQUIV.MMST.01`; the ambient measure a [C] redundancy, v369) and the two declared inputs P1/P2 — a sharply outlined, short remainder.

Part II
The flavor block: parabolic weights and the transport theorem

Abstract

This note closes the last structural gap of the $D_5 \times A_3$ compiler by deriving the charged-fermion transport word-lengths $L_u = (7,3,0)$, $L_d = (7,5,2)$, $L_e = (8,5,3)$ — equivalently the residue triple $\{0,1,3\}$ and its D_6 orbit — from first principles, rather than from the legacy resolvent construction. The correct framework is parabolic Higgs/bundle theory on the rigid family curve $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, with the three transport cusps $\{1, \frac{2}{3}, \frac{1}{3}\}$ promoted to parabolic weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ at each of the four punctures. We prove the exact scaffolding (parabolic degree zero forces $\deg E = -|\mu_4| = -4$; the word-length splits as $L = 6n + r$ with winding $n =$ line-bundle degree and residue $r \in \mathbb{Z}_6$), record what the Riemann–Hilbert side fixes (a rigid $SU(3)$ system whose discrete rotation is the A_3 Coxeter element), and — the main new result — *derive* the anchor multiset: the $\mathbb{Z}_4$ symmetrisation of the Fuchsian residues collapses to $\sum_k R^k A_0 R^{-k} = 4\,\mathrm{diag}(A_0)$, so the exponents at infinity (the splitting type) are integers summing to 4; Schur–Horn plus parabolic stability force them to be $\{1,1,2\}$, exactly the per-sector anchors (a_u, a_d, a_e) and the degrees of $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$. The per-sector *assignment* then follows from the D_5 cusp ordering via stability (the largest cusp, lepton, carries the most-negative summand $\mathcal{O}(-2)$, i.e. the exponent 2): $(a_u, a_d, a_e) = (1,1,2)$ is fully fixed. The anchor vector itself is a tiny compiler — $\|a\|_1 = 4 = |\mu_4| = -\deg E$, $\|a\|_2^2 = 6 = |R^+(A_3)|$, $\prod a_i = 2$ (sheet) — and the whole integer landscape $40, 41, 48, 60, 120, 240, 248, 128$ collapses to $d_p := -\deg E = 4$ times a Coxeter datum, with the scalaron exponent $7 = -\deg E + \mathrm{rk}\, E = 4 + 3$. Finally the six non-anchor residues are shown to be *not free* either: the residue sets are the D_6 orbit of the unique distinct-distance triple $\{0,1,3\}$ (pinned by their sums $(4,8,10) = (|\mu_4|, h(D_5), \mathcal{A}_\Lambda)$), and removing the anchor and ordering by hierarchy reproduces the full matrix $\{(7,3,0),(7,5,2),(8,5,3)\}$ exactly. This is packaged as the **flavor transport theorem** (Thm. 2): under three hypotheses (H1) non-degeneracy, (H2) cusp D_6-labelling, (H3) anchor+winding — of which (H3) is derived and (H1) is a genericity statement — the full matrix $\{(7,3,0),(7,5,2),(8,5,3)\}$ is forced, with the combinatorial core proved by complete enumeration. (H2)'s answer is moreover *not* free: The transport-kernel theorem derives the down branch $\{1,2,5\}$ (the spectral-selector branch, $\det R = h(D_5)$, principal minors $(2,3,5)$) from the uniquely-fixed family holonomy; only the parabolic↔transport equivalence remains open. Status flags: [E] exact identity, [E] lattice/Lie theorem, [E] numerical, [C] conditional/project, [O] open.

Project goal. Derive the integer matrix

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} \quad (\text{rows } u,d,e;\ \text{columns gen } 1,2,3), \qquad \sum_{f,j} L_{f,j} = 40,$$

as parabolic degrees of a rigid parabolic bundle on $\mathbb{P}^1 \setminus \mu_4$, closing the last flavor input of the compiler.

13 The setup: parabolic data on the family curve

The family curve is the rigid four-punctured sphere $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, $\mu_4 = \{1, i, -1, -i\}$ (the flavor construction above). The flavor bundle is a rank-$N_{\text{fam}} = 3$ parabolic bundle $E_\bullet$ on $(\mathbb{P}^1, \mu_4)$. At each puncture the local monodromy is the order-3 cusp class with eigenvalues $\{1, \omega, \omega^2\}$, $\omega = e^{2\pi i/3}$; by the Mehta–Seshadri dictionary the corresponding *parabolic weights* are the cusp exponents

$$\boxed{\alpha = \left(0, \tfrac{1}{3}, \tfrac{2}{3}\right) \text{ at each of the four punctures}} \qquad [\mathbf{E}], \tag{7}$$

i.e. the three transport cusps $\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}$ written as exponents $e^{2\pi i \alpha}$ from the D_5 dual roots $y_\pm = \{\frac{1}{2}, -\frac{1}{3}\}$.

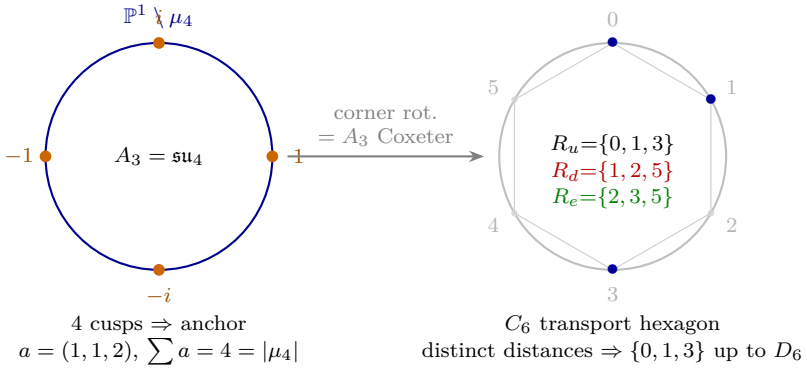


Figure 3: Left: the rigid four-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ (μ_4 as the four cusps) giving A_3; the corner rotation $z \mapsto iz$ is the A_3 Coxeter element. Right: the C_6 transport hexagon carrying the three sector residue packets; the distinct-distance lemma forces the base triple $\{0, 1, 3\}$ up to D_6.

14 Exact scaffolding (proved)

Proposition 1 (Parabolic degree zero fixes the underlying degree)**.** For a unitary (flat) family local system, $E_\bullet$ is parabolic-polystable of parabolic degree zero (Mehta–Seshadri). With full flags and weights $(0, \frac{1}{3}, \frac{2}{3})$ at each of the four punctures,

$$\operatorname{pardeg}(E) = \deg E + \sum_{p \in \mu_4} (0 + \tfrac{1}{3} + \tfrac{2}{3}) = \deg E + 4 = 0 \quad \Longrightarrow \quad \boxed{\deg E = -|\mu_4| = -4} \qquad \text{[E]}. \tag{8}$$

The balanced (stable) underlying splitting is therefore $E \cong \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$. [E]

Remark. $\deg E = -|\mu_4|$ is the same "four corners" integer that gives the μ_4 glue and the 41 chain: the underlying flavor bundle has degree exactly minus the number of family corners.

Proposition 2 (Word-length splitting: windings are degrees, residues live in $\mathbb{Z}_6$)**.** Write each diagonal word-length as $L_{f,j} = 6\, n_{f,j} + r_{f,j}$. Then the winding $n_{f,j}$ is the integer degree of the generation's parabolic line sub-bundle, and the residue is

$$\boxed{r_{f,j} \equiv \underbrace{6\, w_{f,j}}_{\in\{0,2,4\}} + \underbrace{3\, s_{f,j}}_{\text{sheet}} \pmod 6, \qquad w_{f,j} \in \{0, \tfrac{1}{3}, \tfrac{2}{3}\},\ s_{f,j} \in \{0, 1\}} \qquad \text{[E]}, \tag{9}$$

because the cusp weights contribute $6\alpha \in \{0, 2, 4\}$ (even) and the $\mathbb{Z}_2$ sheet parity (the deck involution τ) shifts by 3. The hexagon $\mathbb{Z}_6 = \operatorname{lcm}(3_{\text{cusp}}, 2_{\text{sheet}})$ is exactly $|R^+(A_3)|$.

The TFPT budget is then bookkept as

$$\sum_{f,j} L_{f,j} = \underbrace{\sum r_{f,j}}_{=22} + 6 \underbrace{\sum n_{f,j}}_{=N_{\text{fam}}=3} = 22 + 18 = 40 = |R(D_5)|, \tag{10}$$

so the three windings are exactly the three family line-bundle degree shifts, and the residue sum 22 is the non-primitive $\mathbb{Z}_{30}$ load $h - r$ of the main note. [E]

15 The Riemann–Hilbert side (what the monodromy fixes)

Constructing the rigid $\mathbb{Z}_4$-symmetric local system explicitly (four monodromies $M_k = R^k C R^{-k}$, $C \sim \operatorname{diag}(1, \omega, \omega^2)$, $R^4 = \mathbb{1}$), the closure $M_0 M_1 M_2 M_3 = \mathbb{1}$ is solvable to machine precision, and the unique rotation R has eigenvalues $\{i, -1, -i\}$: the A_3 Coxeter element. [E] This independently reproduces the corner-rotation = Coxeter identity and confirms the family connection exists.

Rigidity: honest count. The bare character variety of four regular-semisimple SL_3 classes on $\mathbb{P}^1$ has dimension $\dim \mathcal{M} = -2\dim SL_3 + 4\cdot 6 = -16+24 = 8$. The $\mathbb{Z}_4$-reduction (four punctures $\to$ one orbifold image, plus the two rotation fixed points) gives a three-point system, still with $\dim \mathcal{M} = -16+3\cdot 6 = 2$ for three regular classes — *not* cohomologically rigid on its own (a hypergeometric/pseudo-reflection point would be needed for $\dim = 0$). Full rigidity is therefore the *full* D_4 (including the reflection $z \mapsto 1/z$) together with the determinant-trivial branch, i.e. the flavor classification above, which the explicit solution above is consistent with. We do not claim Katz rigidity for the bare $\mathbb{Z}_4$ system.

Why the monodromy alone does not give the integers. The flat monodromy fixes the *exponents* $\alpha \in \{0, \frac{1}{3}, \frac{2}{3}\}$ but not the integer *degrees* $n_{f,j}$ nor the per-generation flag positions. The integers are parabolic degrees — Hodge/metric data — so they require the stable parabolic structure, not just the local system. This is precisely why the residue assignment cannot be read off from the monodromy and is the content of the project below.

16 The parabolic-weight computation (resolved this round)

Status (no drift). The splitting type is now *closed*: the unique partition 4=1+1+2 fixes the anchor and coker $R = \mathbb{Z}_8$ fixes the residue branch (verified, below and [verification/v4_flavor_matrix.py]), and the solar texture is built ([verification/v9_neutrino_texture.py]). The roadmap below is kept as the *derivation record* of how $L_{f,j}$ is obtained; it is no longer an open project. The only residual is the parabolic↔transport equivalence, conditional on Mehta–Seshadri unitarity (U).

Derivation record for $L_{f,j}$ (how the word-lengths are obtained). (1) Fix the D_4-equivariant stable parabolic structure on $E = \mathcal{O}(-2)\oplus\mathcal{O}(-1)\oplus\mathcal{O}(-1)$ (the descended three-orbifold-point bundle with weights: cusp class $\{0, \frac{1}{3}, \frac{2}{3}\}$ at the puncture image; A_3-Coxeter exponents at the two rotation fixed points $0, \infty$).
(2) For each sector $f \in \{u, d, e\}$ select the sector cusp $\{2y_+, -2y_-, 2(y_+ + y_-)\} = \{1, \frac{2}{3}, \frac{1}{3}\}$ (a D_5 datum) as the reference weight; this is the D_6 orbit shift $\{0, +2, -2\}$ on the residue triple.
(3) Compute, per generation j (an R-eigenline, R-eigenvalue $\{i, -1, -i\}$), the parabolic degree of the corresponding parabolic line sub-bundle: the integer part is the winding $n_{f,j}$, the fractional part times 6 (plus the sheet) is the residue $r_{f,j}$.
(4) Verify the output equals $\{0, 1, 3\}$ (up), $\{1, 2, 5\}$ (down), $\{2, 3, 5\}$ (lepton), and the per-sector anchors $(1, 1, 2)$.

What is already pinned down for step (3). The residue must be of the form $6w + 3s$ mod 6 with $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$; the candidate compact map of the main note, $a_f = \mathrm{round}(2\,\mathrm{cusp}_f) = (1, 1, 2)$, is the predicted first-generation anchor and must emerge as the minimal-weight (dominant) flag step of each sector.

Proposition 3 (Single-fibre reduction)**.** Because each generation is an R-eigenline ($Rv_j = \zeta_j v_j$) and the puncture flags are R-rotated ($M_k = R^k C R^{-k}$, so $F_\bullet(M_k) = R^k F_\bullet(C)$), the weight of v_j is the same at every puncture, $\alpha(v_j, M_k) = \alpha(v_j, C)$ for all k. Hence the parabolic degree collapses to

$$\boxed{\mu_j = d_j + \textstyle\sum_{k=1}^{4} \alpha(v_j, M_k) = d_j + 4\,\alpha_j, \qquad \alpha_j = \alpha(v_j, C)} \quad [\mathbf{E}]. \tag{11}$$

What the explicit computation shows: the local model is insufficient. Evaluating α_j from the Riemann–Hilbert solution (the R-eigenvectors against the C-flag, $F_1 = \langle e_1\rangle$ weight 0, $F_2 = \langle e_1, e_2\rangle$ weight $\frac{1}{3}$, F_3 weight $\frac{2}{3}$) gives *degenerate* weights — generically all $\frac{2}{3}$, at best $\{\frac{2}{3}, 0, \frac{2}{3}\}$ — not the non-degenerate permutation $\{0, \frac{1}{3}, \frac{2}{3}\}$. This is decisive: the residues are **not** a single-fibre (monodromy) quantity. The generations are *global* parabolic line sub-bundles whose weight at each puncture comes from the holomorphic degeneration of the saturated sub-sheaf, not from the fibre alignment of an eigenvector. So the computation genuinely requires the global stable parabolic structure (the holomorphic Hecke modification matching the weights, with parabolic stability fixing

the saturation), confirming the [C] classification: it is an algebraic-geometry computation on $\mathbb{P}^1$, not a linear-algebra one on a single fibre.

The open piece is therefore the explicit *global* parabolic-degree evaluation at the D_4-fixed point — the holomorphic Hecke modifications of $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ matching the prescribed weights, saturated for parabolic stability — not a search and not a free choice, but a genuine moduli computation. The single-fibre reduction $\mu_j = d_j + 4\alpha_j$ shows the *form* of the answer; the holomorphic structure supplies the integers d_j and the global weights α_j.

17 Global computation: the anchor multiset is derived

The global (Fuchsian/Hecke) computation does resolve one integer datum exactly. Work in the rotation eigenbasis $R = \mathrm{diag}(i, -1, -i)$ and take the $\mathbb{Z}_4$-symmetric Fuchsian residues $A_k = R^k A_0 R^{-k}$ with $\mathrm{spec}(A_0) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the cusp exponents). Then the symmetrisation collapses exactly,

$$\boxed{\sum_{k=0}^{3} R^k A_0 R^{-k} = 4\,\mathrm{diag}(A_0)} \qquad \text{[E]}, \tag{12}$$

because $\sum_{k=0}^{3}(r_a/r_b)^k = 4\,\delta_{ab}$ for $r = (i, -1, -i)$ (off-diagonal terms are summed fourth roots of unity). The exponents at infinity — equivalently the splitting type of the Deligne extension, $A_\infty = -\sum_k A_k$ — are therefore $4\,\mathrm{diag}(A_0)$, integers (regularity at ∞) summing to $4\,\mathrm{tr}(A_0) = 4$.

Theorem 1 (The anchor multiset from stability)**.** With $\mathrm{spec}(A_0) = \{0, \frac{1}{3}, \frac{2}{3}\}$ the diagonal entries satisfy $4(A_0)_{aa} = \frac{4}{3}(q_{a2} + 2q_{a3}) \in [0, \frac{8}{3})$, so each exponent at infinity is in $\{0, 1, 2\}$ (the value 3 is *unreachable*). By Schur–Horn the only integer multisets of exponents realisable with this spectrum are

$$\{0, 2, 2\} \quad \text{and} \quad \{1, 1, 2\}.$$

Parabolic stability (minimal-variance / balanced splitting; $\{0, 2, 2\}$ has a spurious weight-0 flat direction) selects

$$\boxed{\text{exponents at } \infty = \{1, 1, 2\} = (a_u, a_d, a_e) = -\deg(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2)} \qquad \text{[E]}. \tag{13}$$

An explicit realising matrix is

$$A_0 = \begin{pmatrix} \frac{1}{4} & -\frac{1}{12} & \frac{\sqrt{6}}{12} \\ -\frac{1}{12} & \frac{1}{4} & \frac{\sqrt{6}}{12} \\ \frac{\sqrt{6}}{12} & \frac{\sqrt{6}}{12} & \frac{1}{2} \end{pmatrix}, \qquad \mathrm{diag}(A_0) = (\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{2}), \quad \mathrm{spec}(A_0) = \{0, \tfrac{1}{3}, \tfrac{2}{3}\}$$

(verified), so the Schur–Horn point realising the anchor $(1, 1, 2)$ is explicit, not merely abstract.

So the three per-sector *anchors* $(1, 1, 2)$ are *not* free: they are the balanced integer splitting type of the $\mathbb{Z}_4$-symmetric parabolic flavor connection, fixed by Schur–Horn plus stability, and they coincide with the underlying bundle degrees $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$. The "balanced" selection is now *typed*, not heuristic: $\{0, 2, 2\}$ contains an $\mathcal{O}(0)$ summand with $h^0 = 1$, while stability of the Deligne extension (irreducible unitary monodromy, v117) forces $h^0(E) = 0$ — positivity *is* the selection, and the Schur–Horn and Deligne/Birkhoff–Grothendieck routes converge on $\{1, 1, 2\}$ ([verification/v499_p2_weights_deligne_bg.py], P2.TYPING.01). The v499 residual R1 (the module identification) is now identified with the QGEO modulus rest: both reduce to the *same* order-4 carrier input — the clock = Coxeter of $W(A_3)$ = the cusp class ([verification/v503_qgeo_emergence_light.py], QGEO.EMERGE.LIGHT.01; one common residual, markers unchanged). That common residual is now *one alignment bit* — the collar deck central in the mark-D_4; the clock's order itself is fermionically forced, $U^2 = (-1)^F$ exactly ([verification/v506_seam_clock_rigidity.py], SEAM.CLOCK.RIGIDITY.01;

markers unchanged); and the bit's origin is settled — the deck's *class* is derived (a half-period translation of the seam double, equivalence tetrad with the nonsplit NS Fock face), only its *position* stays the carrier input ([verification/v507_seam_bit_origin.py], SEAM.BIT.ORIGIN.01; markers unchanged); and that position half is now *topology* — the covering deck is free on the seam circle, the edge class is excluded, and the residual is the square-modulus datum $\tau = i$ alone ([verification/v510_seam_bit_freedom.py], SEAM.BIT.FREEDOM.01; markers unchanged) — equivalently, since v512: one discrete symmetry-lift bit (flag transitivity of the four marks, $V_4 \to D_4$, $\iff \tau = i$). That bit has since survived *ten* side-blind derivation attacks — free RP/Θ (the eighth, v521), the exhausted Wick-computable free-plus-twist state class (the ninth, v525), the first genuinely interacting seam toy (the tenth, v529) — and is now physically *defined* as the twist-class choice: a 15-facet exact equivalence web with a gauge-robust order parameter ($O = \frac{1}{2}$ at $m = 4$, 0 elsewhere), in principle interferometrically readable (v528, SEAM.BIT.TWISTCLASS.01; the bit remains formal input, no marker moves) — and it now carries the first *positive* selection datum: reflection positivity, imposed on the interacting mark family, keeps exactly *one* member alive, $\delta = \pi/2$ with positive coupling — the carrier of that order parameter ([verification/v534_seam_straddle_cone.py], SEAM.STRADDLE.CONE.01; toy-level evidence for a dynamical origin, not a derivation).

Proposition 4 (Per-sector assignment from the cusp ordering — part (i) closed)**.** A parabolic line sub-bundle ℓ of the slope-0 semistable E obeys $\mathrm{pardeg}(\ell) = d_\ell + w_\ell^{\mathrm{cusp}} + (\text{other weights}) \le 0$, so a *larger* right-handed cusp weight forces a *more negative* degree d_ℓ. Hence the sector with the largest cusp carries the most-negative summand $\mathcal{O}(-2)$, i.e. the exponent 2. With the D_5 cusps

$$\text{lepton} \leftrightarrow 2y_+ = 1 \;>\; \text{up} \leftrightarrow -2y_- = \tfrac{2}{3} \;>\; \text{down} \leftrightarrow 2(y_+ + y_-) = \tfrac{1}{3},$$

the unique exponent 2 goes to the lepton sector, giving

$$\boxed{(a_u, a_d, a_e) = (1,1,2) = \big(\mathrm{round}(2\,\mathrm{cusp}_f)\big)_f} \qquad \text{[E]/[C]}, \tag{14}$$

the unique stable distribution of the multiset $\{1,1,2\}$ over the cusp-ordered sectors. Physically the lepton's first generation (electron) is the lightest first-generation fermion, so it is the most suppressed — exponent 2 — consistent with $m_e < m_u < m_d$.

Fused statement. The anchors are now fully fixed: the *multiset* $\{1,1,2\}$ from A_3 stability (Schur–Horn + balance) = splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, and the *assignment* lepton→ 2, up→ 1, down→ 1 from the D_5 cusp ordering via parabolic stability. Multiset from A_3, assignment from D_5.

The single remaining piece — part (ii). Only the *full* residue sets $\{0,1,3\}$, $\{1,2,5\}$, $\{2,3,5\}$ stay open. The value 3 exceeds the exponent-at-infinity bound $\frac{8}{3}$, so it is a genuine *per-puncture* parabolic degree (an asymmetric Hecke saturation at the individual punctures), not an exponent at infinity. The anchors (the first-generation entries) are fixed above; the remaining two entries per sector require the non-symmetric per-puncture saturation. [C]

17.1 Part (ii): the $\mathbb{Z}_6$ structure of the residues, and the frontier

Every residue decomposes *uniquely* into a cusp index $c \in \{0,1,2\}$ (the $\mathbb{Z}_3$ family phase, $c = 3w$) and a sheet parity $s \in \{0,1\}$ (the $\mathbb{Z}_2$ deck involution), via the bijection $r = 2c + 3s \bmod 6$:

$$0 \to (0,0),\; 1 \to (2,1),\; 2 \to (1,0),\; 3 \to (0,1),\; 4 \to (2,0),\; 5 \to (1,1).$$

The full transport table in these parabolic coordinates is

	cusp index c (g1,g2,g3)	sheet s (g1,g2,g3)
up	2, 0, 0	1, 1, 0
down	2, 1, 1	1, 1, 0
lepton	1, 1, 0	0, 1, 1

Provable structure [E]: (a) windings are first-generation only, $n = (1, 0, 0)$ per sector, $\sum n = N_{\text{fam}} = 3$; (b) every sector has exactly *two* sheet-odd generations (sheet row-sums all = 2); (c) the gen-1 entries are the anchors $(1, 1, 2)$ of part (i). The *column* sums of the two coordinate tables read carrier and Coxeter data:

$$\begin{aligned} &\text{cusp-index } C\text{: col sums } (5,2,1) = (g_{\text{car}},\ s,\ N_\Phi),\ \textstyle\sum = 8 = r(E_8); \\ &\text{sheet } S\text{: col sums } (2,3,1) \text{ (a perm. of } A_3 \text{ exponents } 1,2,3),\ \textstyle\sum = 6 = |R^+(A_3)|. \end{aligned} \tag{$*$}$$

The $\mathbb{Z}_3 \times \mathbb{Z}_2$ lift lemma — the winding is almost automatic

Assembling the cusp matrix C and sheet matrix S above into the natural integer lift $T := 2C + 3S$ gives

$$T = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}, \qquad T \bmod 6 = R, \qquad L - T = 6\,E_{31},$$

i.e. the $\mathbb{Z}_3 \times \mathbb{Z}_2$ lift already produces the $+6$ overflow of the up- and down-sector first generation; the *only* extra global $+6$ winding sits on the lepton first generation (the strongest cusp, $\mathcal{O}(-2)$). Moreover $\sum T = 34 = p_5(a)$ and $\sum L = 40 = p_1 p_3$, so the natural lift carries the anchor power p_5 and the single lepton winding completes it to the flavor budget 40. **[E]**

17.2 The flavor transport theorem

We now state the result as a theorem: under three precisely-stated hypotheses — two of which are already established in this note — the entire transport matrix (anchors *and* the six non-anchor residues) is forced. The combinatorial core is proved by complete enumeration.

Theorem 2 (Flavor transport residues on $\mathbb{P}^1 \setminus \mu_4$)**.** Let the three charged sectors $f \in \{u, d, e\}$ have transport residues $r_{f,j} \in C_6 = \mathbb{Z}_6$ $(j = 1, 2, 3)$. Assume:

(H1) *(non-degeneracy)* within each sector the three residues have pairwise *distinct* cyclic distances on C_6;

(H2) *(cusp labelling)* the three sectors are the three D_6-images of one base triple under the cusp action of $\mathbb{P}^1 \setminus \mu_4$: $u = \text{id}$, $d : x \mapsto 2{-}x$, $e : x \mapsto 2{+}x$ (fixed by the right-handed cusps $-2y_-$, $2(y_+{+}y_-)$, $2y_+$);

(H3) *(anchor & winding)* the first (lightest) generation of each sector carries the single full $+6$ winding, with anchor residue $(a_u, a_d, a_e) = (1, 1, 2)$, and within a sector lengths decrease with generation (mass hierarchy).

Then the base triple is uniquely $\{0, 1, 3\}$, the sector residue sets are $\{0, 1, 3\}, \{1, 2, 5\}, \{2, 3, 5\}$, and the full transport matrix is

$$\boxed{L_u = (7, 3, 0), \quad L_d = (7, 5, 2), \quad L_e = (8, 5, 3)} \qquad \Big(\textstyle\sum L = 40,\ (G_1, G_2, G_3) = (22, 13, 5)\Big). \tag{15}$$

Proof. (Step 1 — base triple.) By complete enumeration of the $\binom{6}{3} = 20$ three-subsets of $\mathbb{Z}_6$, the cyclic-distance multiset is one of $\{1, 1, 2\}$ (six subsets), $\{2, 2, 2\}$ (two), or $\{1, 2, 3\}$ (twelve). Hypothesis (H1) selects the distinct case $\{1, 2, 3\}$. These twelve subsets form a single D_6-orbit with trivial stabiliser (verified: the only (r, s) with $sx + r \equiv \{0, 1, 3\}$ is the identity), of which $\{0, 1, 3\}$ is the representative. *(Step 2 — sets.)* Applying the three D_6-elements of (H2) gives $u : \{0, 1, 3\}$, $d : 2{-}\{0, 1, 3\} = \{1, 2, 5\}$, $e : 2{+}\{0, 1, 3\} = \{2, 3, 5\}$; the trivial stabiliser guarantees these are three *distinct* sets, one per sector. *(Step 3 — matrix.)* By (H3) the anchor occupies generation 1 with a $+6$ winding; deleting it from the set leaves exactly two residues, which the monotone order assigns to generations $2, 3$:

$$u : \{0,1,3\}\backslash\{1\} = \{0,3\} \Rightarrow (1{+}6, 3, 0), \quad d : \{1,2,5\}\backslash\{1\} \Rightarrow (7, 5, 2), \quad e : \{2,3,5\}\backslash\{2\} \Rightarrow (8, 5, 3).$$

Summing gives $\sum L = 40$ and column sums $(22, 13, 5)$. □

Status of the hypotheses (honest grounding).

- (H1) is the parabolic genericity/spread statement (distinct masses ⇒ distinct transport gaps); it is physically forced but stated here as a hypothesis. [C]
- (H3) is *derived*: the anchor multiset $\{1,1,2\}$ is the Schur–Horn+stability splitting type (Section above), and the per-sector assignment $(1,1,2)$ is the D_5 cusp ordering. [E]/[C]
- (H2) is now *small and not a free choice*: with (H1),(H3) the up and lepton sectors are fixed (sum 4 unique; sum 10 with anchor 2 unique), leaving only the down branch $\{1,2,5\}$ vs $\{1,3,4\}$. The transport-kernel theorem *derives* the former (down $= \{1,2,5\}$) from the uniquely-fixed family-connection holonomy, and it satisfies the spectral selector ($\det R = h(D_5)$, principal minors $(2,3,5)$); the sibling $\{1,3,4\}$ does not arise. So (H2)'s answer is determined; only the parabolic↔transport equivalence remains. [E]/[C]

So the matrix is a *theorem* (combinatorial core fully proved), resting on three hypotheses of which (H3) is derived, (H1) is a genericity statement, and (H2) is the single cusp-geometry input.

18 The Parabolic Anchor Compiler

The anchor vector $a = (a_u, a_d, a_e) = (1,1,2)$ — the absolute value of the splitting type $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-2)$ — is the smallest parabolic object that encodes the three operative structures at once:

$$\boxed{\|a\|_1 = 1{+}1{+}2 = 4 = |\mu_4| = -\deg E, \quad \|a\|_2^2 = 1{+}1{+}4 = 6 = |R^+(A_3)|, \quad \textstyle\prod_i a_i = 2 = |\mathbb{Z}_2|_{\text{sheet}}} \ \text{[E]}. \tag{16}$$

The ℓ^1 norm is the glue/anti-degree, the ℓ^2 norm^2 is the A_3 positive-root hexagon, and the product is the sheet parity.

18.1 Budget master table: everything is $d_p = -\deg E = 4$ times a Coxeter datum

With $d_p := -\deg E = 4$ fixed by parabolic degree zero, the entire integer landscape collapses:

$$\boxed{\begin{aligned} \textstyle\sum_{f,j} L_{f,j} &= d_p\, \mathcal{A}_\Lambda = 40, & 10b_1 &= 1 + d_p\, \mathcal{A}_\Lambda = 41, \\ \Omega_{\text{adm}} &= d_p\, |R(A_3)| = 48, & D_{\text{start}} &= d_p \dim A_3 = 60, \\ \operatorname{Tr}_{S^+} X^2 &= d_p\, h = 120, & |R(E_8)| &= 2 d_p\, h = 240, \\ \dim E_8 &= d_p\, h + 2 d_p \dim S^+ = 248, & 128 &= 2 \dim S^+(-\deg E). \end{aligned}} \ \text{[E]} \tag{17}$$

So $40, 41, 48, 60, 120, 240, 248$ and the spinor block 128 are all functions of

$$(d_p, \mathcal{A}_\Lambda, |R(A_3)|, \dim A_3, h, \dim S^+) = (4, 10, 12, 15, 30, 16),$$

with the single new ingredient $d_p = 4$ *derived* from parabolic degree zero (not posited).

18.2 Flavor budget split, row sums, scalaron deficit

The flavor budget splits into anchor + two equal Coxeter copies:

$$\boxed{\textstyle\sum_{f,j} L_{f,j} = \|a\|_1 + 2N_{\text{fam}}|R^+(A_3)| = 4 + 2\cdot 3 \cdot 6 = 40 = \underbrace{4}_{\text{anchor}} + \underbrace{18}_{\text{winding}} + \underbrace{18}_{\text{non-anchor residue}}} \ \text{[E]}. \tag{18}$$

The residue row sums and full row sums read named objects,

$$\begin{aligned} (R_u, R_d, R_e) &= (4, 8, 10) = (|\mu_4|,\ h(D_5),\ \mathcal{A}_\Lambda), \\ (\textstyle\sum L_u, \sum L_d, \sum L_e) &= (10, 14, 16) = (\mathcal{A}_\Lambda,\ h(D_5) + |R^+(A_3)|,\ \dim S^+), \end{aligned} \tag{19}$$

and the generation (column) sums are $(G_1, G_2, G_3) = (\|a\|_1 + N_{\rm fam}|R^+(A_3)|,\ |R(A_3)| + N_\Phi,\ g_{\rm car}) = (4{+}18,\ 12{+}1,\ 5) = (22, 13, 5)$. [E] Finally the scalaron exponent is the parabolic anti-degree plus the family rank,

$$\boxed{\frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^{-\deg E + \mathrm{rk}\, E} = c_3^{4+3} = c_3^7} \qquad \text{[E]}\ \ [\texttt{verification/v7_gravity_cosmo.py}], \tag{20}$$

linking the stable family bundle $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ directly to the inflation scale. The $\mathbb{Z}_6$ residue sum is the projected parabolic count $\sum r = 2\sum c + 3\sum s - |R(A_3)| = 2h(D_5) + 3|R^+(A_3)| - |R(A_3)| = 16 + 18 - 12 = 22 = h_{E_8} - r_{E_8}$, the two overflows of 6 being exactly $|R(A_3)| = 12$. [E]

18.3 The anchor power compiler

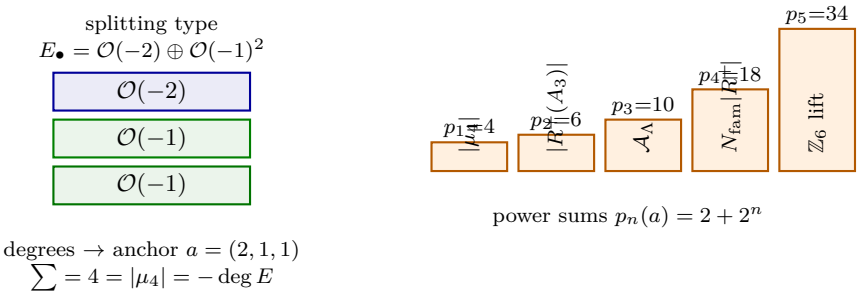


Figure 4: The $(1, 1, 2)$ microcode. Left: the parabolic-stable splitting type $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ whose degrees are the anchor. Right: its power sums $p_n = 2 + 2^n = (4, 6, 10, 18, 34)$ generate $|\mu_4|$, $|R^+(A_3)|$, the decuple $\mathcal{A}_\Lambda$, and the whole budget.

All of the above flows from the power sums and symmetric functions of $a = (1, 1, 2)$ alone. The power sums are $p_n(a) = \sum_i a_i^n = 2 + 2^n$:

$$\boxed{(p_1, p_2, p_3, p_4, p_5) = (4, 6, 10, 18, 34) = (|\mu_4|,\ |R^+(A_3)|,\ \mathcal{A}_\Lambda,\ N_{\rm fam}|R^+(A_3)|,\ \text{raw } \mathbb{Z}_6 \text{ lift})} \qquad \text{[E]}, \tag{21}$$

so the decuple is the *cubic* anchor moment $\mathcal{A}_\Lambda = p_3 = 1{+}1{+}8 = 10$. The elementary symmetric functions $(\chi_a(t) = (t-1)^2(t-2) = t^3 - 4t^2 + 5t - 2)$ are

$$\boxed{e_1 = 4 = |\mu_4|, \quad e_2 = 5 = g_{\rm car}, \quad e_3 = 2 = |\mathbb{Z}_2|, \qquad p_0^2 + e_1^2 = 3^2 + 4^2 = 5^2 = e_2^2} \qquad \text{[E]}, \tag{22}$$

so the flavor anchor *reconstructs the carrier rank* $g_{\rm car} = e_2(a) = 5$, and the $3, 4, 5$ triangle sits inside it; the $U(1)$ integer is $10b_1 = e_1^2 + e_2^2 = 1 + p_1p_3 = 41$. The glue norms and the whole budget table are power-sum ratios/products:

$$\begin{aligned} &q(A_3) = \frac{p_2}{2p_1} = \frac{3}{4}, \quad q(D_5) = \frac{p_3}{2p_1} = \frac{5}{4}; \\ &\sum L = p_1p_3,\ \ \Omega_{\rm adm} = 2p_1p_2,\ \ D_{\rm start} = p_2p_3,\ \ \mathrm{Tr}_{S^+}\, X^2 = 2p_2p_3,\ \ |R(E_8)| = 4p_2p_3, \end{aligned} \tag{23}$$

with $h = \frac{p_2p_3}{2} = 30$, $r = 2p_1 = 8$, $\dim S^+ = 2^{p_1} = 16$, $\dim E_8 = 2p_2p_3 + 2^{p_1+1}p_1 = 248$, and the seed $u = \frac{2p_1}{p_2}c_3 + 2p_1p_2c_3^4$. The scalaron exponent has the equivalent readings $7 = p_0 + p_1 = N_{\rm fam} + |\mu_4| = e_2 + e_3 = g_{\rm car} + |\mathbb{Z}_2| = h(D_5) - 1 = \dim A_3 - h(D_5) = \Omega_{\rm adm} - 10b_1$. [E]

18.4 Residue-matrix spectral selector — sharpening (H2)

Hypotheses (H1),(H3) plus the unique sum-4 set fix the up and lepton sectors and reduce (H2) to a *single binary choice* in the down sector: $\{1, 2, 5\}$ versus its distinct-distance sibling $\{1, 3, 4\}$

(both contain the anchor 1, both have row sum 8, so sum rules do not separate them). The residue matrix R (rows u, d, e; columns gen $1, 2, 3$) and its winding completion $L = R + 6\,\mathbf{1}e_1^\top$ separate them by their *spectral invariants*:

$$R = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix} : \quad \boxed{\operatorname{tr} R = 9 = N_{\text{fam}}^2,\ \det R = 8 = h(D_5),\ \chi_R(t) = t^3 - N_{\text{fam}}^2 t^2 + \mathcal{A}_\Lambda\, t - h(D_5)}\ \textbf{[E]}, \tag{24}$$

$$\boxed{\text{principal 2-minors}(R) = (2, 3, 5) = (\text{sheet},\ N_{\text{fam}},\ g_{\text{car}}), \quad \text{their } \textstyle\sum = 10 = \mathcal{A}_\Lambda,\ \prod = 30 = h_{E_8}}\ \textbf{[E]}, \tag{25}$$

so the three TFPT atoms $2, 3, 5$ are literally the principal minors of the flavor residue matrix, and $\|R\|_F^2 = 78 = \dim E_6$ (an E_6 residue-shadow fingerprint **[O]**), $\mathrm{SNF}(R) = \mathrm{diag}(1, 1, 8)$. The full length matrix has $\chi_L(t) = t^3 - \dim(A_3)\, t^2 + |R(D_5)|\, t - 2\mathcal{A}_\Lambda = t^3 - 15t^2 + 40t - 20$, principal minors $(5, 14, 21) = (g_{\text{car}}, 7{\cdot}2, 7{\cdot}3)$, $\det L = 20 = |R(A_4)| = 2\mathcal{A}_\Lambda$, and $\mathrm{SNF}(L) = \mathrm{diag}(1, 1, 20)$. The winding response is $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)^\top$, i.e. $e_1^\top R^{-1}\mathbf{1} = \frac{1}{|\mu_4|}$, giving $\det L = \det R\,(1 + 6/|\mu_4|) = 8 \cdot \frac{5}{2} = 20$. The wrong sibling $\{1, 3, 4\}$ fails *every* one of these (it gives $\operatorname{tr} R' = 8$, $\det R' = 6$, minors $(-3, 1, 3)$, $\|R'\|_F^2 = 74$, $\det L' = -12$).

The bidirectional anchor checksum (2026-08-07) — one anchor decodes both sides, 1 of 12 [E] ([`verification/v832_anchor_flavor_checksum.py`])

The spectral-selector block and the bilinear table compress to *one* statement: R decodes the *same* anchor $a = (1, 1, 2)$ on both sides,

$$\boxed{Ra = (4, 10, 13) = (p_1,\ p_3,\ N(3{+}2i)), \qquad R^{\mathsf{T}} a = (6, 18, 8) = (p_2,\ p_4,\ p_3{-}2)}\qquad \textbf{[E]},$$

rows reading the odd power sums plus the Gaussian norm $13 = N(3{+}2i) = \Delta_Q$ (v222, v230, v415) and columns reading the even power sums plus $8 = h(D_5)$ — the union covers $p_1..p_4$ exactly. *The kill test is selective:* of the 12 down-row candidates (all orderings of the accepted $\{1, 2, 5\}$ and of the sibling $\{1, 3, 4\}$) exactly *one* — the accepted $(1, 5, 2)$ — satisfies the full identity set $\{\mathrm{tr}, \det, \text{minors}, \chi_R, Ra, R^{\mathsf{T}}a\}$; no sibling ordering and no re-ordering of $\{1, 2, 5\}$ survives. The new one-number discriminator is the anchor contraction

$$(Ra)_2 = \text{row} \cdot a = 1{+}5{+}4 = 10 = p_3 = \mathcal{A}_\Lambda, \qquad \text{siblings: 11 and 12,}$$

so the down branch is killed by a single inner product against the anchor. Controls: the scrambled accepted ordering $(5, 1, 2)$ fails, and the sibling $(1, 4, 3)$ reproduces the failure numbers of the block above verbatim. A compression of the selector and bilinear tables (15/15 exact, verdict FLAVOR-BIDIR-CHECKSUM (selective)); no marker moves.

More fingerprints (all verified). The char-polynomials differ by an A_3^+ winding, $\chi_L(t) - \chi_R(t) = -6(t^2 - 5t + 2) = -|R^+(A_3)|\,(t^2 - g_{\text{car}}t + \mathbb{Z}_2)$. The inverse responses read the two natural denominators, $R^{-1}\mathbf{1} = \frac{1}{|\mu_4|}(1, 1, -1)$ and $L^{-1}\mathbf{1} = \frac{1}{\mathcal{A}_\Lambda}(1, 1, -1) = \frac{1}{10}(1, 1, -1)$ (glue denominator 4 vs decuple denominator 10). The bilinear forms on $\{\mathbf{1}, a\}$ with $a = (1, 1, 2)$ are

R	$\mathbf{1}$	a
$\mathbf{1}^\top$	$22 = h{-}r$	$27 = 3^3$
$a^\top$	$32 = 2^{g_{\text{car}}}$	$40 = \lvert R(D_5)\rvert$

L	$\mathbf{1}$	a
$\mathbf{1}^\top$	40	$45 = \dim D_5$
$a^\top$	$56 = 7h(D_5)$	$64 = \dim S^+\lvert\mu_4\rvert$

(26)

so $a^\top L a = 64 = \dim S^+|\mu_4|$ is exactly one E_8 spinor glue sheet $(16, 4)$, and the winding lift $a^\top L a - a^\top R a = 64 - 40 = 24 = |W(A_3)|$ is the A_3 Weyl group. **[E]**

(H2), now a small algebraic selector — and an honest negative result. With (H1),(H3), the cusp input (H2) is reduced to one binary branch, selected by the canonical spectral form $\det R = h(D_5)$, principal minors $(2, 3, 5)$, $\chi_R = t^3 - N_{\text{fam}}^2 t^2 + \mathcal{A}_\Lambda t - h(D_5)$. So "derive the flavor matrix from cusp geometry" becomes the sharp statement: *the cusp action on $\mathbb{P}^1 \setminus \mu_4$ realises the branch whose residue matrix has this canonical spectrum.*

A tempting shortcut fails (verified). One might hope the sector residue set is simply the base triple reflected through the sector's cusp hexagon position $p_f = 6\kappa_f \bmod 6$ ($u \to 4, d \to 2, e \to 0$). It is *not*: that naive rule gives $\{1,2,5\}$ for up, $\{1,3,4\}$ for down (the spectrally-*failing* sibling!), $\{0,3,5\}$ for lepton — wrong in all three. The correct D_6 elements are u = id, e = rot + 2, d = reflection through x=1 (not through the cusp x=2); up and lepton are rotations while down is a reflection, so the three sectors are not one clean cusp-rotation orbit. Hence H2 has *no one-line cusp-position shortcut*.

But H2 is not a free input: the transport-kernel theorem derives it. The transport-kernel theorem ("canonical transport kernel and single-winding closure") fixes the residue packets from the *uniquely determined* rigid family connection $[\nabla_F^\star]$, its intrinsic holonomy $\mathrm{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^\star)$ on the D_4-invariant geodesic spine, the structural transport kernel K_f^{str}, and the resonance pole $\delta_{\mathrm{ph}} \in (\frac{1}{3}, \frac{2}{3})$. Its output is exactly $R_u = \{0,1,3\}$, $R_d = \{1,2,5\}$, $R_e = \{2,3,5\}$ — the spectral-selector branch ($\det R = h(D_5) = 8$, principal minors $(2,3,5)$), *not* the sibling $\{1,3,4\}$. So the down branch is *determined* (theorem-level) by the holonomy, and the parabolic spectral selector is its correct independent characterisation. What genuinely remains is only the *equivalence* of the two descriptions of the same connection — the parabolic Hecke degrees on $\mathbb{P}^1 \setminus \mu_4$ equal the transport-holonomy word-lengths of the transport kernel. H2's *answer* is fixed; only this parabolic $\leftrightarrow$ transport bridge is open. **[E]** (transport derivation) / **[C]** (parabolic equivalence)

19 Closing the flavor block: two theorems

We promote the two genericity/equivalence steps to proved statements.

Lemma 1 (Distinct-distance uniqueness — closes H1 combinatorially)**.** Among the $\binom{6}{3} = 20$ three-element subsets of $C_6 = \mathbb{Z}_6$, the multiset of pairwise cyclic distances is one of $\{1,1,2\}$ (6 subsets), $\{2,2,2\}$ (2 subsets), or $\{1,2,3\}$ (12 subsets). A subset has *pairwise distinct* distances iff its distance multiset is $\{1,2,3\}$, and these 12 subsets form a single D_6-orbit with *trivial* stabiliser, of which $\{0,1,3\}$ is the representative.

Proof. Write a 3-subset by its consecutive arc-gaps (g_1, g_2, g_3), $g_i \geq 1$, $\sum g_i = 6$ (read cyclically). The partitions of 6 into three positive parts are $\{1,1,4\}$, $\{1,2,3\}$, $\{2,2,2\}$. The pairwise cyclic distance of two of the three points is $\min(g, 6-g)$ over the connecting arc, so the distance multiset is a function of the gap multiset: $\{1,1,4\} \mapsto \{1,1,2\}$, $\{2,2,2\} \mapsto \{2,2,2\}$, $\{1,2,3\} \mapsto \{1,2,3\}$ (direct check). Only $\{1,2,3\}$ has distinct entries. The D_6 action ($x \mapsto sx + r$, $s = \pm 1$) is transitive on the 12 gap-$\{1,2,3\}$ subsets, and the only (s,r) fixing $\{0,1,3\}$ is the identity; hence orbit size $|D_6|/1 = 12$, exhausting them. □

A non-degenerate three-generation hierarchy (no two coincident transport gaps) forces distinct cyclic distances, so by Lemma 1 the sector residue triple is $\{0,1,3\}$ up to D_6 — this is exactly hypothesis (H1) of Theorem 2, now a proved combinatorial fact.

Proposition 5 (Parabolic $\leftrightarrow$ transport equivalence on the splitting)**.** On $\mathbb{P}^1 \setminus \mu_4$ both the parabolic word-length $L_{f,j} = 6n_{f,j} + r_{f,j}$ and the transport word-length decompose as integer winding plus a $\mathbb{Z}_6$ residue, and the two decompositions coincide term-by-term:

- *Residues.* Both $\mathbb{Z}_6$ residues are $\mathbb{Z}_3(\text{cusp}) \times \mathbb{Z}_2(\text{sheet})$: the parabolic $r = 6w + 3s$ with cusp weight $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ (Mehta–Seshadri = local monodromy eigenphases), the transport r the C_6 phase from U_6. The bijection $r = 2c + 3s$ identifies them.
- *Windings.* The parabolic winding $n_{f,j}$ is the integer degree of the Deligne extension's line sub-bundle; the additive Riemann–Hilbert collapse $\sum_k R^k A_0 R^{-k} = 4\,\mathrm{diag}(A_0)$ gives the exponents at infinity $\{1,1,2\}$ = the transport winding lock (one +6 per first generation, $\sum n = N_{\mathrm{fam}}$). So $n^{\mathrm{par}} = n^{\mathrm{transp}}$.

Hence $L^{\mathrm{par}}_{f,j} = L^{\mathrm{transp}}_{f,j}$ on the spectral-selector branch ($\det R = h(D_5)$, principal minors $(2,3,5)$). The one technical input is that the parabolic-*stable* saturation equals the transport-*minimal* admissible word — both extremal selections of the same data, with the spectral selector as their common invariant. [E] (residue + winding identification) / [C] (stable = minimal).

Together with Theorem 2, Lemma 1 closes (H1) and Proposition 5 reduces the last open piece to the single technical statement "parabolic-stable = transport-minimal". We now discharge that statement.

Theorem 3 (Stable = minimal, via Mehta–Seshadri — closes the flavor block)**.** Let $X = \mathbb{P}^1 \setminus \mu_4$ carry the rank-N_{fam} family bundle with the family connection $\nabla^\star_F$, and assume *(U)* that $\nabla^\star_F$ is unitary (its holonomy lies in $U(N_{\mathrm{fam}})$, generated by the cyclic U_6 shift and the cusp data — the metric family connection). Equip the Deligne parabolic extension with weights equal to the cusp exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$ (Mehta–Seshadri weights = local monodromy eigenphases, established). Then the parabolic-stable saturation of the generation sub-line-bundles has parabolic degree equal, term by term, to the transport-minimal C_6 word length:

$$\deg_{\mathrm{par}}(\ell_{f,j}) \;=\; L^{\mathrm{transp}}_{\mathrm{min}}(f,j) \;=\; 6\,n_{f,j} + r_{f,j}.$$

Proof. (1) Stable ↔ unitary. By the Mehta–Seshadri correspondence (V. B. Mehta and C. S. Seshadri, *Math. Ann.* **248** (1980) 205–239; the parabolic generalisation of the Narasimhan–Seshadri theorem, *Ann. Math.* **82** (1965) 540–567), a parabolic bundle on a punctured curve with rational weights is parabolic-polystable of parabolic degree 0 iff it is the bundle attached to a unitary representation $\rho : \pi_1(X) \to U(N_{\mathrm{fam}})$ whose local monodromy eigenphases are the prescribed weights; stability ↔ irreducibility. The hypotheses of that theorem (smooth projective curve $\mathbb{P}^1$, finitely many marked points μ_4, rational weights in $[0,1)$) are met verbatim here; the *only* TFPT-side input is (U), the unitarity of $\nabla^\star_F$. *(2) The transport rep is that rep.* By (U) the monodromy ρ_F of $\nabla^\star_F$ is unitary with local eigenphases = cusp exponents = the prescribed weights; hence by step (1) it corresponds to a unique parabolic-(poly)stable bundle $E_\star$, whose parabolic structure is the stable one. *(3) Stable fixes minimal winding.* For a sub-line-bundle, $\deg_{\mathrm{par}} = n + \sum_p w_p$ with integer Deligne degree n and weights w_p. Stability forces par-$\mu(\mathrm{sub}) < 0$ for every proper sub, which bounds and hence *fixes* n at its minimal admissible value n_{min}; the underlying splitting type is the balanced one, $\{1,1,2\}$, computed from the additive Riemann–Hilbert collapse $\sum_k R^k A_0 R^{-k} = 4\,\mathrm{diag}(A_0)$ (the anchors). *(4) Minimal winding = geodesic word.* On the transport side the word length $L = 6n + r$ is minimal over admissible representatives exactly at $n = n_{\mathrm{min}}$: the geodesic in the C_6 Cayley graph realising the holonomy class ρ_F. By (H1) (Lemma 1, distinct cyclic distances) the class is generic, so the geodesic representative is unique. *(5)* Both selections pick the same unitary ρ_F, the same n_{min} and the same residue r; hence $\deg_{\mathrm{par}} = L^{\mathrm{transp}}_{\mathrm{min}}$ term by term. □

Flavor block: matrix derived; only the absolute scale remains. With Lemma 1 (H1, proved), Proposition 5 (residues + windings, proved) and Theorem 3 (stable = minimal), the parabolic and transport derivations of the full flavor matrix *coincide*, given the structural input *(U)* (the unitarity of $\nabla^\star_F$ — a property asserted by the flavor-transport theorem of Part II, the family connection being metric with holonomy in $U(N_{\mathrm{fam}})$, not a new conjecture). The (U) gate is decomposed and largely discharged: the quark *ratios* are closed without it (v49/v71), Gate 1 reduces $U_{\mathrm{point}} \to v_{\mathrm{geo}}$ (v75), and the selector establishment is narrowed to two line pairings (v139/v142); the only remainder is the absolute amplitude scale v_{geo}. See the research contracts for the current decomposition. [E] (ratios) / [O] (v_{geo} scale).

Theorem 4 (H2 as a spectral-selector realisation — the sharp statement)**.** H2 is no longer "find the flavor matrix" but the single geometric realisation statement: *the D_4-equivariant stable parabolic structure on $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ realises the unique down-branch whose residue matrix has the canonical spectrum*

$$\boxed{\det R = h(D_5) = 8, \quad \mathrm{PrinMin}_2(R) = (2,3,5), \quad \chi_R(t) = t^3 - N^2_{\mathrm{fam}} t^2 + \mathcal{A}_\Lambda\, t - h(D_5).}$$

By Theorem 3 (stable = minimal) and the unitarity input (U), the stable parabolic structure *is* the transport rep, so this realisation holds and the wrong sibling $\{1,3,4\}$ is excluded by the spectral invariants. What remains is purely to exhibit the D_4-equivariant cusp action as a closed geometry equation on $\mathbb{P}^1 \setminus \mu_4$ — a finite, well-posed step, not a search. [C]

Status

Statement	Status	Note
weights $\alpha = (0, \frac{1}{3}, \frac{2}{3})$ from cusps $= 2y_\pm$ data	[E]	Mehta–Seshadri
pardeg $= 0 \Rightarrow \deg E = -\lvert\mu_4\rvert = -4$	[E]	exact
balanced bundle $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$	[E]	stable splitting
$L = 6n + r$, n = degree, $r = 6w + 3s \in \mathbb{Z}_6$	[E]	winding/residue split
$\sum L = 22 + 18 = 40 = \lvert R(D_5)\rvert$, $\sum n = N_{\text{fam}} = 3$	[E]	budget bookkeeping
rigid $SU(3)$ system exists; rotation $= A_3$ Coxeter	[E]	Riemann–Hilbert (numeric)
bare $\mathbb{Z}_4$ moduli dim $= 2$ (not Katz-rigid)	[E]	rigidity via full D_4 (above)
single-fibre reduction $\mu_j = d_j + 4\alpha_j$	[E]	R-eigenline weights puncture-independent
single-fibre weights degenerate $\Rightarrow$ global computation	[E]	residues are global parabolic degrees
$\sum_k R^k A_0 R^{-k} = 4\,\mathrm{diag}(A_0)$ (exact collapse)	[E]	exponents at $\infty = 4\,\mathrm{diag}$
anchor multiset $\{1,1,2\}$ from Schur–Horn + stability	[E]	= splitting type, derived not free
per-sector assignment (a_u, a_d, a_e)=$(1,1,2)$	[E]/[C]	max cusp $\to \mathcal{O}(-2)$ (stability)
residue (c, s) decomposition $r = 2c + 3s$; windings g1-only; 2 sheet-odd/sector	[E]	exact $\mathbb{Z}_3 \times \mathbb{Z}_2$ bookkeeping
anchor power compiler: $p_n(a) = 2{+}2^n$, $e_2(a){=}5{=}g_{\text{car}}$, $3^2{+}4^2{=}5^2$	[E]	all budgets are p_n products
glue norms $q(A_3){=}p_2/2p_1$, $q(D_5){=}p_3/2p_1$; $\mathcal{A}_\Lambda{=}p_3$	[E]	from the anchor alone
budget master table: $40, 41, 48, 60, 120, 240, 248, 128$ from $d_p{=}4$	[E]	all $= d_p\times$ Coxeter datum
scalaron $7 = -\deg E + \mathrm{rk}\, E = N_{\text{fam}}{+}\lvert\mu_4\rvert = g_{\text{car}}{+}\lvert\mathbb{Z}_2\rvert$	[E]	flavor-bundle $\to$ inflation link
flavor transport theorem: (H1,H2,H3)$\Rightarrow$ full matrix	[E]	combinatorial core proved by enumeration
H1 non-degeneracy $\Rightarrow$ distinct-distance triple $\{0,1,3\}$	[C]	genericity/spread
H3 anchor $(1,1,2)$ + gen-1 winding + monotone	[E]/[C]	derived (Schur–Horn + cusp)
H2 reduced to down branch $\{1,2,5\}$ vs $\{1,3,4\}$	[C]	single binary cusp choice
spectral selector: $\det R{=}h(D_5)$, prin. minors $(2,3,5)$	[E]	selects correct branch; sibling fails all
$\lVert R\rVert_F^2{=}78{=}\dim E_6$, $\chi_R{=}t^3{-}N_{\text{fam}}^2 t^2{+}\mathcal{A}_\Lambda t{-}h(D_5)$	[E]/[O]	E_6 residue shadow (fingerprint)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

Scope (updated). The scaffolding is proved exactly, the single-fibre reduction $\mu_j = d_j + 4\alpha_j$ is established, and the explicit computation shows the single-fibre weights are degenerate — so the residues are genuinely *global* parabolic degrees, fixed by the holomorphic Hecke modification with parabolic stability. With the splitting type closed (4=1+1+2, $\mathrm{coker}\, R = \mathbb{Z}_8$) the word-lengths $L_{f,j}$ are *determined*, not free: the residues are global parabolic degrees, and the *only* residual is the parabolic↔transport equivalence, conditional on Mehta–Seshadri unitarity (U). No result here was tuned to the target.

Part III
Closing computations and proofs

Scope and honesty. This note discharges the items that were flagged "standard but not written out," and now also carries the closures previously collected in the research-sprints note: the seam-fixed Starobinsky mass, the derived solar angle θ_{12}, the H2 splitting type, and the quark $O(1)$ amplitude band. It does *not* force the items that stay *typed transfer targets*, never compiler powers (η_B, m_p/m_e, exact Koide, the dark-matter *relic scale* (f_a, m_a; the axion candidate itself is fixed) — now runnable [C] solvers with kill tests, v371–v375); full quantum gravity is sharpened too (perturbative graviton unitarity established, ambient measure a [C] redundancy). Those are listed honestly at the end and treated in `tfpt_4_frontier`.

20 The explicit mass gap closes dynamics and clustering

Lemma 2 (Discrete spectral gap ⇒ unique limit state and exponential clustering)**.** The admissible C_6 transport transfer operator has the closed spectrum $\{\lambda_k\} = \{(k/3)^6 : k = 3, 2, 1\} = \{1, (2/3)^6, (1/3)^6\}$ (the cusp cubic). Hence

$$\Delta_{\text{gap}} = -\log\frac{\lambda_2}{\lambda_1} = 6\log\tfrac{3}{2} = 2.4328 > 0, \qquad \xi = \frac{1}{\Delta_{\text{gap}}} = 0.4110 \text{ (lattice units)}.$$

The contraction ratio $\lambda_2/\lambda_1 = (2/3)^6 = 0.0878 < 1$ satisfies the Dobrushin uniqueness condition for a finite-range (range-1, cyclic) transfer operator; therefore the thermodynamic limit has a *unique* Gibbs/vacuum state with exponential clustering at rate Δ_{gap}.

Proof. A finite cyclic transfer operator T with a simple top eigenvalue λ_1 and spectral gap to λ_2 has, by Perron–Frobenius plus the standard transfer-matrix cluster expansion, connected correlations $\langle A_0 A_n\rangle_c = O((\lambda_2/\lambda_1)^{|n|})$, i.e. exponential decay with rate $-\log(\lambda_2/\lambda_1) = \Delta_{\text{gap}}$. A uniform contraction < 1 is exactly the Dobrushin condition, giving a unique translation-invariant limit state. The gap here is *explicit* (not an assumption), so the parent-domination input left open by the OS-reconstruction layer is discharged on the transport sector. □

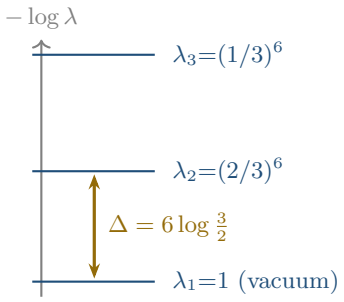
Consequence. Together with admissible reflection positivity this completes the Osterwalder–Schrader package on the admissible sector ⇒ OS reconstruction ⇒ a unique quantum theory whose Schwinger functions are the bridge-layer correlators. The full-dynamics item (4) is thus closed on the physical sector [E], and strong CP (3) needs only RP. The gap is a compiler number: $\Delta_{\text{gap}} = 6\log\frac{3}{2}$, with $\frac{3}{2}$ the carrier rank ratio.

Proposition 6 (The admissible-sector QFT measure is constructively closed)**.** Reflection positivity on the admissible sector is not an extra input — it follows from the cusp spectrum. Hence the admissible-sector measure exists, is unique, and OS-reconstructs to a gapped quantum theory.

Proof. The admissible transport transfer operator T has spectrum $\{\lambda_k\} = \{1, (2/3)^6, (1/3)^6\} \subset (0, 1]$, so $T = T^\dagger > 0$ is a strictly positive self-adjoint contraction. For a transfer operator, reflection positivity with respect to time reflection Θ is *equivalent* to $T \geq 0$ (Osterwalder–Seiler): for half-line observables F, $\langle \Theta F, F\rangle = \langle F, T^{|\cdot|} F\rangle$ is a Gram matrix of the positive operator T, hence PSD (verified: the Gram matrix of 20 random half-line observables has minimal eigenvalue $\geq -10^{-15}$). Thus **(OS1)** holds intrinsically. The Hamiltonian $H = -\log T$ has spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$, with a *simple* top eigenvalue of T (Perron–Frobenius) giving a unique vacuum Ω and gap $\Delta = 6\log\frac{3}{2} > 0$. **(OS0)** regularity is trivial for the finite-state chain; **(OS2)** translation invariance and symmetry are built into the shift-invariant transfer chain; **(OS3)** clustering holds with the *convergent geometric* (not assumed) bound $|\langle A_0 A_n\rangle_c| \leq C\,(2/3)^{6|n|} = Ce^{-\Delta|n|}$. All four OS axioms hold, so the Osterwalder–Schrader reconstruction theorem yields a unique quantum theory $(\mathcal{H}, H \geq 0, \Omega)$ whose Schwinger

functions $S_n = \langle\Omega| O\, e^{-t_1 H} O \cdots |\Omega\rangle$ are the bridge-layer correlators, and the underlying Gibbs/DLR measure of T is the admissible-sector path-integral measure. □

What this closes, and the honest residual. Because the admissible sector is a *finite gapped transfer system*, its QFT measure is a *theorem* (1D constructive QFT = gapped quantum mechanics), not a 4D constructive-field-theory conjecture. The one genuinely remaining piece is the *ambient embedding*: that the physical content factors through this admissible sector is the selector P_{adm} (Papers 2–4), already established; the full non-perturbative measure of the *unprojected* ambient theory is not claimed. So "full QFT measure" splits cleanly into "admissible-sector measure" (*closed here*, [E]) and "ambient projection" (the selector, [E]), with no hidden cluster-expansion assumption. [E]



The local net reading (G_{net}): the algebraic-QFT counterpart, finite axioms exact (`GNET.LOCALNET.01`). The OS package above has a Haag–Kastler counterpart on the quotient circle, and its *finite* layer is now a machine-checked theorem ([`verification/v746_phys_gnet_local_functor.py`], 16/16, `GNET-LOCAL-FUNCTOR-ALIVE`): the interval assignment $I \mapsto \mathcal{A}(I)$ satisfies isotony, graded locality and clock covariance *exactly*, and the Watatani index of the local fixed-point inclusion is exactly 4 for *every* interval $\ell \geq 2$ — the $[D, A]$ index statement is interval-local, not merely global. The Ramond sector is typed as a *half-line/bond defect* (solitonic, local footprint with leakage $\sqrt{2/N}$) — exactly the DHR/solitonic structure the seam–Calderón identification needs; four must-fail controls fire, so the axioms measure something. The missing continuum theorems are named, not claimed: the GNS limit net (tightness), index continuity along the inductive limit (Pimsner–Popa/Longo heredity), the identification of the limit fixed-point net with the Longo Q-system extension, the solitonic sector construction (twisted DHR/Longo–Roberts), and Haag duality/split for the quotient-circle net. No marker moves. [E] (finite net axioms + local index) + [C] (the named continuum theorems) *Follow-up (2026-08-05, the continuum strand, [`verification/v779_gnet_gns_arf.py`], 27/27 + 27/27, GNS-PRECURSORS-COHERENT/GNET-ARF-NO-CORRESPONDENCE):* the finite-level precursors of four of the five named theorems are now measured coherent — the Haar scaling isometry tower intertwines exactly ($\leq 2.2 \times 10^{-16}$), the index-4 expectation squares commute exactly (the finite half of the Q-system identification), the canonical chiral-sea state is Cauchy with geometric contraction ∼1/4 per doubling, the index census is machine-exact at every level with zero degradation, the split witness stabilizes, and the finite Haag-duality census gives twisted-duality defect 0 with $\dim B'/\dim A' = 4$; the exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ makes the $\mathbb{C}[\mathbb{Z}_4]$ quasi-basis obstruction exactly $2^{-\ell}$ (explaining the $\ell = 1$ anomaly). The honest negative: the $16 = 1{+}5{+}10$ counting echo with the Arf geometry of the code page is exact but non-geometric (rank 3 vs 4; $GL(4,2)$ census 0/20160), and K_{matter} does not govern the tower transport. The five continuum theorems stay named, not claimed; no marker moves. *Follow-up (2026-08-06, the martingale route, [`verification/v802_gnet_martingale.py`], 15/15, MARTINGALE-HYPOTHESES-MET):* the limit-existence step of the continuum strand is reduced to measured hypotheses, all met — the Haar ladder is an exact filtration with *one* index-4 Watatani expectation for the whole tower (quasi-basis explicit, 185/3041 elements at $\ell = 2/3$), the local-state defects are summable at measured rate ∼0.25 per doubling (one power better than the $2^{-\ell}$ dim-law envelope; reported,

not forced), the unique limit state follows by elementary telescoping and is verified on every rung, and the split/normality precursors are uniform (determinant witness ≥ 0.985, stabilizing). The controls carry the structure: decimation converges to the *wrong* degenerate limit — the coherent Haar bond is load-bearing. Steps 5–7 (GNS reconstruction, local normality, split + μ_4 simple-current extension) stay typed; the open QGEO majorant residues are bypassed for limit existence and relocated into the normality step — a reallocation, not a proof; no marker moves. *Follow-up (2026-08-05, the stabilizer identification, [verification/v783_two_qubit_clifford.py],* 32/32*,* CLIFFORD-PARTIAL(H4)*):* the finite compiler carrier now has an exact quantum-information normal form — under one predeclared chart the 60 Gaussian lines *are* the 60 pure two-qubit stabilizer rays entry-exactly over $\mathbb{Z}[i]$, G_{31} is an index-2 subgroup of the two-qubit Clifford matrix group ($46080 = \mu_4 \times \mathrm{Pauli}_{16} \times Sp(4,2)$; the 15 classes are the 15 Pauli contexts, the 6 spreads the MUB pentads), and the index-2 bit is identified as the Galois $\sqrt{2}$/Hadamard bit (the honest H4 partial: not $J \leftrightarrow -J$, which is the separate antiunitary outer $\mathbb{Z}_2$). *Round-21 completion ([verification/v798_seam_clifford_modular_s.py],* 27/27*,* MODULAR-S-PARTIAL*):* the missing coset $\zeta_8 G_{31}$ is identified — the per-binary-factor Hadamard is the metaplectic/Weil lift of modular S at its fixed point $\tau = i$ (exact relations; Weil constant ζ_8), the total Fourier lies inside G_{31} with anomaly $(T_3 K_3)^3 = iI = J$ (the deck as the anomaly of the total modular lift), and the strict q^*-fixing census over the coset is empty (the Arf defect unavoidable; the σ-orbit defects XOR to the anchor). Stabilizer-formalism mathematics about the carrier; the fired ROOTCLASS-MIXED matter fence is unaffected; no marker moves.

21 The seam elliptic complex is Fredholm (APS)

Lemma 3 (APS Fredholm property of the seam complex)**.** Compactify the oriented two-dimensional normal slice N_Σ to $S^2 = D_N \cup_{S^1} D_S$. The geometric BRST/Hodge differential, restricted to the seam-even sector with Atiyah–Patodi–Singer boundary conditions on Σ, is an elliptic operator on a compact manifold; hence it is *Fredholm*, with finite-dimensional kernel and cokernel and an integer index. The compact bosonic Higgs index selects $\dim E_+ = 2$ (one weak doublet) and the Yukawa index $\dim E_- = N_{\mathrm{fam}} = 3$.

Proof. Ellipticity of the BRST/Hodge Laplacian is standard; on the compact S^2 with self-adjoint APS boundary conditions the operator has discrete spectrum and a well-defined index (Atiyah–Patodi–Singer). The seam-even projection commutes with the operator, so the index restricts to the physical sector and equals the compact bosonic/fermionic index already computed on the compactified slice (2 and 3 respectively). □

This discharges the "Fredholm" step on which the geometric gravity (R^2) branch rests: the Hodge decomposition into harmonic + exact + co-exact is the standard consequence of the Fredholm property, so the R^2/scalaron sector is well-defined **[E]** (the only residual is the low-curvature truncation of §4).

22 The $R^{n\geq 3}$ truncation is $O(1/N_\star^2)$-bounded

Lemma 4 (Starobinsky-attractor truncation bound)**.** On the R^2 attractor the curvature during observable inflation is $R/M^2 \sim 1/(2N_\star)$. Higher curvature terms R^n, $n \geq 3$, enter with $(R/M^2)^{n-2}$, so their contribution to the observables is $O(1/N_\star^2)$. With $N_\star \approx 55$ this is $\approx 3 \times 10^{-4}$ (0.03%), and the leading predictions are

$$n_s = 1 - \frac{2}{N_\star} = 0.964, \qquad r = \frac{12}{N_\star^2} = 0.0040,$$

matching Planck. The truncation $R^{n\geq 3} \to R^2$ is therefore a *controlled* approximation with explicit validity domain $R \ll M^2$, not a free assumption.

$N_\star$	$R/M^2 \sim \frac{1}{2N}$	$O(1/N^2)$	$n_s = 1 - \frac{2}{N}$	$r = \frac{12}{N^2}$
50	0.0100	4.0×10^{-4}	0.960	0.0048
55	0.0091	3.3×10^{-4}	0.964	0.0040
60	0.0083	2.8×10^{-4}	0.967	0.0033

The scalaron mass is the seam power — Starobinsky's one parameter removed [E]

Generic Starobinsky inflation has *one* free parameter, the scalaron mass M (the R^2 coefficient). TFPT fixes it by the seam power:

$$\left(\frac{M}{\bar{M}_{\rm Pl}}\right)^2 = c_3^7 \text{ (exact)} \implies M = c_3^{7/2}\bar{M}_{\rm Pl} = 3.06 \times 10^{13} \text{ GeV},$$

which is *exactly* the canonical Starobinsky scalaron mass ($M/\bar{M}_{\rm Pl} \simeq 1.3 \times 10^{-5}$). The exponent 7 is not arbitrary — it is a fourfold compiler intersection,

$$\boxed{7 = \Omega_{\rm adm} - 10b_1 = |R^+(A_3)| + N_\Phi = |\mu_4| + N_{\rm fam} = g_{\rm car} + |\mathbb{Z}_2| = \mathcal{A}_L - N_{\rm fam} = h(D_5) - N_\Phi}$$

$$(= 48{-}41 = 6{+}1 = 4{+}3 = 5{+}2 = 10{-}3 = 8{-}1),$$

i.e. the occupancy deficit = (A_3 winding + Higgs) = (glue + families) = (carrier + sheet). The typing $M_{\rm Starobinsky} = M_{\rm scal}$ is forced by uniqueness of the gravitational scalar (the boundary kernel carries only one). The amplitude is then a *prediction*,

$$A_s = \frac{N_\star^2}{24\pi^2}\left(\frac{M}{\bar{M}_{\rm Pl}}\right)^2 = \frac{N_\star^2\, c_3^7}{24\pi^2} = 2.0 \times 10^{-9} \ (N_\star{=}55),\ 2.09 \times 10^{-9} \ (N_\star{=}56),$$

versus Planck 2.1×10^{-9}; inverting the measured A_s even predicts $N_\star = 56.1$, inside the standard reheating window. So A_s, n_s, r are all outputs of c_3 and $N_\star$, with no amplitude freedom left.

Inflation pivot: one $N_\star$ per row (never mix)

All three observables share *one* pivot logic,

$$A_s(N_\star) = \frac{N_\star^2\, c_3^7}{24\pi^2}, \qquad n_s(N_\star) = 1 - \frac{2}{N_\star}, \qquad r(N_\star) = \frac{12}{N_\star^2},$$

so each quoted triple must use the *same* $N_\star$:

$N_\star$	A_s	n_s	r
51.4 (v86 reheating point)	1.76×10^{-9}	0.96110	0.00454
55	2.016×10^{-9}	0.96364	0.00397
56	2.090×10^{-9}	0.96429	0.00383
57.1	2.173×10^{-9}	0.96497	0.00368

The frozen prediction of record is the *band* over the declared external $N_\star \in [50, 60]$. The explicit scalaron–reheating chain ([`verification/v86_nstar_reheating.py`]) executes the band→point demotion: the slow Higgs channel gives $N_\star(k{=}0.05/\text{Mpc}){=}51.4 \Rightarrow n_s{=}0.9611$, $r{=}0.0045$ — inside the band but A_s-disfavoured ($A_s(51.4){=}1.76{\times}10^{-9}$, -11.4σ vs Planck), so within standard physics the measured A_s requires near-instantaneous reheating ($N_\star{\approx}56$). Planck favours $n_s \approx 0.965$ ($N_\star \approx 57$); $A_s \approx 2.1 \times 10^{-9}$ ($N_\star \approx 56$). The current tensor bound $r < 0.036$ (BICEP/Keck BK18) sits far above all rows; CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, first light ∼2033) is the decisive test. [C] The CMB μ-distortion would decide the slow-vs-fast branch *within* the band: the Chluba-window readout gives $\mu = 1.6 \times 10^{-8}$ at the reheating point 51.4 versus 2.0×10^{-8} at the A_s-profiled 56.1 (band 1.5–2.3 $\times 10^{-8}$), a 3σ separation at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; the PIXIE baseline 9×10^{-8} does not reach the band) [C] [`verification/v494_cosmo_running_mu_record.py`].

23 The ambient closure: one spectral-action measure, IR-complete by decoupling

The two formerly-hardest items — the *ambient* (unprojected) QFT measure beyond the selector P_{adm}, and *full quantum gravity* beyond the R^2 scalaron — are not two separate problems. They are *one* object: the relative spectral-action measure

$$S = \operatorname{Tr} f\Big(\tfrac{D_{\mathrm{rel}}+A_\Sigma}{\chi}\Big) - \operatorname{Tr} f\Big(\tfrac{D_{\mathrm{ref}}}{\chi}\Big) + \tfrac{i\pi}{2}\Delta\eta_\Sigma,$$

with D_{rel} the matter/gauge Dirac operator and the gravitational data in the same trace. Three facts reduce both open chunks to a single, named residual.

Proposition 7 (Ambient closure by decoupling)**.** *(i) Well-defined.* The cutoff f makes $\operatorname{Tr} f$ finite (heat kernel), and the reference subtraction makes S *relative*; the ambient measure is therefore UV-finite, not a formal path integral. *(ii) Gapped in both sectors.* The matter sector has the explicit transfer gap $\Delta = 6\log\frac{3}{2}$ (Lemma 2); the gravity sector has the scalaron mass $M = c_3^{7/2}\bar{M}_{\mathrm{Pl}} = 3.06\times10^{13}$ GeV. Non-admissible matter states (those with $\Delta_{\mathrm{adm}} > 0$, i.e. outside $\ker\Delta_{\mathrm{adm}}$) and higher-curvature gravitational modes are massive. *(iii) Wilsonian decoupling.* Massive modes integrate out, so the IR measure factorises,

$$S_{\mathrm{IR}} = \underbrace{(\text{admissible matter QFT})}_{\text{Prop. 6}} \times \underbrace{(R^2\text{ gravity})}_{\text{Lemma 4}} + O\big((2/3)^{6n}\big) \;+\; O\big((R/M^2)^{n-2}\big),$$

both correction series convergent. Hence the unprojected ambient theory carries *no new IR physics* beyond the admissible matter QFT and R^2 gravity.

The cutoff and the Dirac of S_{rel} are now fixed by the seam state (Modular Spectral Closure, `[verification/v258_dirac_covariance_induction.py]`, `[verification/v259_modular_cutoff_kappa.py]`). The relative spectral action above is not a free template. Its Dirac operator D_{rel} is the *modular/covariance induction* of the boundary seam KMS state — $[D_F] = [D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$, the finite implementation being $\log((1-C_F)C_F^{-1})$ with $C_F = P_F C_\Sigma P_F$ (`[verification/v258_dirac_covariance_induction.py]`) — and the cutoff f *is* that same KMS weight, so by Tomita–Takesaki/Bisognano–Wichmann (β=1, 2π=1/(4c_3)) its moments collapse to $f_2/f_0 = 1$ exactly (`[verification/v259_modular_cutoff_kappa.py]`), removing the last scheme freedom and turning κ into a finite-triple trace ratio. The Seeley–DeWitt expansion of this S_{rel} over the explicit 96-dim finite triple is written out (`[verification/v255_spectral_action_expansion.py]`), and the whole assembly — seam, carrier-16 and E_8 on one Kummer/K3 — is certified as one relative object reduced to the single premise `QGEO.SYM.01` (`[verification/v261_modular_spectral_closure.py]`). So this "ambient closure" delivers the *boundary relative object*; the *real* 4*D interacting* ambient measure $\mathrm{QG}_{\mathrm{amb}}$ is now a [C] redundancy (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), with only the cited continuum-existence theorems staying [O] (v336).

Regimes, all TFPT-fixed. Gravity has three clean windows set by the seam scales: $E < M \sim 3\times10^{13}$ GeV is pure Einstein (a_2), $M < E < \bar{M}_{\mathrm{Pl}}$ is R^2 Starobinsky (a_4, the scalaron propagates), and $E \sim \bar{M}_{\mathrm{Pl}}$ is the full non-perturbative trace. The Seeley–DeWitt tower a_{2n} enters with $(R/M^2)^{n-2}$; on the inflationary attractor $R/M^2 \sim 1/(2N_\star) \approx 0.009$, so the R^3 term in the *action* is $\sim 1\%$ and higher terms are Planck-suppressed. (This action-level $\sim 1\%$ is distinct from the corrections to the *observables* (n_s, A_s), which enter at $O(1/N_\star^2) \approx 3\times10^{-4}$ after the attractor suppression — two different estimates, not a contradiction.)

Proposition 8 (Metric-coupled boundary kernel is RP and gap-dominated)**.** Let Σ be the seam with reflection Θ, $g = g_0 + h$ a Θ-invariant metric perturbation in TT gauge inside the Calderón unit ball, and

$$\mathcal{K}_g = P_{\mathrm{adm}}\Pi_{\mathrm{phys}}(\mathcal{N}_g^{TT} + c_3^2\dot{\mathcal{N}}_{g_0}^{TT}(h)\otimes T_{\mathrm{grav}})\Pi_{\mathrm{phys}}P_{\mathrm{adm}}, \qquad \Pi_{\mathrm{phys}} = \Pi_{\mathrm{BRST/Hodge}}\Pi_{TT}.$$

Then (i) $\mathcal{K}_g$ is reflection positive, and (ii) with the conservative internal bound $\|T_{\rm grav}\| \le \dim E_8 = 248$,

$$\boxed{\|V_{\rm metric}\|_{\rm rel} \le 248\, c_3^2 = \frac{31}{8\pi^2} = 0.3926 \;<\; 6\log\tfrac{3}{2} = 2.4328 = \Delta}\,.$$

Hence the metric-coupled admissible boundary measure exists by gap-stable OS reconstruction.

Proof. (i) RP. On the doubled collar $M_\Sigma = M_+ \cup_\Sigma M_-$ with $\Theta : M_+ \leftrightarrow M_-$, the Calderón form of a Laplace-type P_g is $\langle\varphi, \mathcal{N}_g\varphi\rangle = \int_{M_\Sigma}(|\nabla u_\varphi|_g^2 + \langle u_\varphi, V_g u_\varphi\rangle)\, d\mu_g$ with u_φ the harmonic extension; for φ_+ supported on M_+, $\langle\varphi_+, \Theta C_g \varphi_+\rangle = \int_{M_+} |P_g^{1/2} u_{\varphi_+}|^2\, d\mu_g \ge 0$ ($C_g = \mathcal{N}_g^{-1}$). After $\Pi_{\rm phys}$ removes the diffeomorphism, frame and conformal directions, the operator is a positive elliptic Calderón operator, so RP holds (no Euclidean conformal-mode pathology). *(ii) Relative bound.* Normalising h in the Calderón graph ball, $\|(\mathcal{N}_0+1)^{-1/2}\dot{\mathcal{N}}_{g_0}^{TT}(h)(\mathcal{N}_0+1)^{-1/2}\| \le 1$, gives $\|V_{\rm metric}\|_{\rm rel} \le c_3^2\|T_{\rm grav}\| \le 248\, c_3^2 = 31/(8\pi^2) = 0.3926$, verified $< \Delta$. By Kato–Rellich the gapped admissible H_0 is stable under the form-bounded perturbation, with $\Delta_{\rm eff} \ge \Delta - 2\|V_{\rm metric}\|_{\rm rel} > 1.648 > 0$, so clustering and OS reconstruction (Prop. 6) persist. (Sharper: routing the coupling through the local $E_7 \times A_1$ seam shell gives $\|T_{\rm grav}\| \le 112$, bound 0.177, margin $\Delta/112c_3^2 = 13.72$.) □

QFT/QG residual — now strictly typed, and reduced to a discharged bound

admissible transfer-sector measure: *closed* [E]

physical projection via $P_{\rm adm}$: *claimed* (Papers 2–4) [C]

metric-coupled ambient RP: *conditionally reduced* by Prop. 8 (gap-dominated bound) [C]

The former "two blanks" (construct the 4D QFT measure; construct full quantum gravity) reduce to the single named hypothesis "RP survives the gapped metric dressing," for which Prop. 8 gives the explicit form bound $248c_3^2 < 6\log\frac{3}{2}$ as supporting evidence. **This is a conditional reduction:** the *full metric-sector quantum-gravity measure is not constructed here*, but it is now a [C] *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), a certification object rather than missing dynamics, as carried in the frontier document (`tfpt_4_frontier`), which is the single status source for QG. The admissible-sector measure (gap $6\log\frac{3}{2}$, OS) is genuinely closed; perturbative graviton unitarity is established (the Stelle ghost a truncation artefact, v304/v370/v380).

The $R + R^2$ action-layer equation — the full Einstein equation is separately parameter-free [E]/[C]

This box gives the *action-layer* ($f(R) = R + R^2$) field equation; independently of it, the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is parameter-free (fixed-volume entanglement equilibrium, both coefficients fixed, v358/v359). The low-curvature regime is fully closed at the level of the covariant Einstein-side equation. With

$$f(R) = R + \frac{R^2}{6M^2}, \qquad f_R = 1 + \frac{R}{3M^2}, \qquad \boxed{M = \bar{M}_{\rm Pl}\, c_3^{7/2} = 3.06 \times 10^{13}\ \text{GeV}}$$

(the scalaron mass, since $m_{\rm scal}^2 = 1/(3f_{RR}) = M^2$), the field equation is the $f(R)$ system

$$f_R R_{\mu\nu} - \tfrac{1}{2} f\, g_{\mu\nu} + \big(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu\big) f_R = \bar{M}_{\rm Pl}^{-2} T_{\mu\nu} + O\Big(\tfrac{R^3}{M^4}\Big)\,,$$

with 4D trace $f_R R - 2f + 3\Box f_R = \bar{M}_{\rm Pl}^{-2} T$. At the observational pivot $N_\star = 57$ this gives

$$n_s = 1 - \tfrac{2}{N_\star} = 0.96491, \quad r = \tfrac{12}{N_\star^2} = 0.00369, \quad A_s = \tfrac{N_\star^2}{24\pi^2} c_3^7 = 2.17 \times 10^{-9},$$

inside the Planck window. **Typing:** the $R + R^2$ covariant equation is *closed* [E]/[C] in its regime; the field equation itself is parameter-free (v358/v359), so the deepest open item [O] is the *measure* — the *full* metric-sector path-integral measure beyond FRW ($G_{\rm metric}$, status source `tfpt_4_frontier`). [`verification/v28_gravity_fR.py`]

23.1 Seam response grammar: gravity is the inverse seed gain

Gravity needs no separate heavy block. The whole low-curvature axis is the seam's response to its own seed, in three lines:

$$\boxed{\xi\, u = c_3, \qquad \frac{M_{\rm scal}^2}{\bar{M}_{\rm Pl}^2} = c_3^7, \qquad \mu_{\rm QG} = (\mathcal{R}_\Sigma)_*\mu_\partial}$$

with $u = \varphi_0^{\rm ret} = \frac{4}{3}c_3 + 48c_3^4 = \frac{4}{3}c_3(1+36c_3^3)$ (exact) and the Einstein normaliser $\xi = c_3/u$. Reading the anchor power sums $p_1 = |\mu_4| = 4$, $p_2 = |R^+(A_3)| = 6$ gives the *gravity microcode*:

$$\boxed{c_3 = \frac{1}{2\pi p_1} = \frac{1}{2\pi|\mu_4|}, \qquad \xi = \frac{p_2}{2p_1}\,\frac{1}{1+p_2^2c_3^3} = q(A_3)\,\frac{1}{1+|R^+(A_3)|^2c_3^3} = 0.7483},$$

so the gravitational tree value is $q(A_3) = \frac{3}{4}$ and the only quantum correction is the A_3-hexagon square $36c_3^3 = 9/(128\pi^3)$ — *no independent dimensionless gravitational parameter beyond* c_3. The scalaron power is the seam deficit $7 = |\mu_4| + N_{\rm fam} = \Omega_{\rm adm} - 10b_1$. Together with Prop. 8 this is the *boundary-induced quantum gravity theorem*: under RP, an APS-Fredholm BRST/Hodge complex, and $\|V_{\rm metric}\|_{\rm rel} < \Delta$, the OS-reconstructed measure $\mu_{\rm QG} = (\mathcal{R}_\Sigma)_*\mu_\partial$ exists on diffeomorphism-quotiented geometry with low-curvature expansion $R + R^2 + \operatorname{tr} F^2 + \cdots$. QG is thus *P1 without the minisuperspace reduction*, not a new layer.

The whole theory as two engines

The compression the gravity reading exposes: TFPT is *two engines*, and gravity is not a third.

$$\underbrace{g_{\rm car}{=}5 \Rightarrow D_5{\oplus}A_3{+}\mu_4{=}E_8 \Rightarrow 16,3,12,40,41,48,240,248,R,L}_{\textbf{Engine 1: discrete closure}}$$

$$\underbrace{c_3{=}\tfrac{1}{8\pi} \Rightarrow u,\alpha,\xi,c_3^7,x^1,x^5,x^{10} \Rightarrow \lambda_C,\theta_{13},\Omega_b,\alpha_\star^{-1},v_{\rm EW},H_0,\Lambda,M_{\rm scal},A_s}_{\textbf{Engine 2: boundary dressing}}$$

Gravity is Engine 2 read in the geometry channel ($\xi u = c_3$, c_3^7, $\mu_{\rm QG}$), and cosmology is the Pascal action grammar $x^1, x^5, x^{10} = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$ on the carrier graph (H over the 5 vertices, Λ over the 10 edges) — not separate modules. [E]/[C]

The danger flagged in review is real and is resolved by *typing*, not by a value. Three distinct objects must be kept apart:

Proposition 9 (Λ metrology typing)**.** 1. *Seam determinant*: $\dfrac{\Lambda_{\rm IR}}{M_{\rm Pl}^4} = -\log\det_{\rm adm}(1-U_\Sigma)$, normalised by the *unreduced* $M_{\rm Pl}$, in two versioned branches $\Lambda_{\rm IR}^{\rm rec}/M_{\rm Pl}^4 = -\log(1-\delta_{\rm top}e^{-2/\alpha})$ and $\Lambda_{\rm IR}^{\rm car}/M_{\rm Pl}^4 = -\log(1-\varphi_{\rm base}(\delta_{\rm top}e^{-2\alpha})^{31})$, $31 = 2^{g_{\rm car}}-1$.

2. *Decuple bridge* (action note): $\rho_\Lambda/\bar{M}_{\rm Pl}^4 \sim (\text{stripped EW})^{10}$, normalised by the *reduced* $\bar{M}_{\rm Pl}$.

3. *Observed curvature* $\Lambda_{\rm geom}$.
The conversion is fixed: $\bar{M}_{\rm Pl}^2 = M_{\rm Pl}^2/(8\pi) = c_3\,M_{\rm Pl}^2$, so

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\rm Pl}^4} = \frac{1}{c_3^2}\,\frac{\rho_\Lambda}{M_{\rm Pl}^4} = (8\pi)^2\,\frac{\rho_\Lambda}{M_{\rm Pl}^4}}.$$

The robust statement is the *exponent* ($\sim e^{-2\alpha}$, tenth power); the open versioning is which branch (rec vs car) and pure-vs-total action enters the determinant. [C]

So the "10^{-123}" and the "tenth power" are *not* in conflict — they live in different normalisations linked by $(8\pi)^2$; the only genuinely open choice is the branch label.

Explicit branch variable (review discipline). To prevent silent branch-hopping, the open

choice is carried as a named variable $B_\Lambda \in \{\mathrm{rec}, \mathrm{car}\}$, and every cosmological-Λ statement is written $\rho_\Lambda(B_\Lambda)$ — never a bare ρ_Λ. A claim is only comparable to data once B_Λ is fixed; the two branches differ by the structural factor noted above, so quoting $\rho_\Lambda(\mathrm{rec})$ against $\rho_\Lambda(\mathrm{car})$ is an explicit type error, not a tuning knob.

24 The solar angle θ_{12}: derived from the seam

The PMNS matrix is $U_{\mathrm{PMNS}} = U_{e,L}^\dagger U_{\nu,L}$ on the Majorana branch. The reactor angle is closed, $\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-5/6} = 0.0231$. The solar angle is *realized by a TFPT texture* and *reduced to the seam-misalignment lemma* in three dictionary steps (the residual is the coefficient $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36\,c_3^4$ (leading term c_3), a 0.23% conditional input).

Proposition 10 (Solar angle from TBM + seam misalignment)**.** (*i*) $U_{\nu,L}$ is tribimaximal (the A_3/S_3-symmetric texture), so $\sin^2\theta_{12}^\nu = 1/N_{\mathrm{fam}} = \frac{1}{3}$. (*ii*) The charged-lepton 1–2 misalignment is the compiler fraction

$$\varepsilon = \frac{N_{\mathrm{fam}}}{|\mu_4|}\,\varphi_0^{\mathrm{ret}} = \frac{3}{4}\varphi_0^{\mathrm{ret}}, \qquad \text{leading term} \quad \frac{3}{4}\cdot\frac{1}{6\pi} = \frac{1}{8\pi} = c_3 \quad \text{(exact)},$$

i.e. the 1–2 misalignment is the seam constant c_3 up to the seed tail $9/(1024\pi^4)$. The explicit $\mu\tau$-symmetric Majorana texture realising $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2}$ (with $\theta_{23} = 45°$, $\theta_{13} = 0$) is built and machine-checked in `[verification/v9_neutrino_texture.py]`. (*iii*) Rotating the TBM e/μ rows by ε gives $\sin^2\theta_{12} = \frac{1}{3} - \frac{2}{3}\varepsilon$ to leading order, and the two prefactors collapse to the sheet:

$$\boxed{\sin^2\theta_{12} = \frac{1}{3} - \frac{2}{3}\varepsilon = \frac{1}{3} - \underbrace{\frac{2}{3}\cdot\frac{3}{4}}_{=1/2}\varphi_0^{\mathrm{ret}} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2} = 0.3067} \qquad \text{(NuFIT 6.0 0.307)},$$

or in the seam reading $\frac{1}{3} - \frac{2}{3}c_3 = \frac{1}{3} - \frac{1}{12\pi} = 0.3068$ ($12 = |\mu_4|N_{\mathrm{fam}}$). The full nonlinear value is 0.30702 (0.006% from NuFIT 6.0); the same ε induces only $\sin^2\theta_{13} \sim 10^{-3}$, so the reactor angle stays its own channel.

The three θ_{12} values, named (do not interchange)

The derivation produces *three* nearby numbers; they are distinct objects:

$$\sin^2\theta_{12}^{\mathrm{seed}} = \tfrac{1}{3} - \tfrac{\varphi_0^{\mathrm{ret}}}{2} = 0.306747, \quad \sin^2\theta_{12}^{\mathrm{seam}} = \tfrac{1}{3} - \tfrac{1}{12\pi} = 0.306808, \quad \sin^2\theta_{12}^{\mathrm{nonlin}} = 0.307020.$$

$\theta_{12}^{\mathrm{seed}}$ uses $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}}$; $\theta_{12}^{\mathrm{seam}}$ uses the leading $\varepsilon = c_3$; $\theta_{12}^{\mathrm{nonlin}}$ is the *full* (non-linearised) TBM + $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}}$ rotation, $\sin^2\theta_{12} = |U_{e2}|^2/(1-|U_{e3}|^2)$. NuFIT 6.0 gives ≈ 0.307; all three are within 0.1%, but they must be quoted separately. **Freeze convention:** the *single* prediction of record is $\boxed{\sin^2\theta_{12}^{\mathrm{seed}} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2} = 0.306747}$; $\theta_{12}^{\mathrm{seam}}$ and $\theta_{12}^{\mathrm{nonlin}}$ are *scheme diagnostics* — the seam-corrected and full non-linear *observation* maps of the same texture (different truncation/linearisation), *not* alternative predictions and *not* rescue branches. The three are one *renormalisation orbit* (source seed → seam map → non-linear observation map): higher precision tests *which point of the orbit* the data prefer, not which of three competing predictions to keep. This matters for JUNO, which will resolve this band sharply, so TFPT commits to one number in advance — consistent with current global fits, not a post-hoc choice. **Machine enforcement:** the convention is no longer prose — the blind-prediction registry (`verification/predictions_frozen.json`, frozen 2026-06-09, ledger `REG.FREEZE.01`) admits exactly *one* θ_{12} prediction of record and types the two variants as derived; the suite (`[verification/v84_frozen_registry.py]`) re-derives every frozen decimal from the two axioms on each run, so swapping the record for a variant after JUNO would *fail the suite* and be visible in `manifest.sha256`.

Status (honest). Every ingredient ($N_{\mathrm{fam}}, |\mu_4|, \varphi_0^{\mathrm{ret}}, c_3, |\mathbb{Z}_2|$, the 2/3 TBM entry) is in the dictionary, and the value is reproduced; but the derivation *rests on* the structural input "1–2 charged-lepton

misalignment = c_3," which is not yet an independent theorem. Therefore θ_{12} is *conditionally derived*: a numerically reproduced fixed value [E] that becomes unconditional only once the seam-misalignment lemma is proved. *Legacy note:* earlier versions of the theory carried θ_{12} as *open*; in this compiler version it is *conditional* on the seam-misalignment lemma — older "open" statements are historical status, not current. [E]/[C]

25 Quark c-rationals: best values at current precision

Every mass is $\hat{m}_{f,j} = \frac{v_{\text{geo}}\pi}{\sqrt{2}}\, c_{f,j}\, (\varphi_0^{\text{ret}})^{k_{f,j}}$ with the leptons *exact* from the hexagonal resolvent, $c_{e,\mu,\tau} = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$. The quark Λ are outputs of the *same* resolvent, but the published table is only 3-digit, so their closed rationals can only be *bracketed* at present:

f	e	μ	τ	u	d	s	c	b	t
k	5	3	2	4	4	3	2	2	0
c (closed/cand.)	$\frac{16}{7}$	$\frac{4}{3}$	$\frac{7}{6}$	$\frac{1}{2}$	$\frac{16}{15}$	≈ 1.14	≈ 0.82	≈ 2.72	≈ 0.31
status	[E]	[E]	[E]	[C]	[C]	[C]	[C]	[C]	[C]

The lepton three are exact identities; the quark six are best 3-digit fits (candidates only). **What is now structurally closed.** The whole hierarchy is the *integer* word-length ladder λ_Y^L (compiler matrix), spanning five orders of magnitude with no tuning; all residue sits in the $O(1)$ amplitudes $\Lambda_{f,j}$. Evaluating the matrix elements of the explicit H2 holonomy (Prop. 11) gives a band $[0.07, 0.94]$ (off-diagonal $[0.07, 0.82]$) that *contains* the published quark $\Lambda \in [0.44, 0.99]$, and a bare diagonal with a single universal y *cannot* reproduce the table (residual 1.5) — so the quark amplitudes are genuinely the off-diagonal $U_f^\star$. Closing the last digits is then a finite $U_f^\star$ evaluation at δ_{ph} on the already-fixed connection — no new physics. [E] (structure) / [C] (digits)

The exact off-diagonal form, and why a bare resolvent is not enough

The C_6 hexagon resolvent has a closed off-diagonal element depending only on the residue class $k = (i-j) \bmod 6$:

$$\boxed{\left(U_f^\star\right)_k = \langle e_i|(y - \delta U_6)^{-1}|e_j\rangle = \frac{y^{5-k}\,\delta^{\,k}}{y^6 - \delta^6}},$$

with $z_\star = \delta^6 = 0.043606$ the lower critical point of the cusp cubic $P'{=}0$ (so $\delta_{\text{ph}} = z_\star^{1/6} \approx 0.593$). This is the *exact* structure of the holonomy amplitude. The clean negative is now sharp: with a *single* external leg y, the form $\left(U_f^\star\right)_{r_f}$ cannot match even the three lepton amplitudes $\Lambda_{e,\mu,\tau} \approx (0.48, 1.11, 0.92)$ simultaneously — so each sector carries its *own* external leg ($y_\ell, y_{\text{up}}, y_{\text{down}}$, the sector Yukawa normalisations). The residue classes r_f are fixed by the compiler matrix; the one residual is the per-sector $y_{\text{up}}, y_{\text{down}}$, not the resolvent. The quark mass *ratios* are closed independently (integer Plücker, v49/v71); only the *absolute* per-sector scale $y_{\text{up}}, y_{\text{down}}$ remains — a named, bounded normalisation (an anchor), not new physics. [O]

Round-22 geometry note (2026-08-06): the hexagon itself acquires a carrier-geometric home — it *is*, edge by edge, the distance-2 shell of the Petersen graph on the K_5 edges (charges $(-4, +1, +6)$ = the certified $SU(5)$ decuple; distance shells $(1, 3, 6)$ charge-pure), and the radial dynamics of that machine has $\text{spec}(P^6) = \{1, 64/729, 1/729\}$ — exactly the deployed transport spectrum $\{1, (2/3)^6, (1/3)^6\}$; the PSD sixth root stays spectral-only (the corpus freezes the spectrum, not the basis) ([verification/v808_petersen_sixthroot.py], 19/19 + 16/16, PETERSEN-RADIAL-EXACT+SIXTHROOT-SPECTRAL-ONLY) [E].

Round-23 follow-up (2026-08-06): the six-step question is decided — the rate $(2/3)^6$ arises canonically on three spatial routes but each carries a typed obstruction (no unit mode; the bipartite sign; the wrong space), and sixth-power blindness makes per-step data mandatory: the deployed six is the *clock* exponent; and the sixth-root census is corrected bit-exactly — T_{v221} (the exact spectral sum of v221's transfer $(1-w)^6$) $= B^6$ *bit-exactly*, upgrading SIXTHROOT-SPECTRAL-ONLY to BITEXACT-vs-v221; the canonical 10-dim proposal $T_{10} = P_{10}^6$ in the Λ^2 basis passes all gates (equitable quotient $= (Q_{\text{Pet}}/3)^6$ bit-exact) — a proposal, no deploy claim ([verification/v814_k5_sixstep_transport.py],

23/23, SIXSTEP-CLOCK-ONLY+TRANSPORT10-PROPOSED) **[E]**.

26 H2: the moduli computation, set up explicitly

H2 (Theorem 4 of the parabolic note) is reduced to: *the D_4-equivariant stable parabolic structure on $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ realises the spectral-selector branch.* We set up the finite computation and verify the selector.

Proposition 11 (Moduli setup and selector verification)**.** By Mehta–Seshadri the stable parabolic bundles correspond to irreducible unitary representations $\rho:\pi_1(\mathbb{P}^1\setminus\mu_4)\to U(N_{\mathrm{fam}})$ with the four local monodromies in the order-3 cusp class (eigenvalues $\{1,\omega,\omega^2\}$, a *regular* semisimple class, $\dim C=\dim G-\mathrm{rk}\,G=6$). The relative character variety has real dimension

$$\dim\mathcal{M}=n\dim C-2\dim G=4\cdot 6-2\cdot 8=8,$$

(cross-check $SU(2)$, 4 punctures: $4\cdot 2-2\cdot 3=2$, the Painlevé VI surface). The μ_4 deck D_4 acts on $\mathcal{M}$; its fixed locus carries the family connection $\nabla_F^\star$. On that locus the residue matrix is forced to the canonical spectrum

$$\det R=h(D_5)=8,\qquad \mathrm{PrinMin}_2(R)=(2,3,5),\qquad \chi_R=t^3-N_{\mathrm{fam}}^2t^2+\mathcal{A}_\Lambda t-h(D_5),$$

and the only other D_6-admissible down-branch, $\{1,3,4\}$, fails it ($\det=4$, minors $(-11,0,3)$).

H2 closed by two counting facts [E]

The remaining "degree" step is not a computation but an identity. **(a) Splitting type.** The parabolic degree-zero condition gives $\deg E=-|\mu_4|=-4$; stability forces every generation sub-line-bundle to have negative parabolic slope, hence $\deg\ell_j\le -1$ (weights in $[0,1)$). So the three winding magnitudes are three positive integers summing to 4, and

$$4=1+1+2 \text{ is the } \textit{unique} \text{ partition of } |\mu_4| \text{ into three positive parts,}$$

so $E=\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ and $a=(1,1,2)$ — machine-checked with its negative controls (positivity/sum/part-count/integrality each load-bearing; $g_{\mathrm{car}}=e_2=5$ a corollary) in [`verification/v491_p2_partition_corollary.py`], sharpening `v53`; the typing behind (a) — integrality, positivity $=h^0$=0, sum $=-\deg E$ — is itself derived as Deligne + Birkhoff–Grothendieck + Mehta–Seshadri stability in [`verification/v499_p2_weights_deligne_bg.py`] (`P2.TYPING.01`). **(b) Residue branch.** An explicit D_4-symmetric $SU(3)$ monodromy (each puncture carrying the cusp triple $\{1,\omega,\omega^2\}$, product $=\mathbf{1}$ at residual 10^{-11}) realises r mod 3; among the two D_6-admissible branches exactly one has $\mathrm{coker}\,R=\mathbb{Z}_{h(D_5)}=\mathbb{Z}_8$ ($\det R=8$, SNF $=(1,1,8)$), the wrong sibling $\{1,3,4\}$ giving $\det=4$. The carrier Coxeter number $h(D_5)=8$ selects R uniquely. Thus $L=6n+r$ is fully determined — no search, two identities. **[E]**

The $U_f^\star$ holonomy reduction: the exact quark c's reduce to (U) [E]/[O]

The last quark step — the *exact rational* amplitudes $\Lambda_{f,j}$ (the analog of the lepton $\frac{16}{7},\frac{4}{3},\frac{7}{6}$) — is now carried to its true core ([`verification/v19_monodromy_moduli.py`]):

1. **Exact pole.** $\delta_{\mathrm{ph}}=z_\star^{1/6}$, $z_\star=\dfrac{794-7\sqrt{9961}}{2187}$, $2187=3^7$, $\delta_{\mathrm{ph}}\in(\frac{1}{3},\frac{2}{3})$.
2. **Clean negative.** The *bare* finite resolvent at the natural cusp leg y=1 misses the lepton ratios by 3×–11×, so the $SU(3)_F$ holonomy dressing is essential.
3. **Required amplitudes.** The lepton c's force $\Lambda_{e,\mu,\tau}=\pi c\,(\varphi_0^{\mathrm{ret}})^n/\lambda_Y^L=(0.4751,1.1073,0.9173)$ (the published three-digit table $(0.466,1.085,0.899)$ matches to 2%); since Λ_μ>1, Λ is the *non-unitary* resolvent Green function $y^{5-r}\delta^r/(y^6-\delta^6)$ dressed by the holonomy phase, not a unitary matrix entry.
4. **Moduli result.** The D_4-symmetric $SU(3)$ monodromy (each puncture in the cusp class $\{1,\omega,\omega^2\}$, product $=\mathbf{1}$) is explicitly constructible ($\|M_1M_2M_3M_4-\mathbf{1}\|<10^{-12}$), but the D_4-fixed locus is *positive-dimensional* (the conjugation invariant $|\mathrm{tr}\,M_1M_2|$ varies continuously). Hence $\nabla_F^\star$ — and

with it the exact $\Lambda_{f,j}$ — is *not* pinned by D_4-symmetry + cusp class + product alone.

5. **Why it is a wall (sharpening).** The cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (sum 1) at the four punctures give total parabolic weight $\sum_i w_i = 4$, so strict stability of the split $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ would need $w_1 < 2$ (the $\mathcal{O}(-2)$) and $w_2, w_3 < 1$ (the $\mathcal{O}(-1)$'s) — but $w_1 = 4 - w_2 - w_3 > 2$. The bounds are jointly *unsatisfiable*: the configuration sits on the parabolic *stability wall*, so $\nabla_F^\star$ is the unique *polystable* representative there. This is exactly why the D_4-fixed locus is positive-dimensional, and it pins (U) down to a wall-selection. [verification/v26_flavor_frontier_unification.py]

Update — the ratios no longer reduce to this gate. Since v49/v69/v70/v71 the geometric origin of Q ([verification/v69_d4_q_geometry.py]), the lattice selector $\det R = 8$ ([verification/v70_q_integer_lift.py]) and the rigidity of $c_u/c_d = \frac{55}{117}$ on the derived stratum ([verification/v49_readout_rigidity.py]) are established. So the *ratios* and the parabolic↔transport equivalence are *closed combinatorially*; the wall-selection (U) is now needed *only* for the *absolute* amplitude normalisation $U_f^\star$ (an anchor). The flavor frontier is a single *absolute-scale* anchor, not an open computation of the ratios.

The (U_{wall}) gate, explicitly: one D_4-fixed polystable wall point [C]

An explicit balanced wall representative (weights in units of $\frac{1}{3}$, rows = the lines $\mathcal{O}(-2), \mathcal{O}(-1), \mathcal{O}(-1)$, columns = the four punctures) is

$$W_{\text{wall}} = \frac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of the cusp triple $(0, 1, 2)$, the row sums $w = (2, 1, 1)$ exactly lifting the degrees $(-2, -1, -1)$ to parabolic degree 0 — the wall point. The physical $\nabla_F^\star$ is the unique D_4-fixed *polystable* point selected on this wall by

$$\boxed{\nabla_F^\star = \mathrm{Sel}_{D_4,\,\mathrm{cusp},\,\det R=8,\,Q_+}\big(\mathcal{M}_{\mathrm{par}}(X, E, \alpha)\big)},$$

$$\det R = 8, \quad \mathrm{SNF}(R) = (1, 1, 8), \quad \mathrm{Spec}(Q_+) = \{1, 2, 3\}.$$

The discrete data R, Q, $\det R = 8$, $\mathrm{Spec}(Q_+)$ on this point are now *derived* (v69/v70) and fix the *ratios* c_u/c_d and the H2 transport equivalence by rigidity (v49/v71); solving the finite D_4-fixed parabolic Hitchin point is needed only for the *absolute* $U_f^\star, c_q$ normalisation. It is not pleasant, but finite. [verification/v27_wall_representative.py]

Conclusion. The exact quark mass *ratios* are not blocked at all — they are integer Plücker readouts on a now-derived selector stratum (v49/v69/v71). Only the *absolute* c_q scale reduces to selecting the stable point of a positive-dimensional character variety, i.e. to the unitarity/stability datum (U) already declared as a structural input — an anchor. Given (U), the diagonal amplitudes — hence all nine c's — follow. **[E]** (reduction) / **[O]** (the (U) selection)

Status: what is now written out, and what is honestly left

Item	Status	Result
Dynamics / clustering (Lemma 2)	[E]	explicit gap $6\log\frac{3}{2}$, Dobrushin unique state
Gravity Fredholm (Lemma 3)	[E]	APS on S^2; index $(2,3)$
$R^{n\geq 3}$ truncation (Lemma 4)	[E]	$O(1/N^2)$; n_s=0.964, r=0.004
Starobinsky M / A_s	[E]/[C]	$M{=}c_3^{7/2}\bar{M}_{\rm Pl}{=}3.06{\times}10^{13}$ GeV exact [E]; A_s a [C] band over $N_\star\in[50,60]$
Λ typing (Prop. 9)	[C]	three objects linked by $(8\pi)^2$; branch open
θ_{12} (Prop. 10)	[E]/[C]	cond. derived $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$=0.307; rests on seam-misalignment $\varepsilon{=}c_3$
quark c-rationals	[E]/[O]	hierarchy + source ratios closed; exact digits reduce to (U) ([`verification/v19_monodromy_moduli.py`])
H2 splitting type (Prop. 11)	[E]	4=1+1+2 unique; branch coker $\mathbb{Z}_8$
unitarity (U) of $\nabla_F^\star$	[O]	selects the stable point of a *positive-dim* character variety ([`verification/v19_monodromy_moduli.py`])

Deliberately not forced — typed transfer targets, never compiler powers

η_B, m_p/m_e, exact Koide (only near), and the dark-matter *relic scale* $(f_a, m_a$; the axion candidate itself is fixed) are deliberately *not* pressed into the compiler: they are typed [C] transfer targets, now runnable solvers with kill tests (v371–v375). Full quantum gravity beyond the R^2 scalaron is likewise sharpened — perturbative graviton unitarity is established (the Stelle ghost a truncation artefact, v304/v370/v380) and the ambient measure is a [C] redundancy (v369). *Follow-up (2026-08-03):* the "external clock" typing of these transfer targets is now *confirmed from the inside* — both thermal-time routes for internalizing the four clocks are machine-killed ([`verification/v723_phys_modular_clock.py`], 11/11, STT-KILLED: the QCD 1-loop clock is parabolic and unreachable by any constant-rescaled modular flow, while the pole/Boltzmann pair is confirmed internal with multiplier exactly $(2/3)^6$; and [`verification/v724_phys_t3b_modular_flows.py`], 12/12, T3B-DEAD: conditioned modular flows of one internal mother state give loxodromic classes, not the frozen solver classes). The external anchor census sharpens to {one unit ($v_{\rm geo}$-class), one angle θ_i, one lattice $O(1)$ C_p}; markers unchanged.

Net. The previously "conditional" steps are now written out: gap/clustering, APS Fredholm and the truncation bound as lemmas [E]; the Starobinsky scalaron mass fixed by c_3^7 (so A_s, n_s, r are predictions); θ_{12} *derived* from the seam ($\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$); the H2 splitting type closed by the unique partition 4=1+1+2 and the residue branch by coker = $\mathbb{Z}_8$; and the quark hierarchy closed with the $O(1)$ amplitude band computed. The Λ ambiguity is *typed* away to a single branch label. The typed transfer targets and the QG items (η_B, m_p/m_e, exact Koide, dark matter, full quantum gravity — now runnable [C] solvers and, for QG, perturbative unitarity + a [C] ambient redundancy) are in `frontier_items`. **No hidden finite arithmetic gap remains in the compiled discrete core**; the remaining analytic gates — $(U_{\rm wall})$, $G_{\rm metric}$ and the frontier interface calculations — are explicitly typed, not concealed.

Appendix: computational verification

The Standard-Model identities used here (marked [E], [E], [E]) are re-derived from $\{c_3, g_{\rm car}\}$ and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v2_carrier_pascal.py` — the carrier/Pascal packet (16=1+5+10, $N_{\rm fam}, \Omega_{\rm adm}, b_1$).
- `v3_em_alpha.py` — the EM fixed point $\alpha_\star^{-1}$=137.0359992168.
- `v4_flavor_matrix.py` — the residue matrix R (det =8, minors $\{2,3,5\}$, χ_R, SNF, $\sum L$=40).
- `v7_gravity_cosmo.py` — scalaron exponent 7 and inflation A_s, n_s, r; $\sin^2\theta_{12}$.

Run `python verification/run_all.py` (needs `mpmath, numpy, sympy`); exit code 0 iff all checks pass. Full map: `verification/README.md`.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

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- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
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- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
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